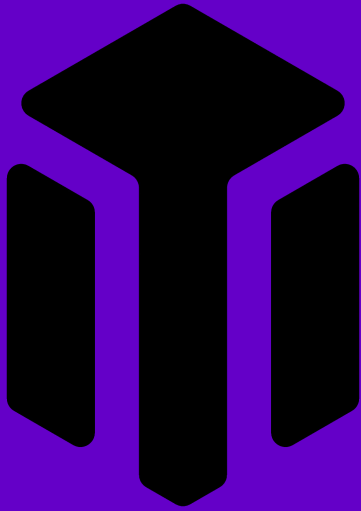




TIER 1

2WD | STOCK FRAME | STOCK PANELS | PRINTED MODULAR TOOLHEAD

If you want to run a printed modular toolhead (even with a “CNC” carriage), and you don’t plan on stiffening up the printer’s frame, then this is the way to go. We strongly recommend you to **build the gantry in the 2WD configuration for better performance** and an overall better experience. If you use AWD in this tier, you won’t see clean input shaper graphs and print quality will suffer.

TIER 2

AWD | STOCK FRAME | STRUCTURAL SIDE PANELS | PRINTED PERFORMANCE TOOLHEAD

This tier is already **capable of utilizing some of the additional effective belt stiffness that 9/10mm AWD offers**. Performance-oriented, non-modular toolheads usually have a better center of mass, and structural side panels stiffen up the printer’s frame where it matters the most.

TIER 3

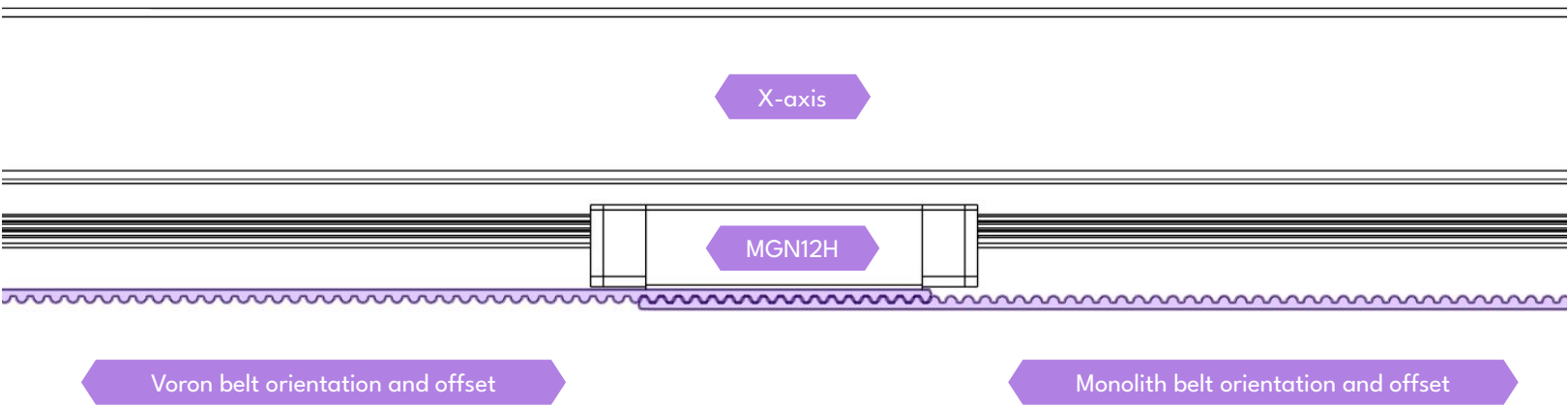
AWD | UPGRADED FRAME | STRUCTURAL PANELS | METAL PERFORMANCE TOOLHEAD | LIGHTWEIGHT ACCESSORIES

The primary objective of this tier is to make the motion system as **close to belt stiffness limited** as possible. Printers in this tier are capable of getting almost twice the recommended acceleration in AWD compared to 2WD with just as clean input shaper graphs.

BELT SPACING AND ORIENTATION

To achieve the short belt path in both 2WD and AWD configuration, the Monolith belt path is flipped compared to the stock one. The belt is also offset by another belt's thickness from the X-beam. This means that wrap-around style belt clips can work without modification.

For the best experience at high belt tension, use toolheads with native Monolith support.



Y-TRAVEL FLEXIBILITY

The Monolith Gantry is designed to get the most Y travel out of a wide range of frame configurations and toolhead nozzle offsets. This means, that the front corner loss with conventional toolheads in AWD can be less than with other AWD solutions.

Depending on your needs, you may not need the whole Y-overtravel this design can offer, which can result in a simpler assembly experience with less supporting mods.

Stock V2: bed size + 10mm

Monolith V2 default config: bed size + 13mm

Monolith V2 no protrusion config in 2WD: bed size + 7mm

Monolith V2 no protrusion config in AWD: bed size + 1mm

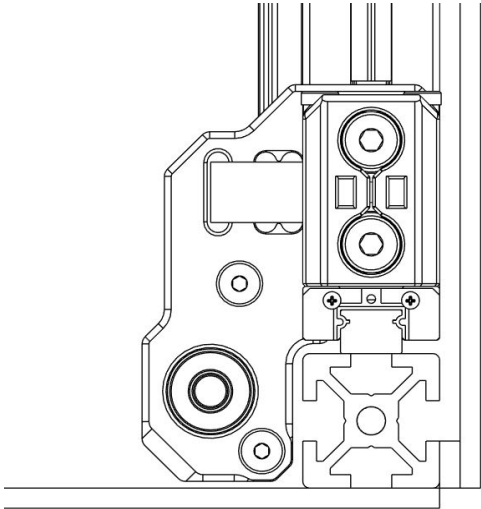
FT = Full Travel

NP = No Protrusion

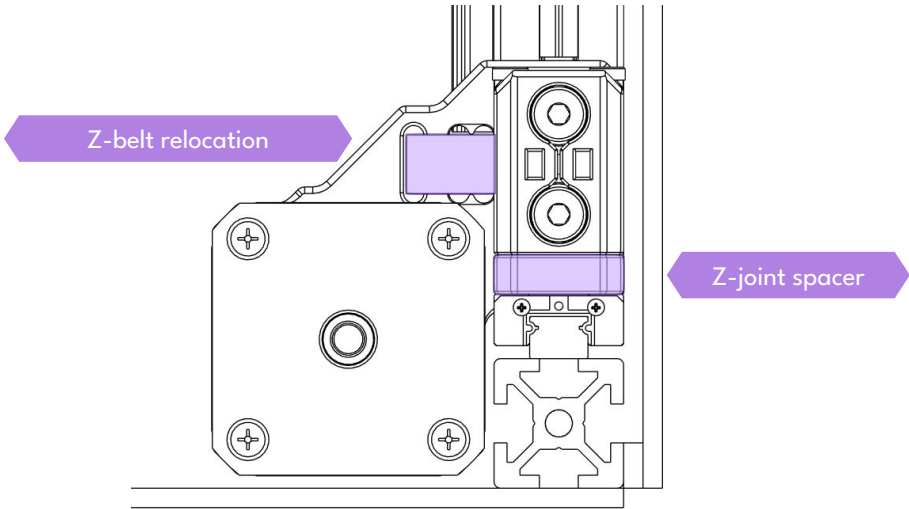
FRONT LAYOUTS COMPATIBLE WITH THE STOCK FRAME AND PANELS

The Monolith 2WD front idlers will always fit inside the stock frame with the stock panels.

In order to fit the front AWD steppers between the Z-belts and the stock door, you have to shift the Z-belt path and the motor mount inwards, which is the NP (No Protrusion) configuration.



Monolith 2WD

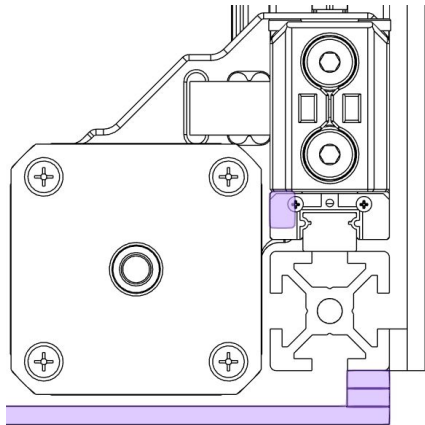


Monolith AWD in NP configuration

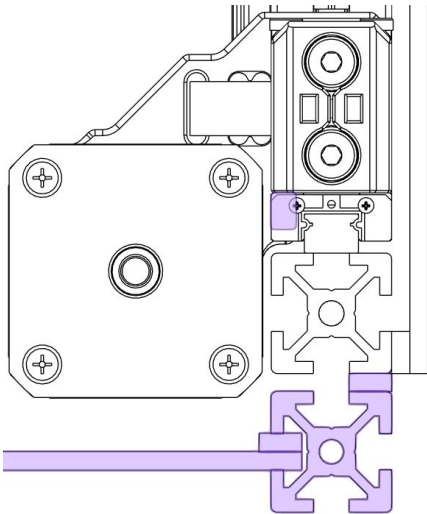
FRONT LAYOUTS NOT COMPATIBLE WITH THE STOCK FRAME AND PANELS

There are 3 possible ways to fit the front AWD mounts in FT (Full Travel) configuration.

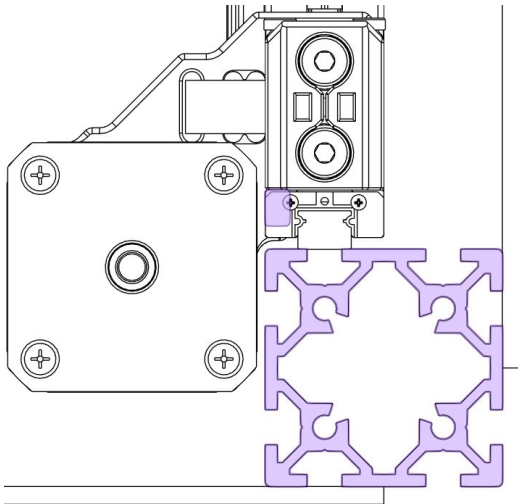
Monolith AWD in FT configuration



Door spaced out by 6mm



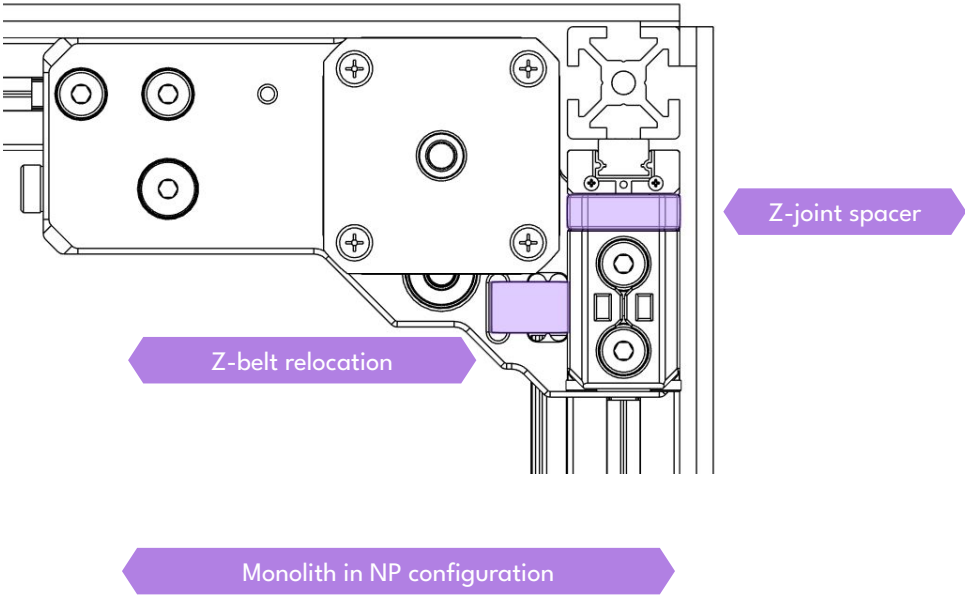
Extrusion door



4040 verticals

REAR LAYOUT COMPATIBLE WITH THE STOCK FRAME AND PANELS

In order to fit the rear steppers between the Z-belts and the stock rear panel, you have to shift the Z-belt path and the mount inwards, which is the NP (No Protrusion) configuration.



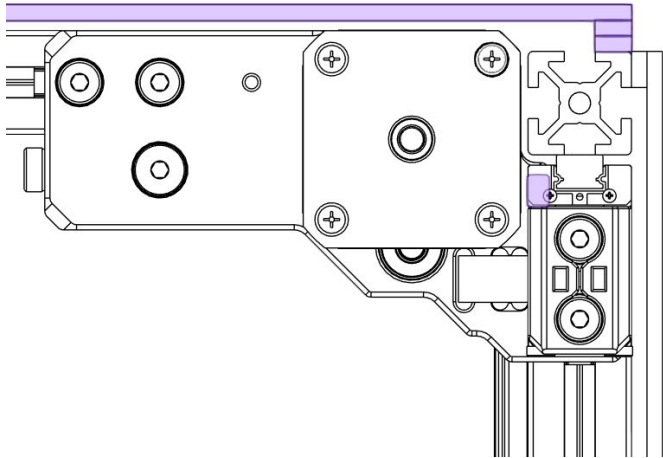
REAR CONFIGURATIONS

MONOLITH

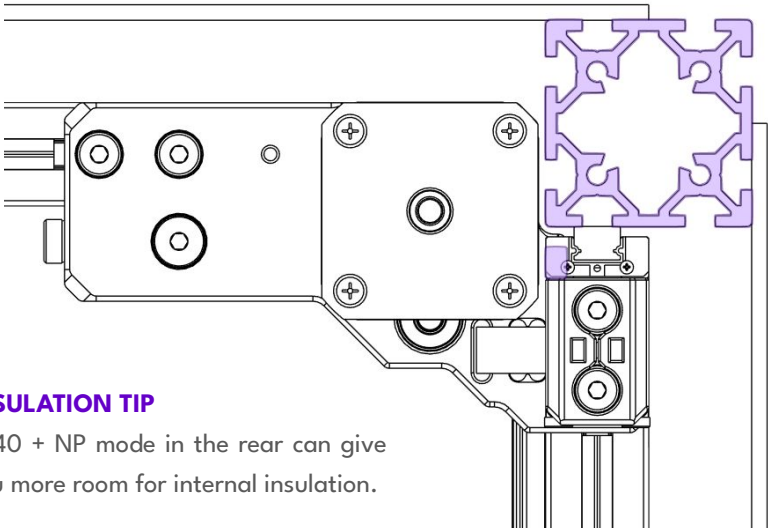
REAR LAYOUTS NOT COMPATIBLE WITH THE STOCK FRAME AND PANELS

There are 2 possible ways to fit the rear mounts in FT (Full Travel) configuration.

Monolith in FT configuration



Rear panel spaced out by 6mm



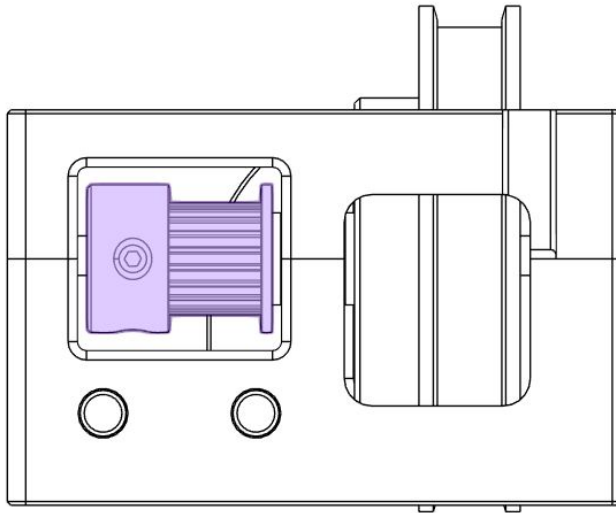
4040 vertical extrusions

INSULATION TIP

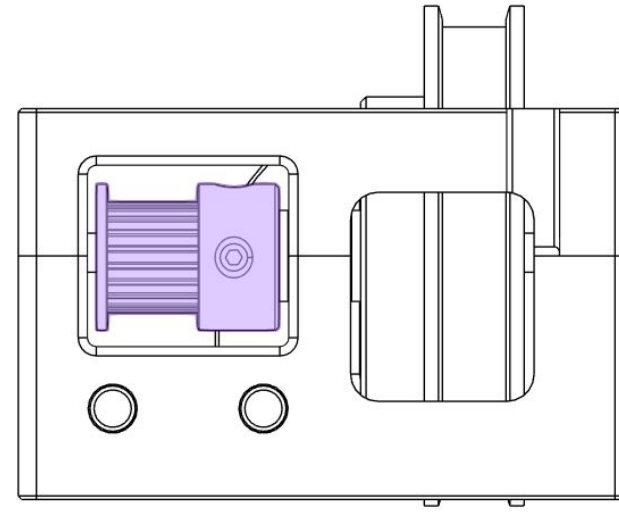
4040 + NP mode in the rear can give you more room for internal insulation.

FLIPPING THE Z-DRIVE PULLEYS

In order to fit the gantry inside the stock 2020 frame's footprint, the XY steppers have to fit between the panels and the Z-belts. Fortunately, the stock Voron Z-drives make it possible to shift the Z-belt path 6mm inwards by simply flipping the 20T pulley on the shaft. **Only do this if you want to build the gantry in No Protrusion mode, and only for the Z-drive pair (front and rear) where it's necessary.**



Stock/Monolith FT configuration

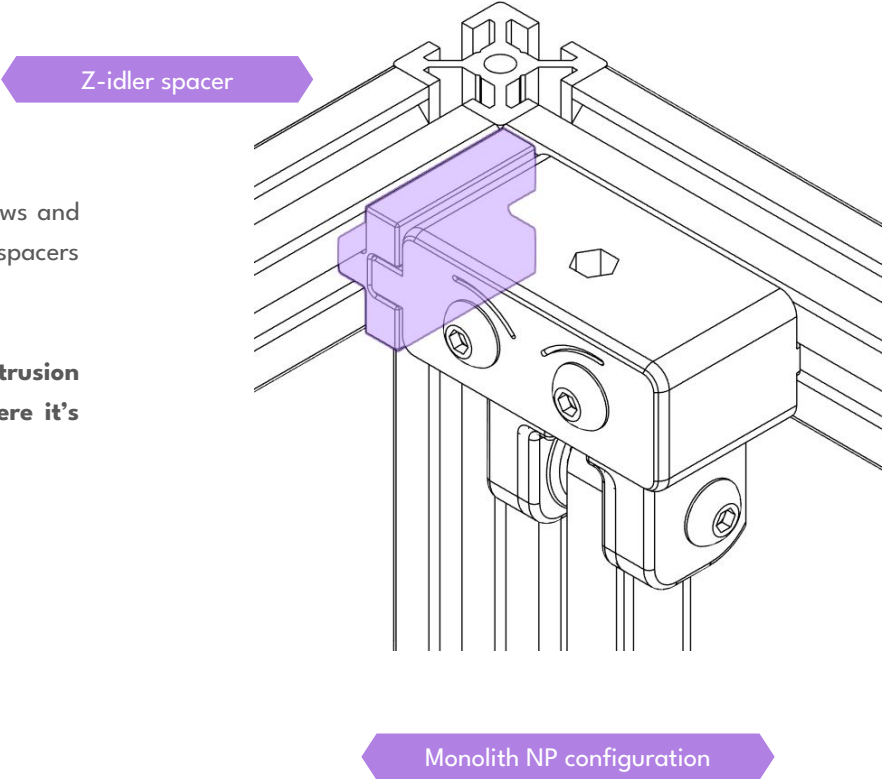


Monolith NP configuration

MOVING THE Z-IDLERS

Relocating the Z-idlers is easier, just loosen the two M5 screws and slide the idler in the extrusion slot. Use the 6mm thick printed spacers to align the idler with the new Z-belt path.

Only do this if you want to build the gantry in No Protrusion mode, and only for the Z-idler pair (front and rear) where it's necessary.

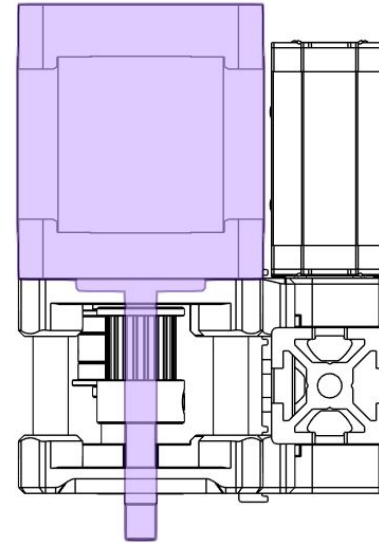


TOP BY DEFAULT

Monolith is designed in a way that allows for top and bottom stepper configurations, but the default layout is motors on top. This puts them farther away from the heated bed and the stepper heat doesn't get trapped as much. This orientation is also better for toolhead clearance and for maintenance, as the steppers won't fall out when half of the mount is removed.

LEADSCREW CLEARANCE

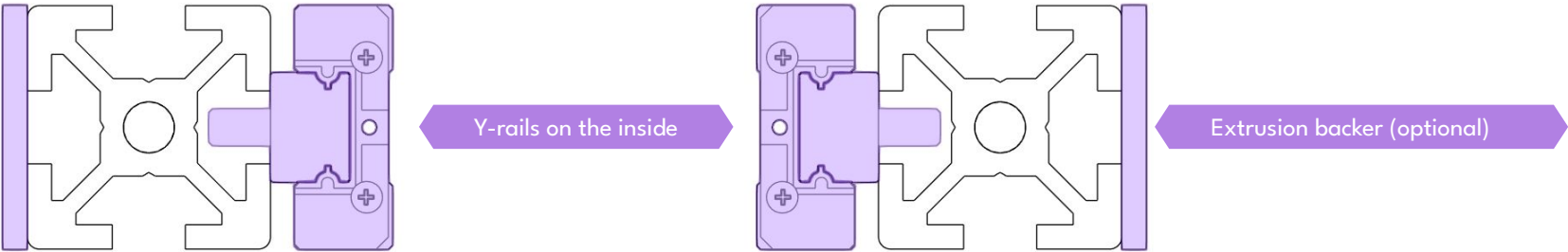
The VT kit front steppers are the exceptions, they have to be mounted on top, otherwise the leadscrews are in the way.



Y-RAILS AND BACKERS

The Monolith Gantry layout has its Y-rails on the inside and not on the top/bottom like in the stock Voron configuration. To switch to this layout on the V2, just rotate the Y-extrusions. On the fixed gantry Trident, it's easier to move the rails to the inside without touching the frame.

Any optional extrusion backers have to be installed on the outside. Please DO NOT put the backers on top for aesthetic reasons.

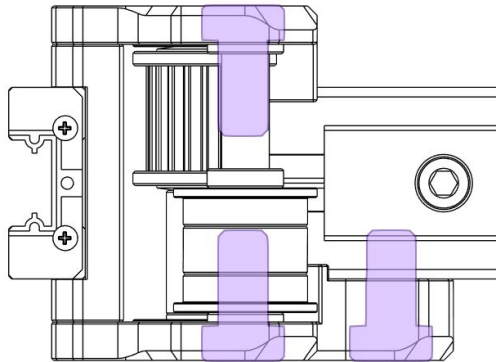


XY-JOINT ORIENTATION

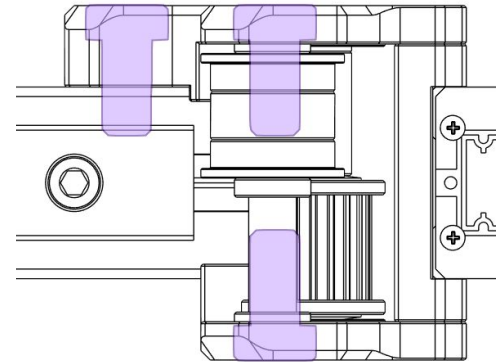
MONOLITH

2 + 1

The Monolith Gantry is compatible with all X-beams made for Voron V2 and Trident, as it uses the same 2 + 1 screw layout. The XY-joints can be flipped the same way as well. Some toolheads may need more room at bottom, while others need more at the top. By installing the right XY-joint on the left (and vice versa) you can opt for the 2 screws at the bottom.



Right XY-joint on the left



Right XY-joint on the right

GANTRY-MOUNTED SWITCHES

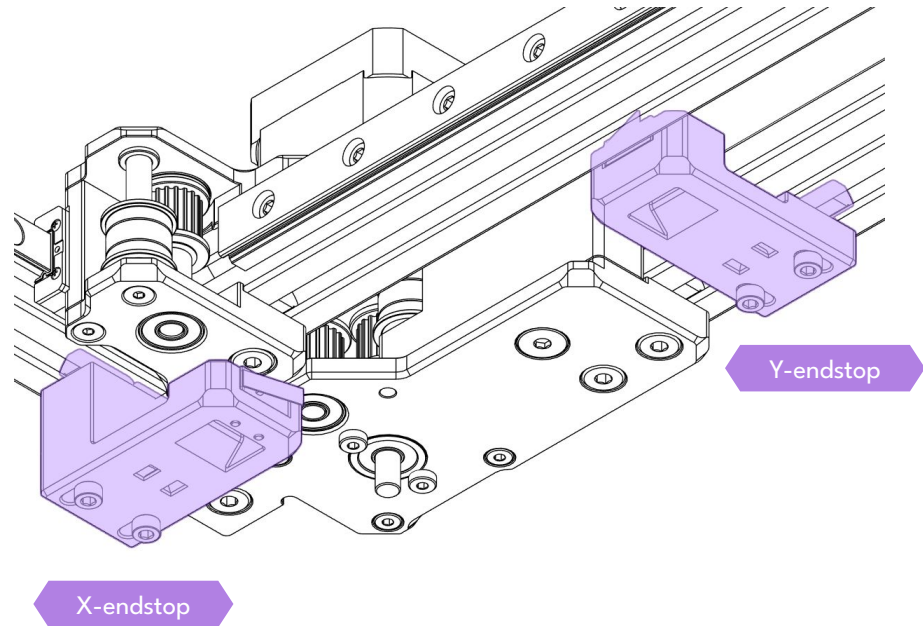
Sensorless homing has come a long way, especially with Kalico, but if you want to use endstop switches, it's supported.

Y: The Y-endstop lives on the rear extrusion, between the rear mounts. It's triggered by the X-beam.

X: Since there's a wide variety of toolheads and XY-joint designs in this ecosystem, there's no universal X-switch location. To offer a potential solution to this problem from the gantry side, there's a Y-extrusion mounted X-switch holder that can work with any toolhead.

HOMING SEQUENCE

To make the gantry-mounted X-endstop work, you'll have to home the Y-axis first

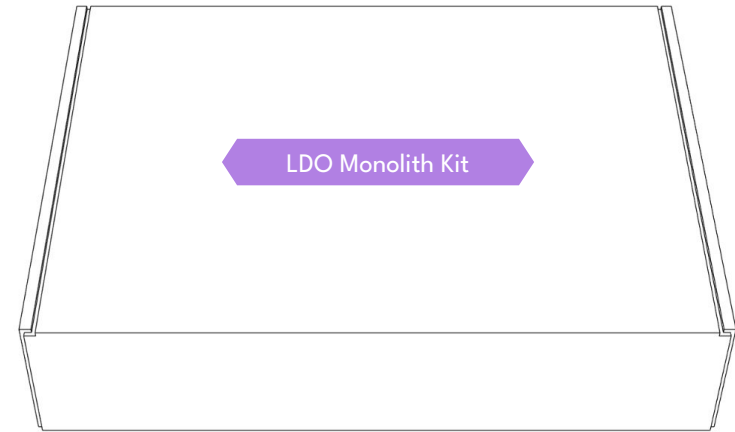


LET'S BUILD!

CHECK THE BOM

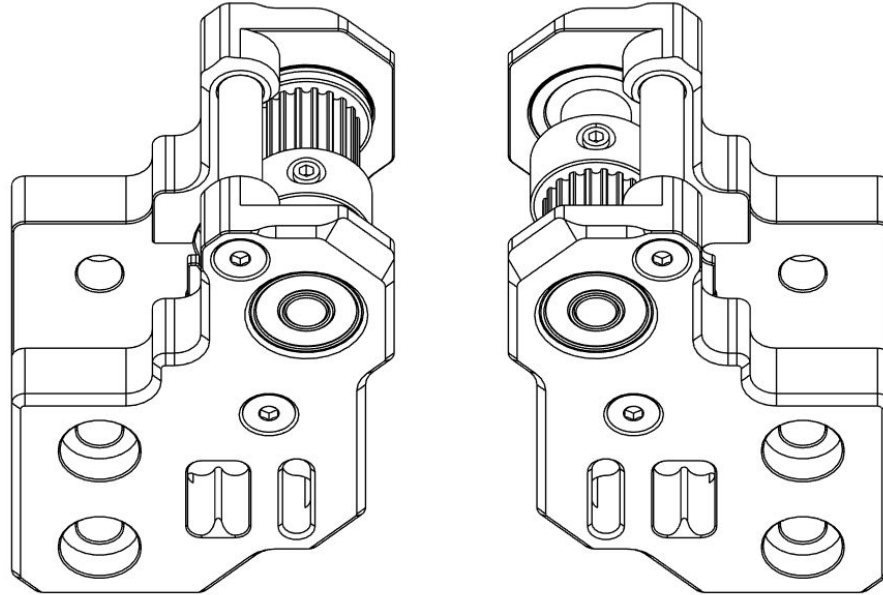
There's a list inside with everything that should be in the package.
Now is a good time to go over this list to see if anything is missing.

The milled parts are also laser engraved, so they're easy to identify.



FRONT IDLERS (2WD only)

MONOLITH



FRONT IDLERS (2WD only)

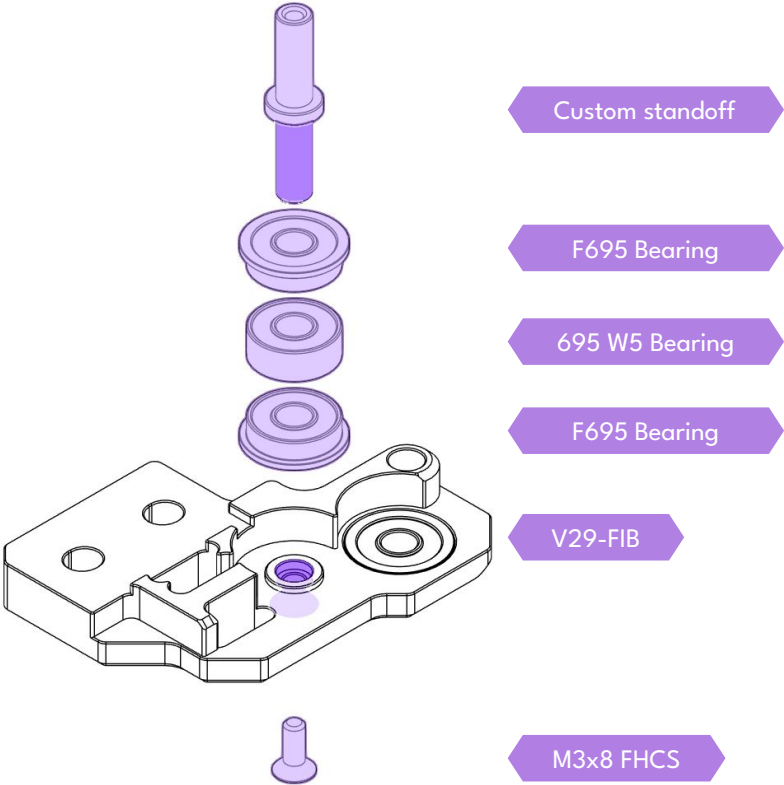
MONOLITH

INSTALL THE IDLER STACK

Slide the 3 bearings onto the standoff, then secure them with a screw.

DO IT TWICE

You may have noticed that there are 2 identical top and bottom front idler parts. This is because the right and the left side are the same, just flipped. Repeat the steps for both sides!

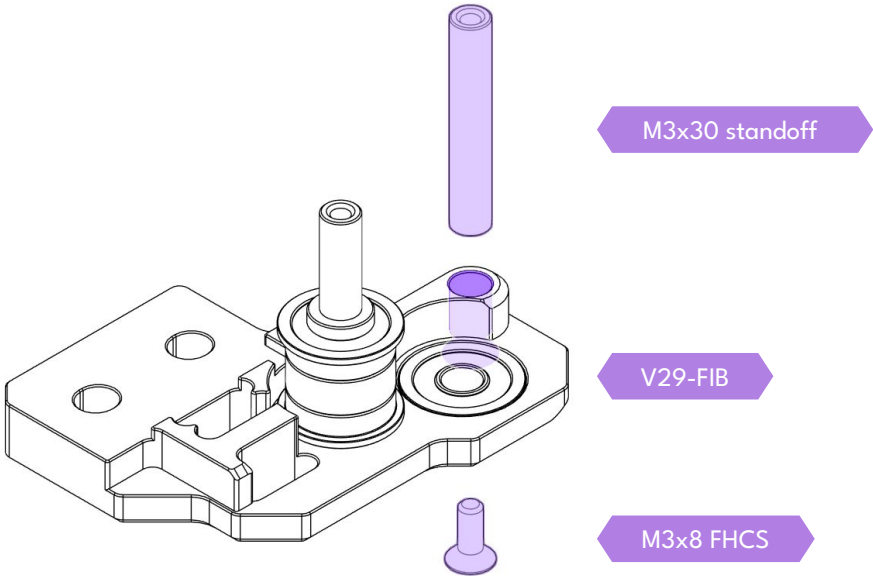


FRONT IDLERS (2WD only)

MONOLITH

INSTALL THE STANDOFF

Secure the standoff with a screw.

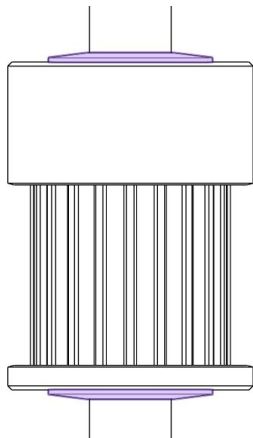


FRONT IDLERS (2WD only)

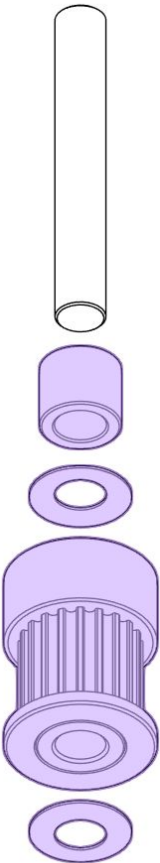
MONOLITH

FRONT LIVE-SHAFT ASSEMBLY

Put together the live-shaft assembly for the front 2WD idlers. Make sure the disc springs are in the correct orientation.



Disc spring orientation



36mm pin

M5x7 spacer

M5 disc spring

20T 9mm pulley

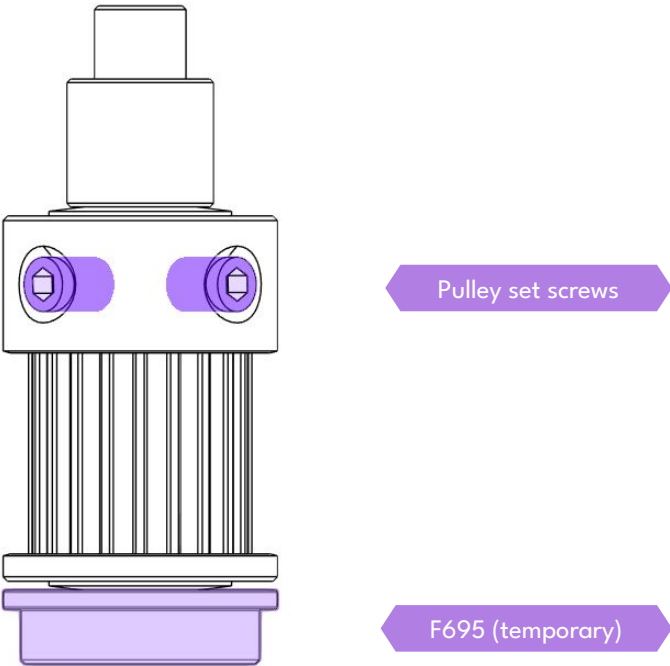
M5 disc spring

FRONT IDLERS (2WD only)

MONOLITH

FRONT LIVE-SHAFT ASSEMBLY

Slide the shaft into an F695 bearing and put the assembly on a flat surface. With everything flush at the bottom, tighten the pulley set screws. The F695 only used for alignment can now be removed.

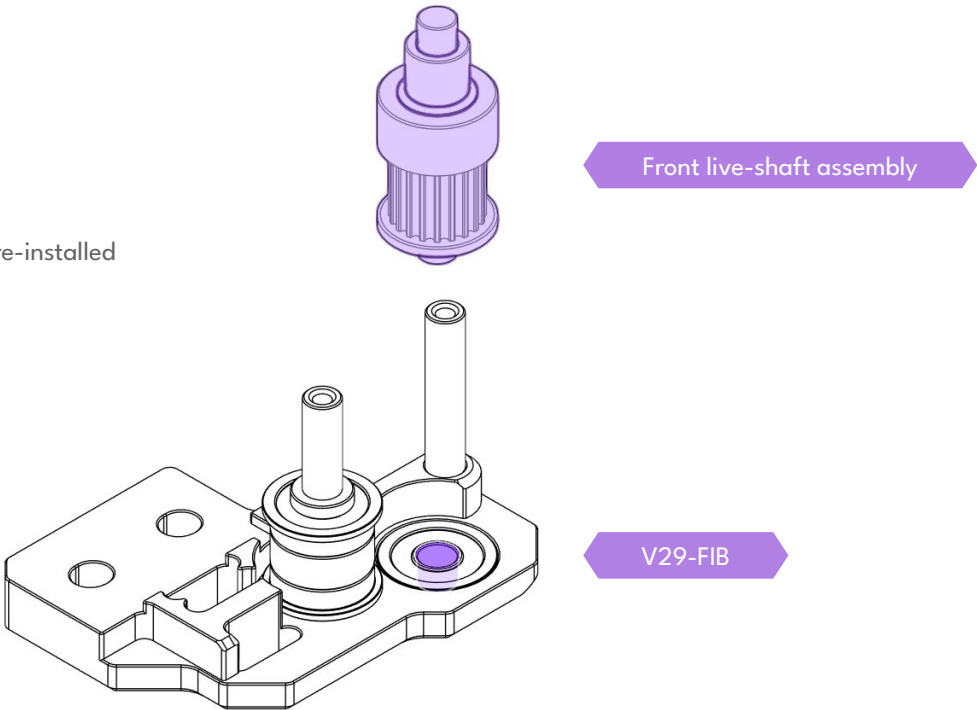


FRONT IDLERS (2WD only)

MONOLITH

LIVE-SHAFT TIME

Insert the previously prepared live-shaft assembly into the pre-installed F695 bearing with the idler on the bottom.



FRONT IDLERS (2WD only)

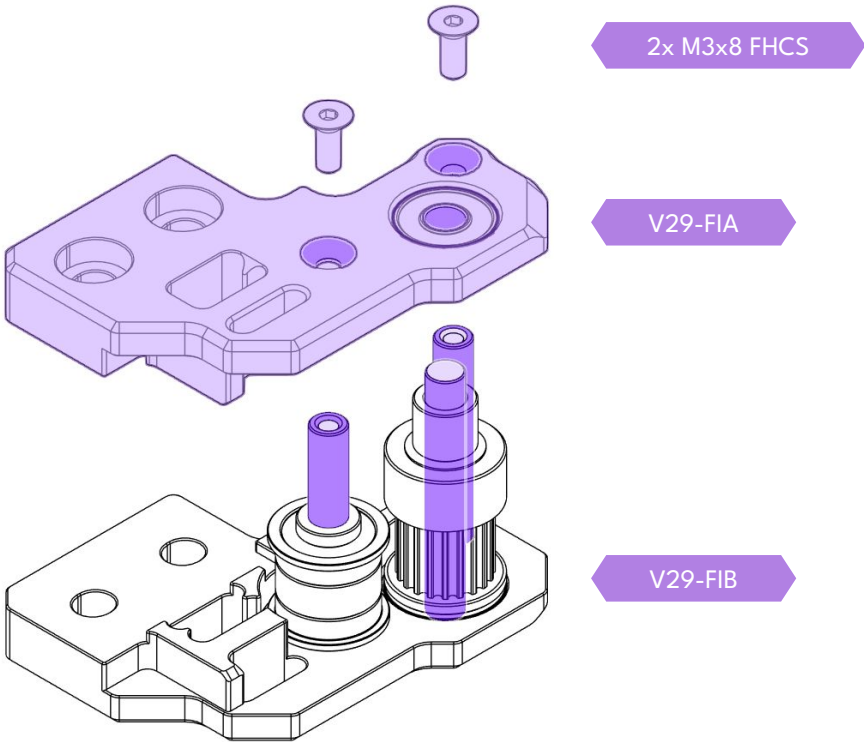
MONOLITH

TOP IT OFF

Get the top part of the front idler and gently put in on the standoffs and the live-shaft. The fit will be the tightest at the bearing, so make sure you're not forcing it at an angle.

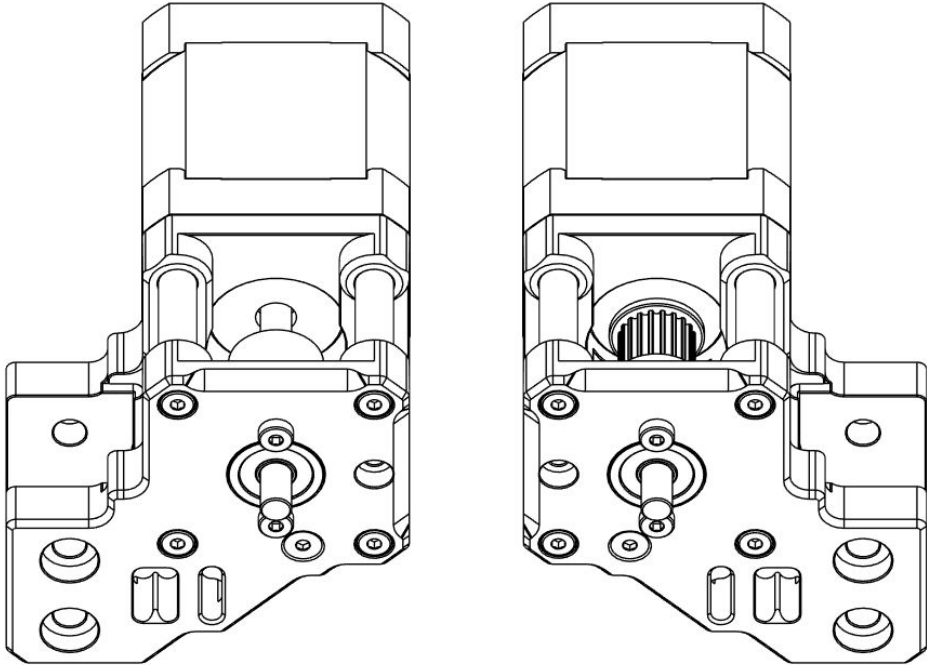
DO IT TWICE

In case you forgot to do the 2WD front idler steps for both sides. Repeat all steps for the other side!



FRONT MOUNTS (AWD only)

MONOLITH

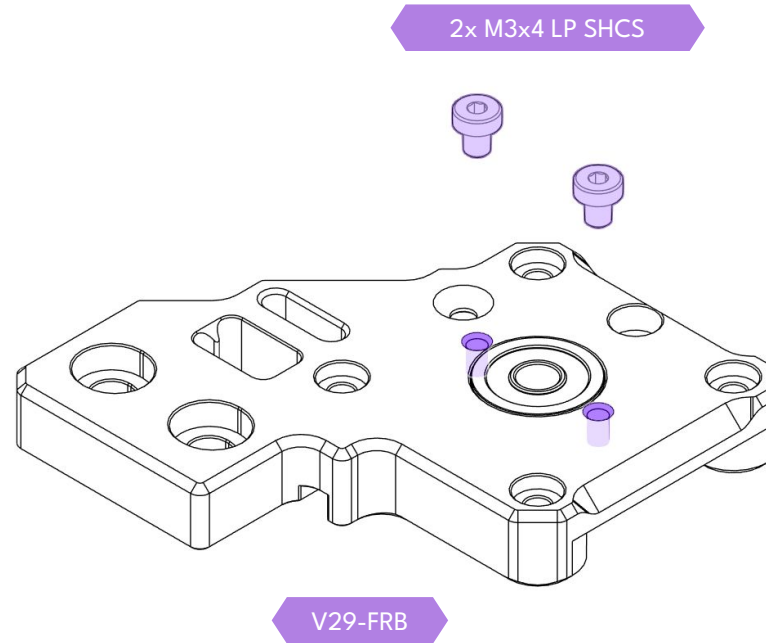


FRONT RIGHT MOUNT (AWD only)

MONOLITH

SECURE THE BEARING

The flanged bearings should come pre-installed. Secure the stepper double-shear bearing with 2 screws.

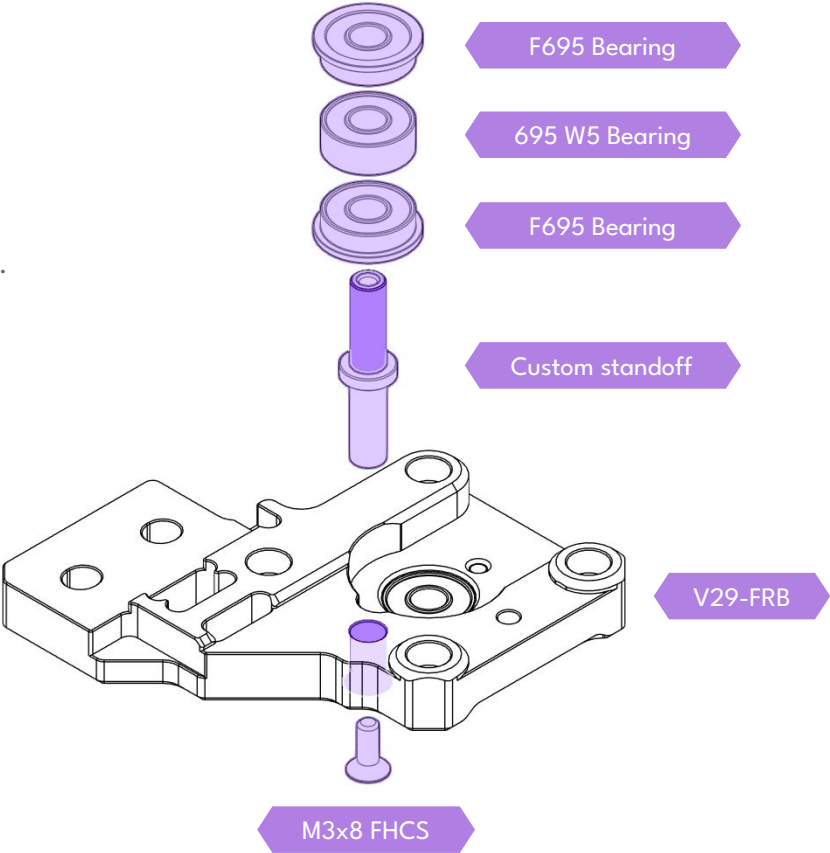


FRONT RIGHT MOUNT (AWD only)

MONOLITH

INSTALL THE IDLER STACK

First insert and secure the standoff, then slide on the 3 bearings.

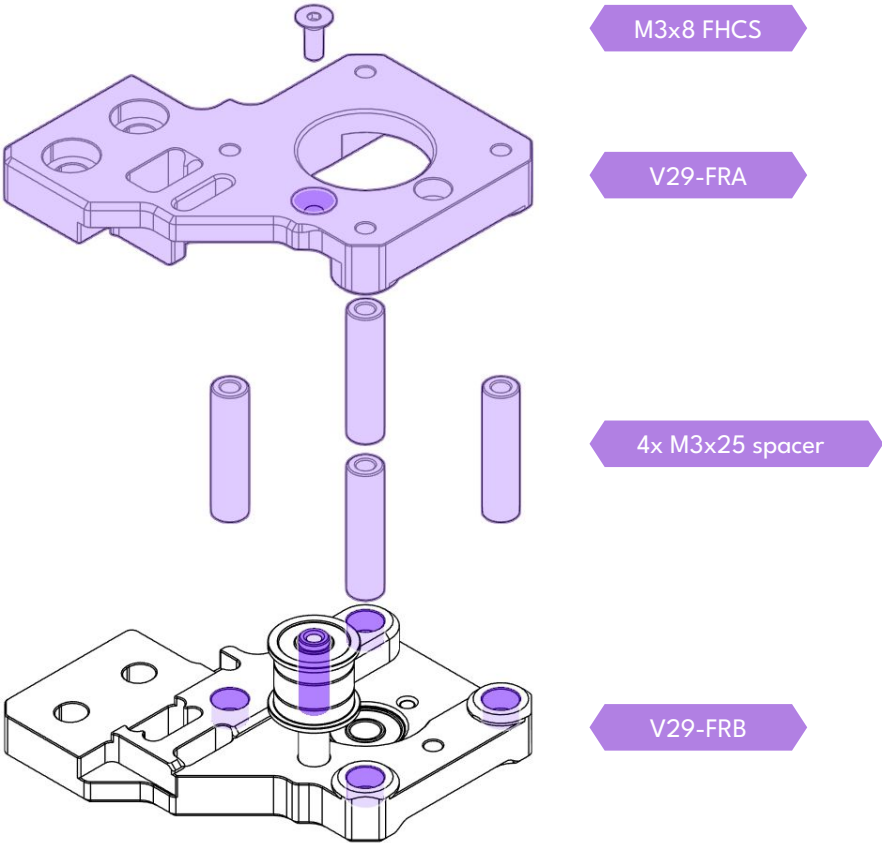


FRONT RIGHT MOUNT (AWD only)

MONOLITH

THE TOP HALF

Place the four M3x25 spacers, then secure the top milled part with the other idler screw.

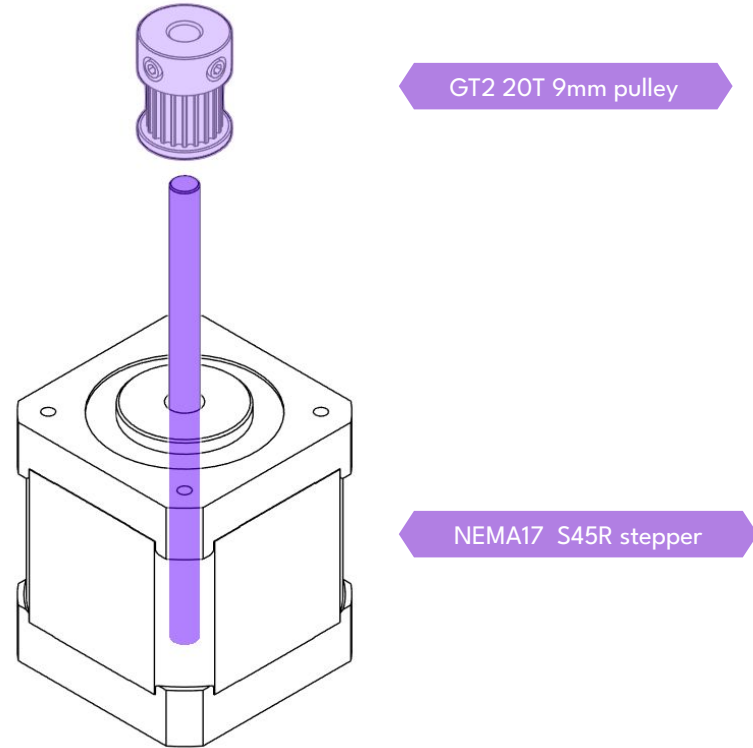


FRONT RIGHT MOUNT (AWD only)

MONOLITH

THE DRIVE PULLEY

Slide the pulley onto the stepper shaft with its hub facing up, but DO NOT tighten the set screws yet.



FRONT RIGHT MOUNT (AWD only)

MONOLITH

THE STEPPER MOTOR

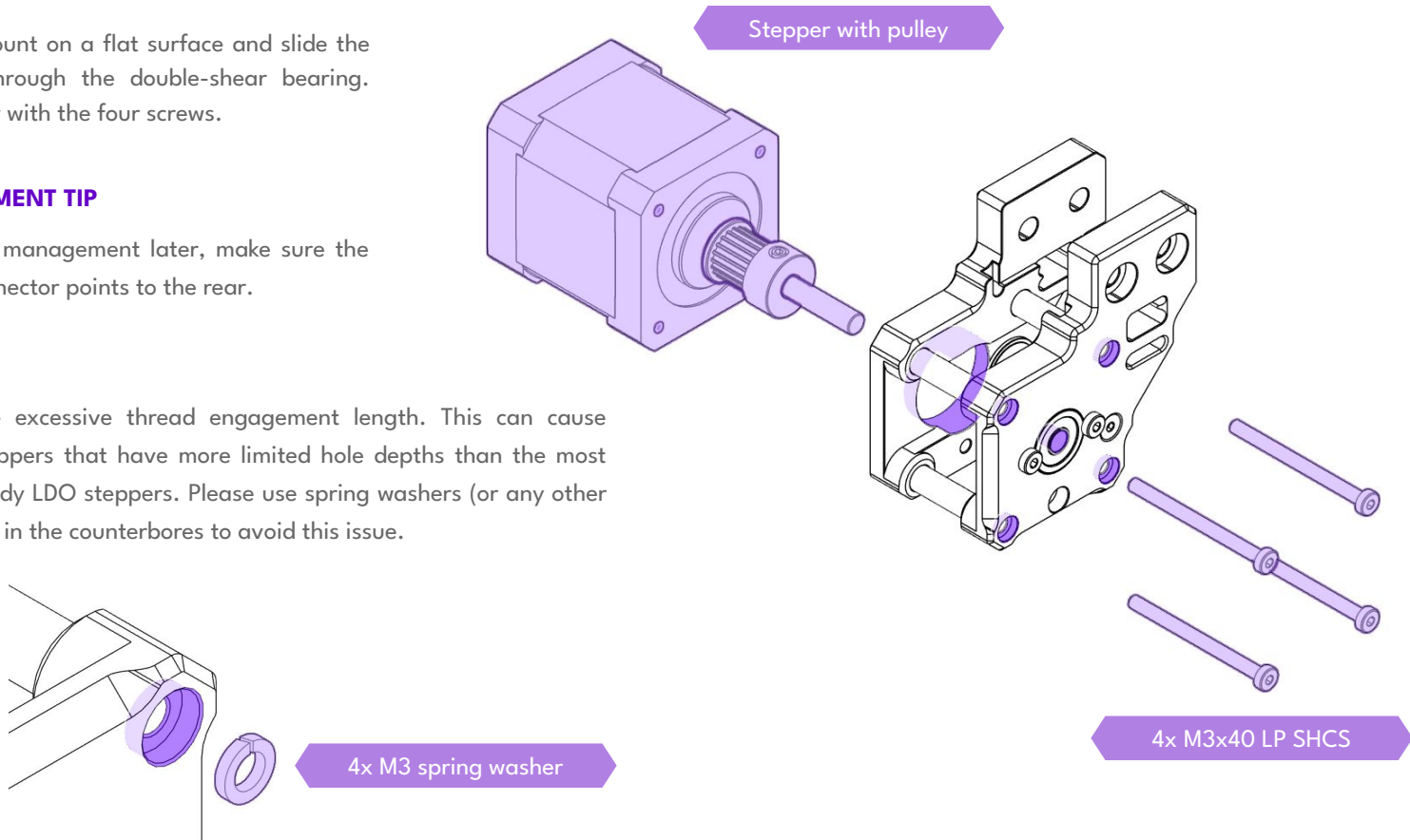
Place the front mount on a flat surface and slide the stepper's shaft through the double-shear bearing. Secure the stepper with the four screws.

CABLE MANAGEMENT TIP

For optimal cable management later, make sure the stepper cable/connector points to the rear.

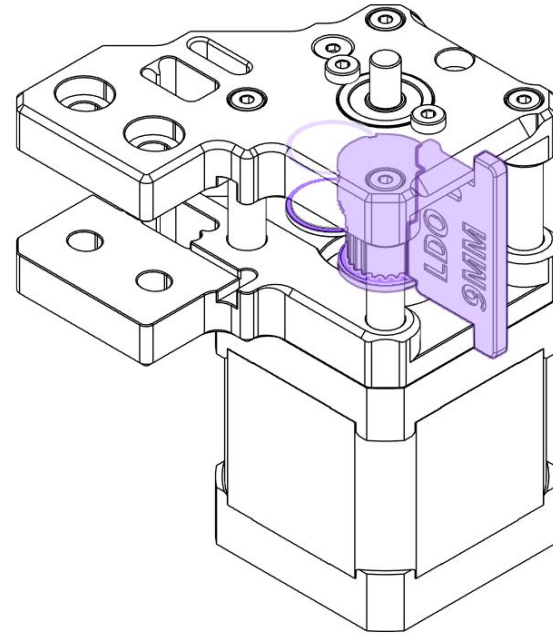
USE WASHERS

Batch 1 kits have excessive thread engagement length. This can cause problems with steppers that have more limited hole depths than the most common 48mm body LDO steppers. Please use spring washers (or any other small OD washers) in the counterbores to avoid this issue.



ALIGN AND SECURE THE PULLEY

The flanges of the pulley should line up with the smooth bearing stack. There's also a printed tool to help with the alignment. It will keep everything in place while you tighten the pulley set screws.

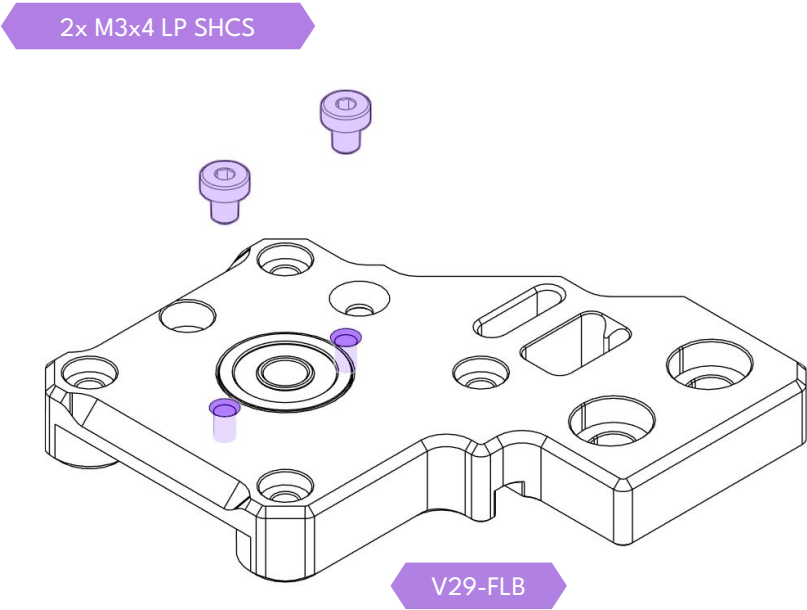


FRONT LEFT MOUNT (AWD only)

MONOLITH

SECURE THE BEARING

The flanged bearings should come pre-installed. Secure the stepper double-shear bearing with 2 screws.

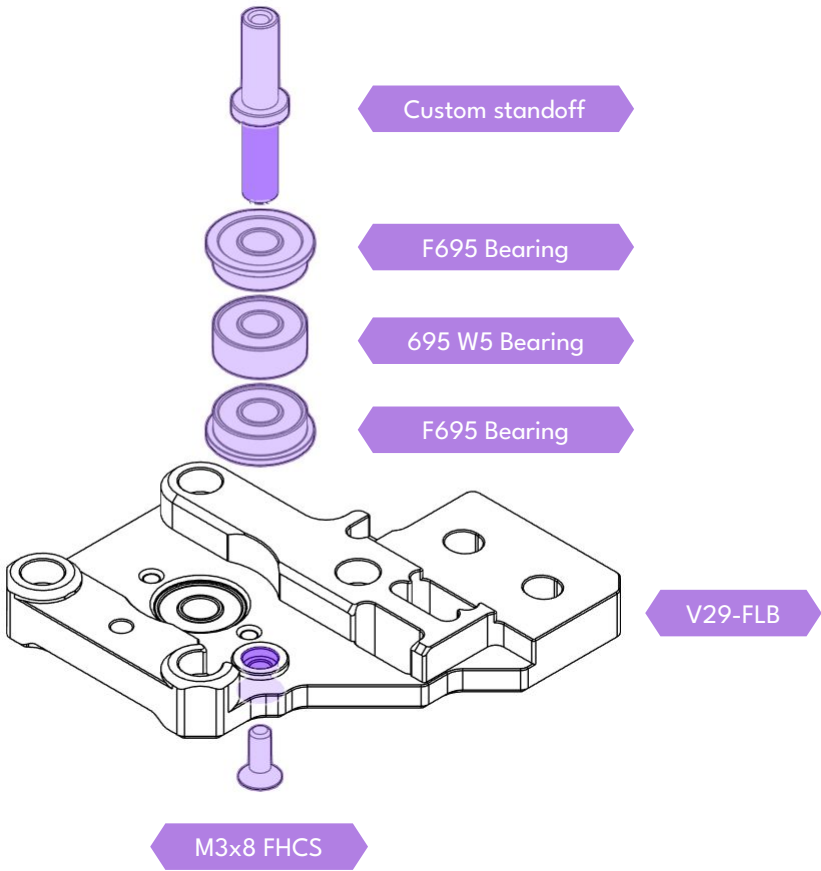


FRONT LEFT MOUNT (AWD only)

MONOLITH

INSTALL THE IDLER STACK

Slide the 3 bearings onto the standoff, then secure them with a screw.

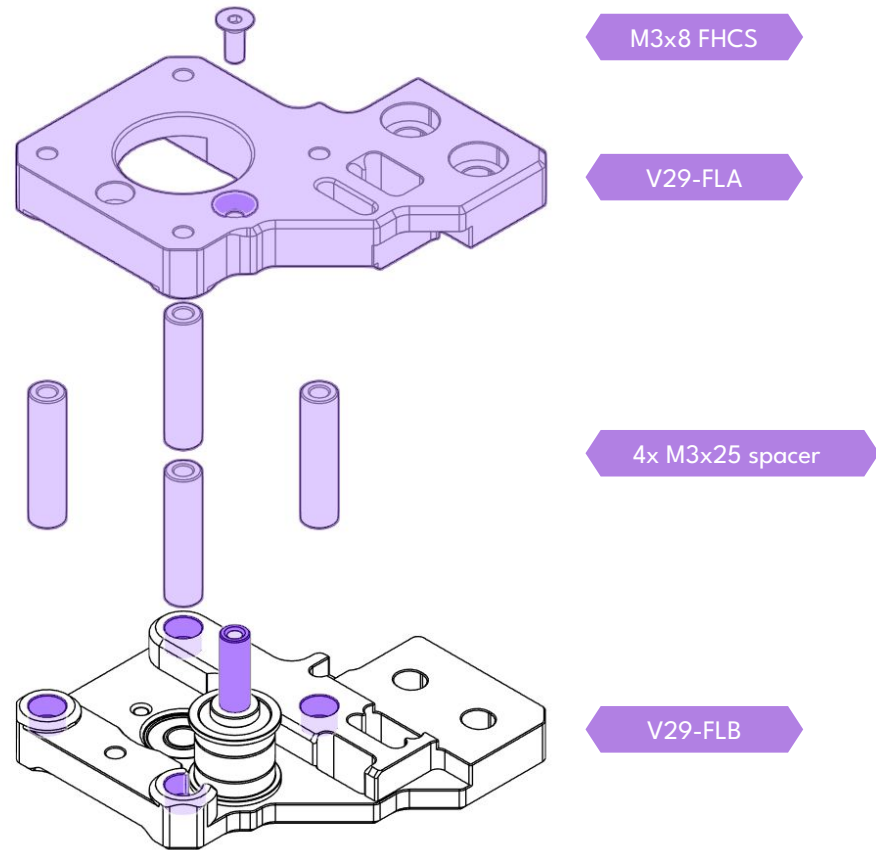


FRONT LEFT MOUNT (AWD only)

MONOLITH

THE TOP HALF

Place the four M3x25 spacers, then secure the top milled part with the other idler screw.

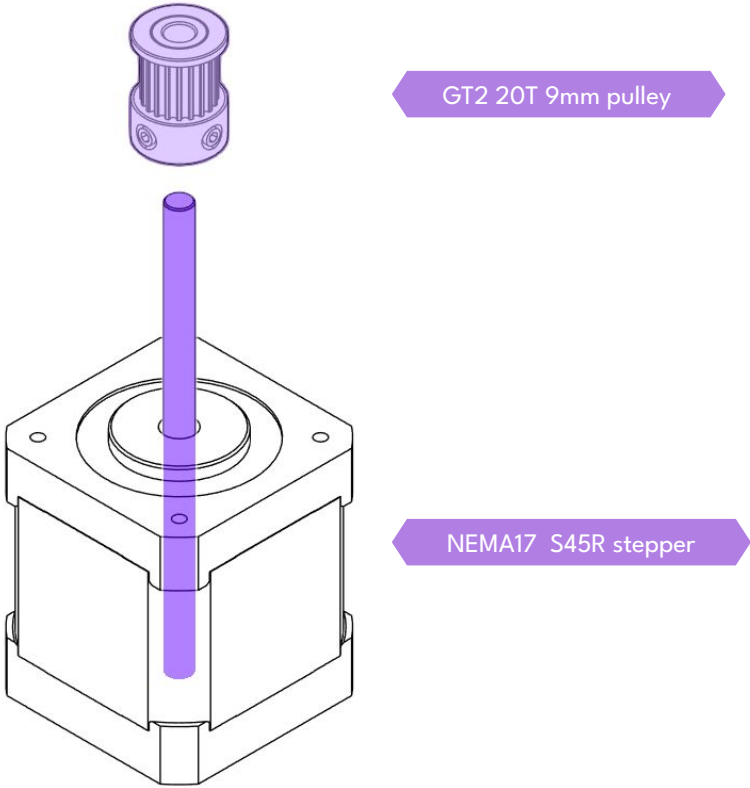


FRONT LEFT MOUNT (AWD only)

MONOLITH

THE DRIVE PULLEY

Slide the pulley onto the stepper shaft with its hub facing down, but DO NOT tighten the set screws yet.



FRONT LEFT MOUNT (AWD only)

MONOLITH

THE STEPPER MOTOR

Place the front mount on a flat surface and slide the stepper's shaft through the double-shear bearing. Secure the stepper with the four screws.

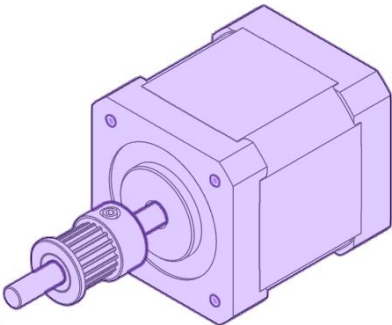
CABLE MANAGEMENT TIP

For optimal cable management later, make sure the stepper cable/connector points to the rear.

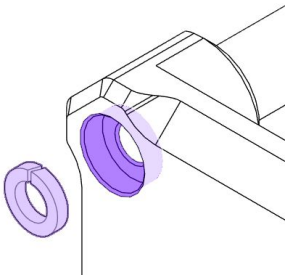
USE WASHERS

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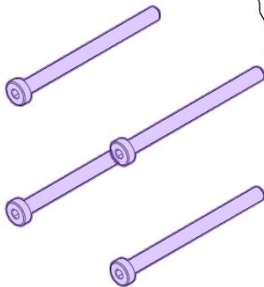
Stepper with pulley



4x M3 spring washer

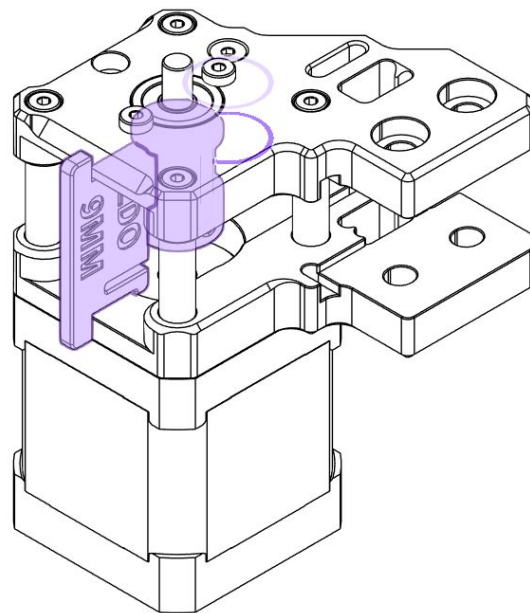


4x M3x40 LP SHCS



ALIGN AND SECURE THE PULLEY

The flanges of the pulley should line up with the smooth bearing stack. There's also a printed tool to help with the alignment. It will keep everything in place while you tighten the pulley set screws.

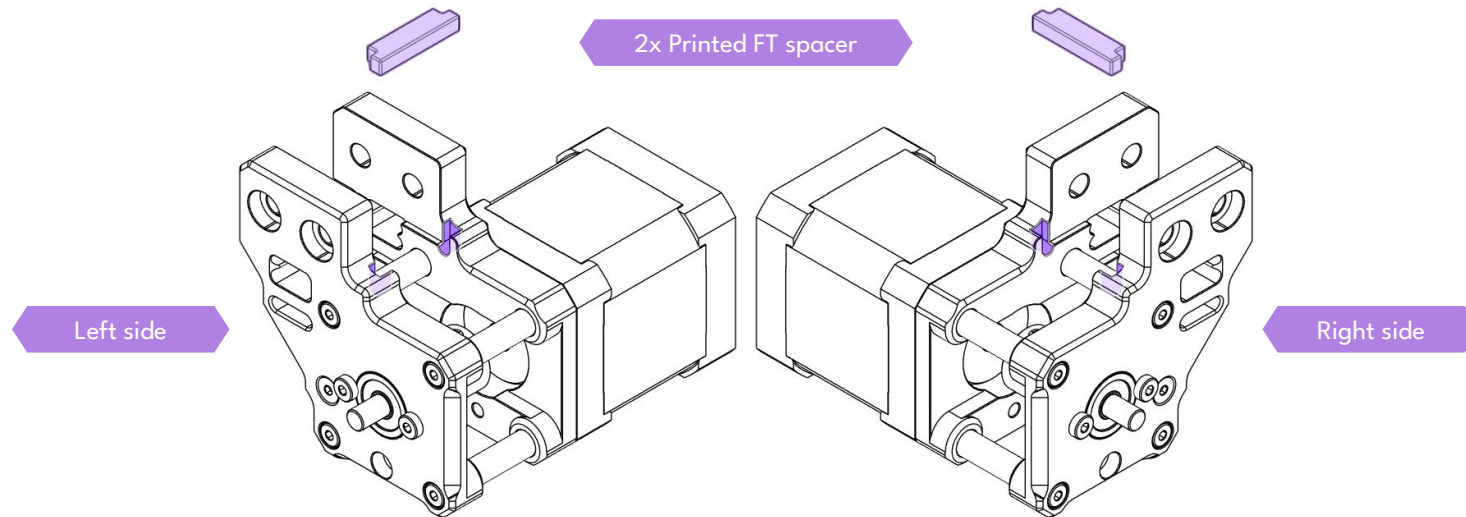


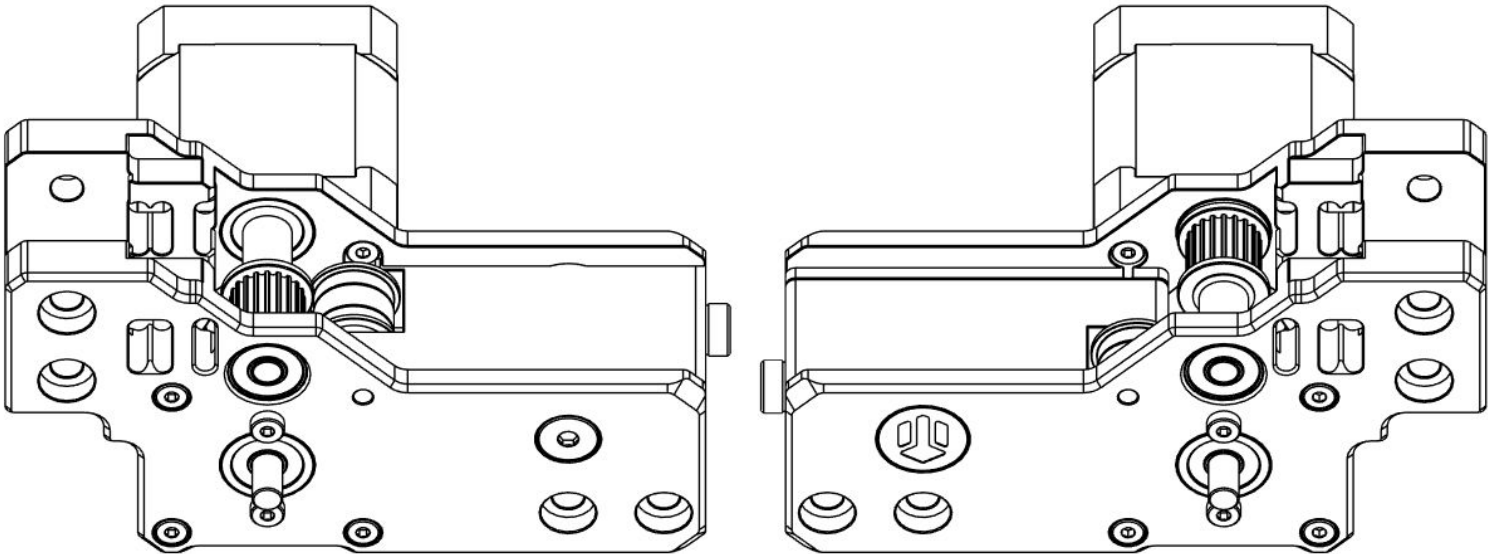
FRONT MOUNTS (AWD only)

MONOLITH

PRINTED FT SPACERS (optional)

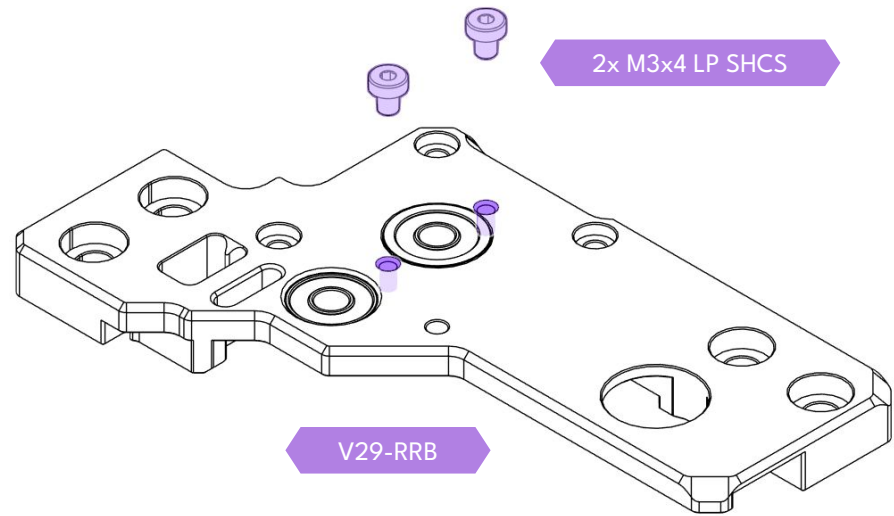
If you chose the FT (Full Travel) configuration for the mounts, now it's time to install the printed spacers which will correctly position the assembly relative to the Y-extrusion. Note the orientation of the printed part. **Skip these if you chose NP (No Protrusion) and want to shift the Z-belt path.**





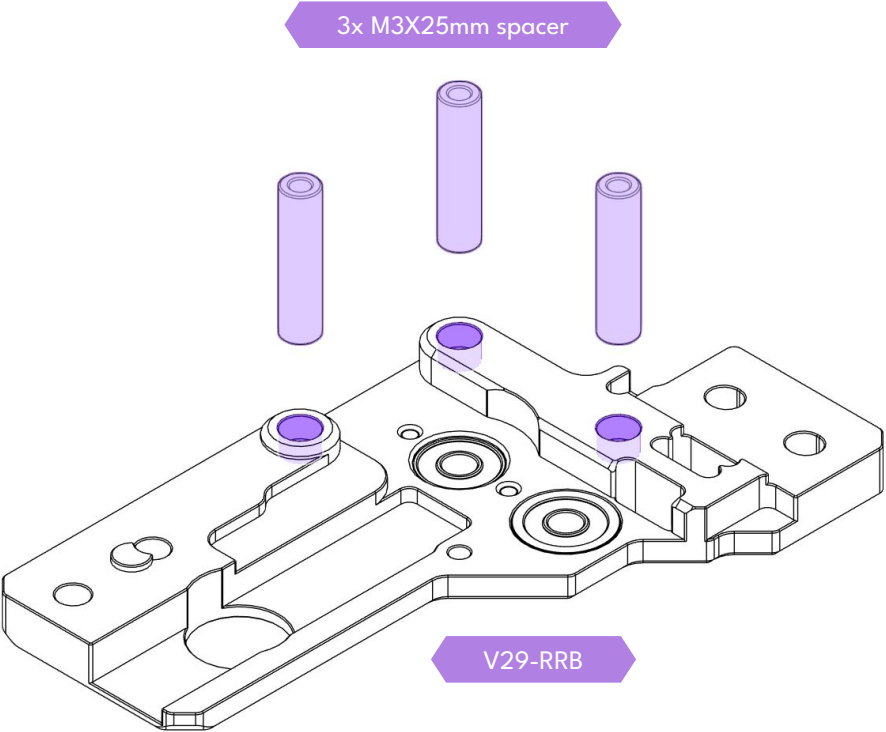
SECURE THE BEARING

The flanged bearings should come pre-installed. Secure the stepper double-shear bearing with 2 screws.



GET THE SPACERS

Place the 25mm long M3 spacers into the holes.

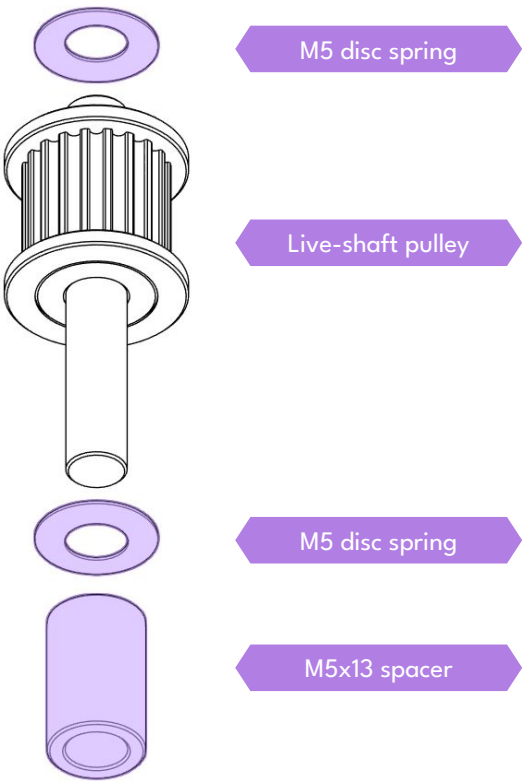
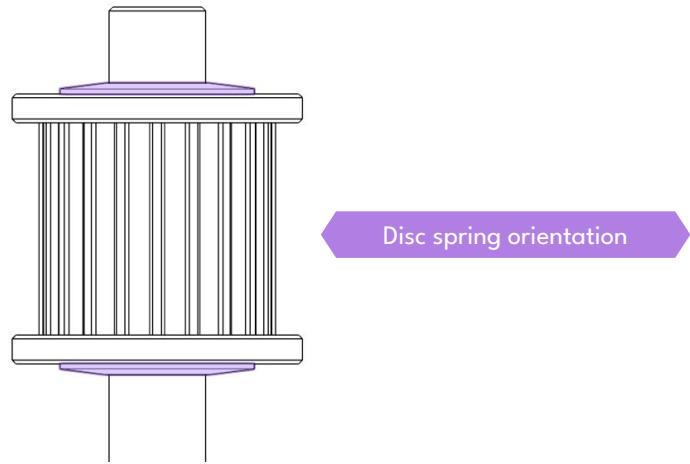


REAR RIGHT MOUNT

MONOLITH

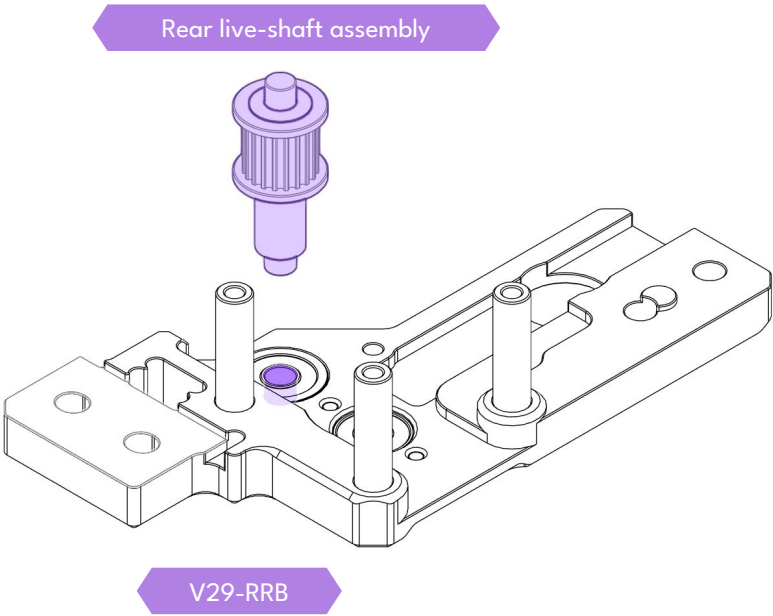
REAR LIVE-SHAFT ASSEMBLY

Put together the live-shaft assembly for the rear mount. Make sure the disc springs are in the correct orientation.



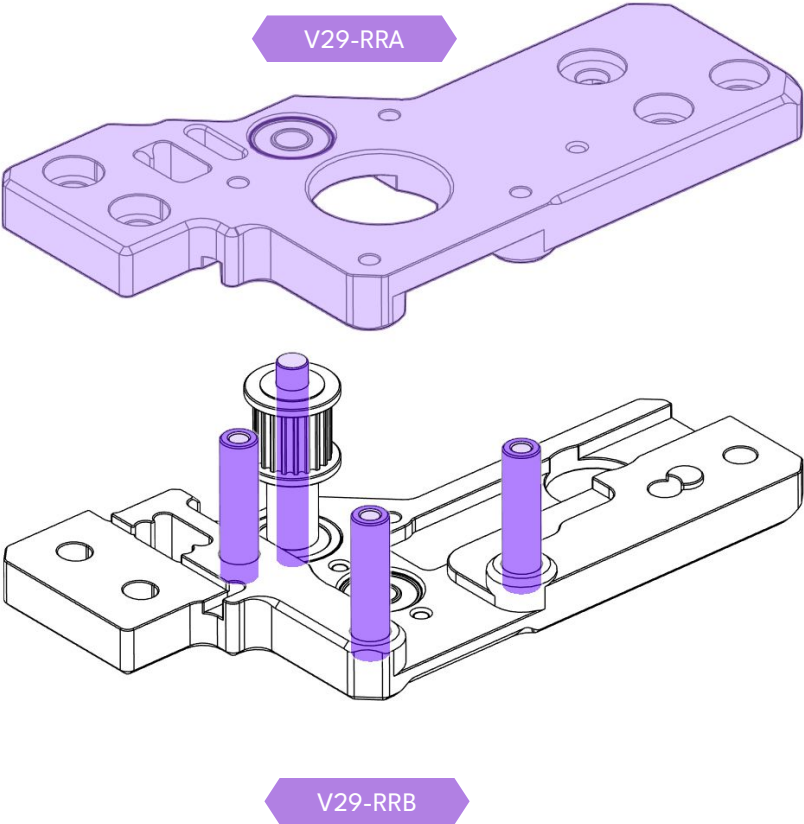
LIVE-SHAFT TIME

Insert the previously prepared live-shaft assembly into the pre-installed F695 bearing with the idler on the top.



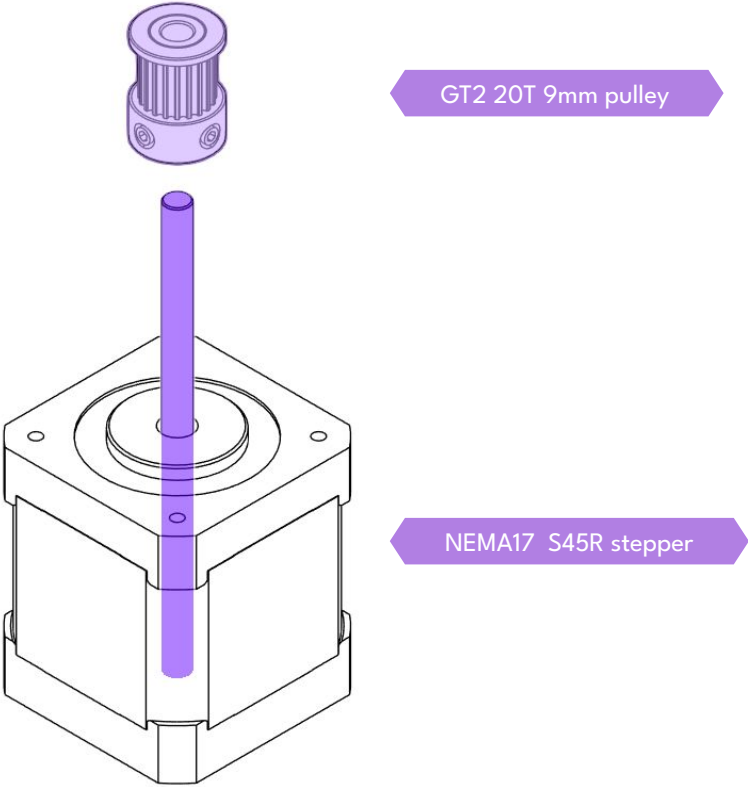
TOP IT OFF

Get the top part of the motor mount and gently put in on the spacers and the live-shaft. The fit will be the tightest at the bearing, so make sure you're not forcing it at an angle.



THE DRIVE PULLEY

Slide the pulley onto the stepper shaft with its hub facing down, but DO NOT tighten the set screws yet.



REAR RIGHT MOUNT

MONOLITH

CABLE MANAGEMENT TIP

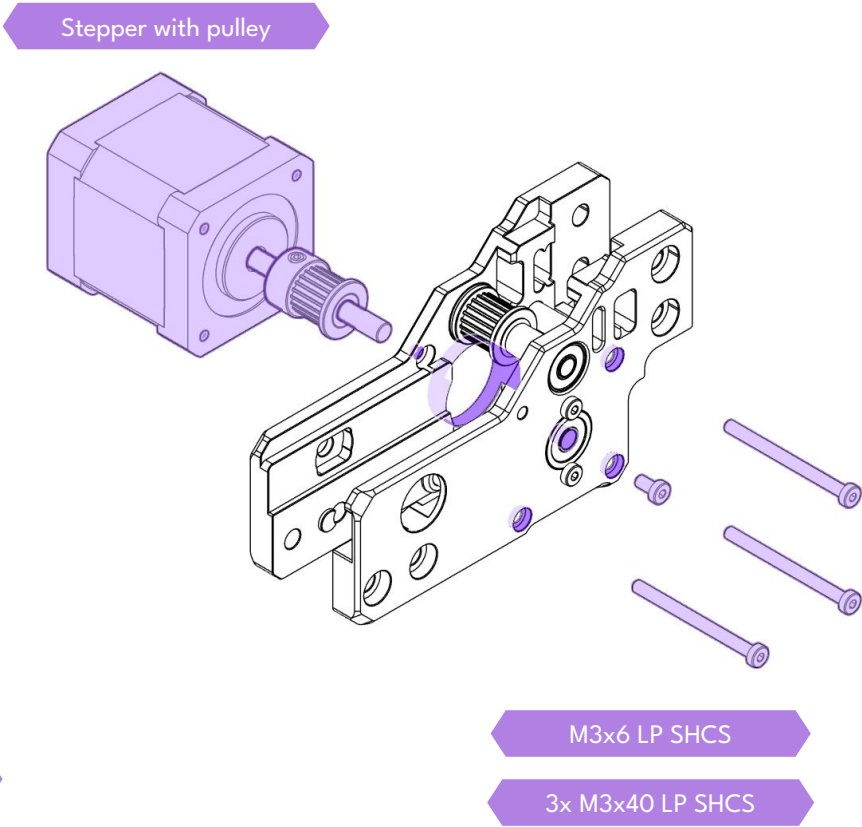
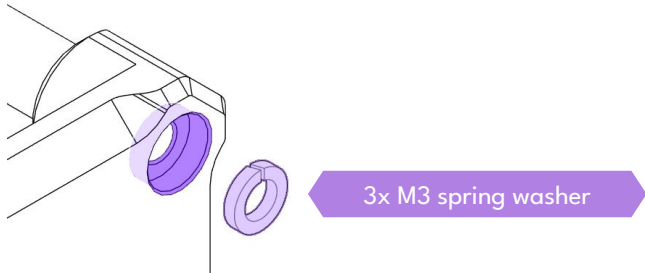
For optimal cable management later, make sure the stepper cable/connector points toward the rear extrusion mounting holes.

MOTORIZED

Put the previous assembly on its side on a flat surface, but be careful, because it can still fall apart. Then secure the stepper with the 4 screws.

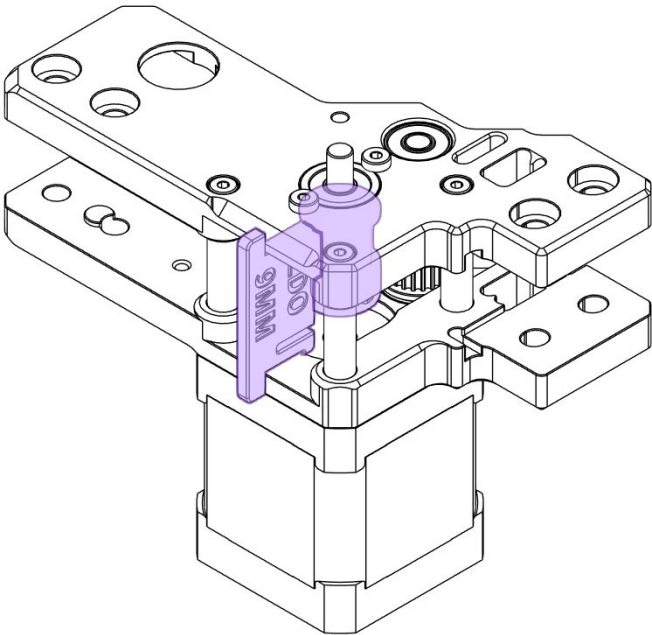
USE WASHERS

Batch 1 kits have excessive thread engagement length. This can cause problems with steppers that have more limited hole depths than the most common 48mm body LDO steppers. Please use spring washers (or any other small OD washers) in the counterbores to avoid this issue.



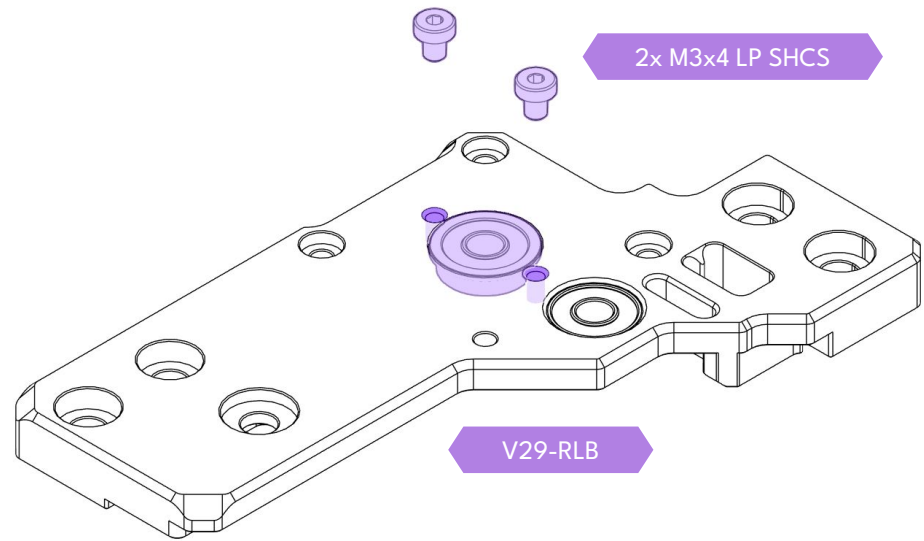
ALIGN AND SECURE THE PULLEY

Use the printed tool to help with the pulley alignment. It will keep everything in place while you tighten the pulley set screws. It's a good idea to double-check the pulley height once the tensioners are also in place.



SECURE THE BEARING

The flanged bearings should come pre-installed. Secure the stepper double-shear bearing with 2 screws.

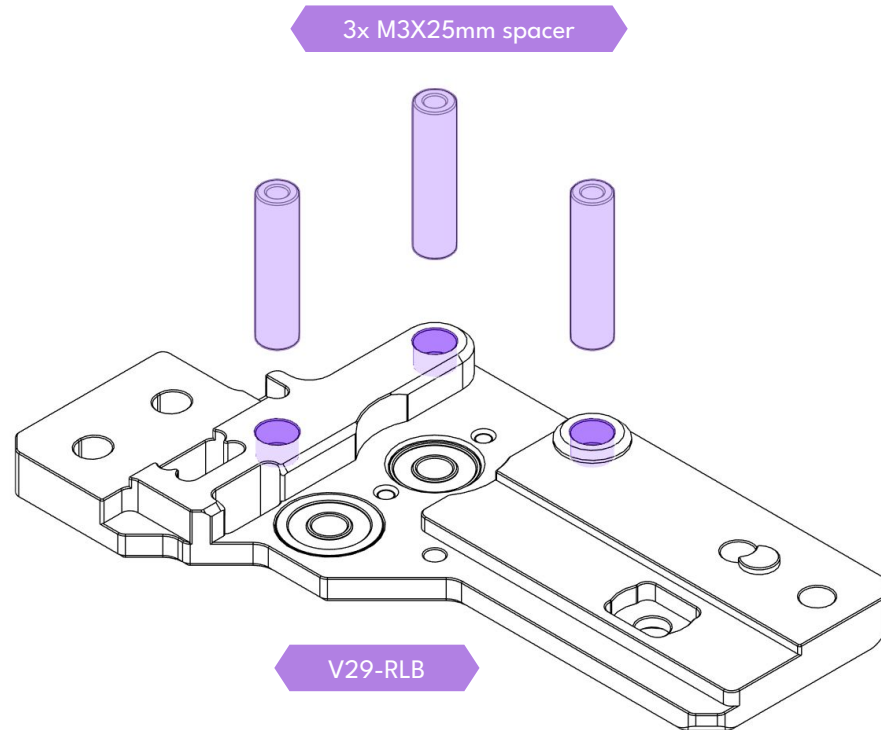


REAR LEFT MOUNT

MONOLITH

GET THE SPACERS

Place the 25mm long M3 spacers into the holes.

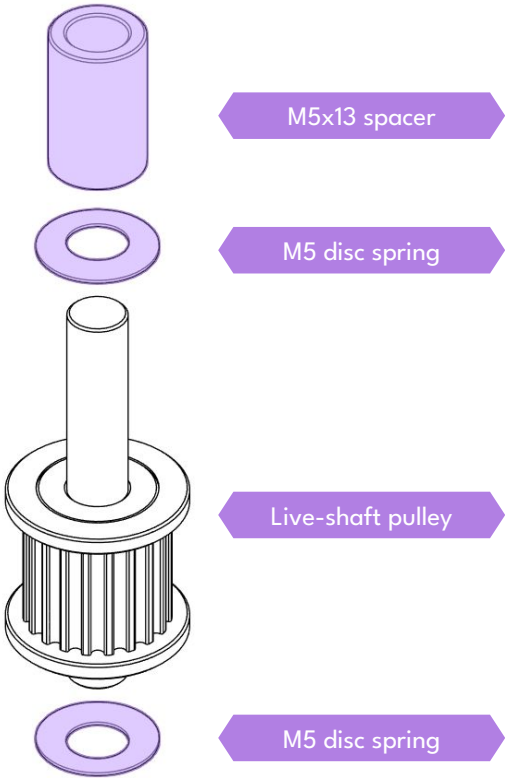
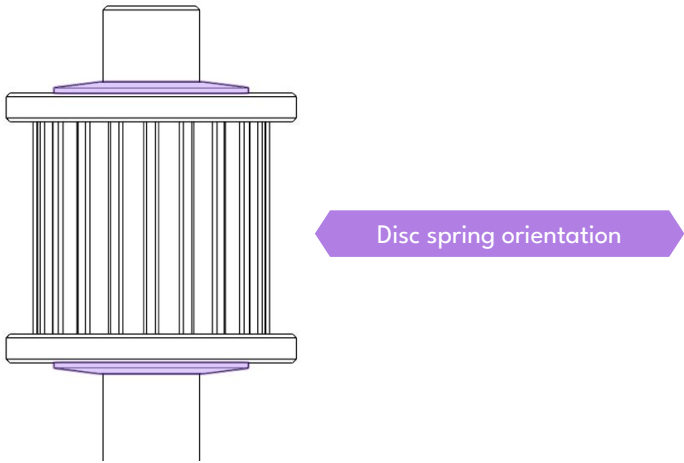


REAR LEFT MOUNT

MONOLITH

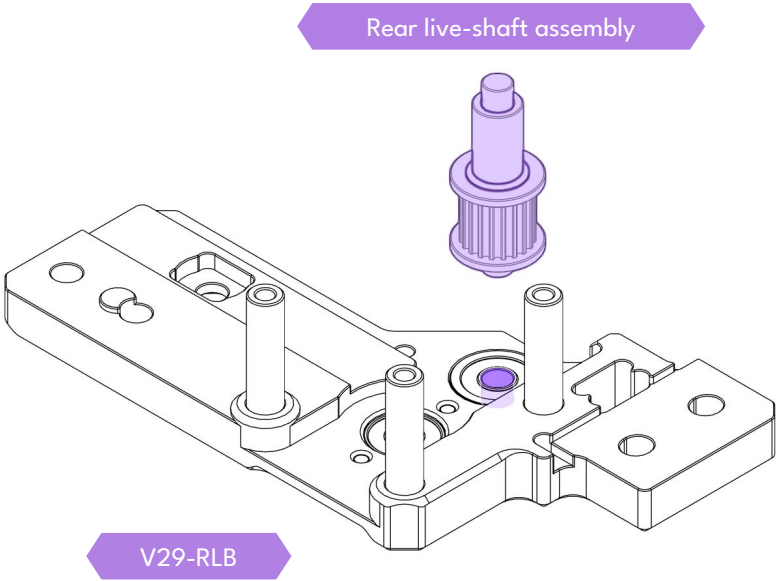
REAR LIVE-SHAFT ASSEMBLY

Put together the live-shaft assembly for the rear mount. Make sure the disc springs are in the correct orientation.



LIVE-SHAFT TIME

Insert the previously prepared live-shaft assembly into the pre-installed F695 bearing with the idler on the top.

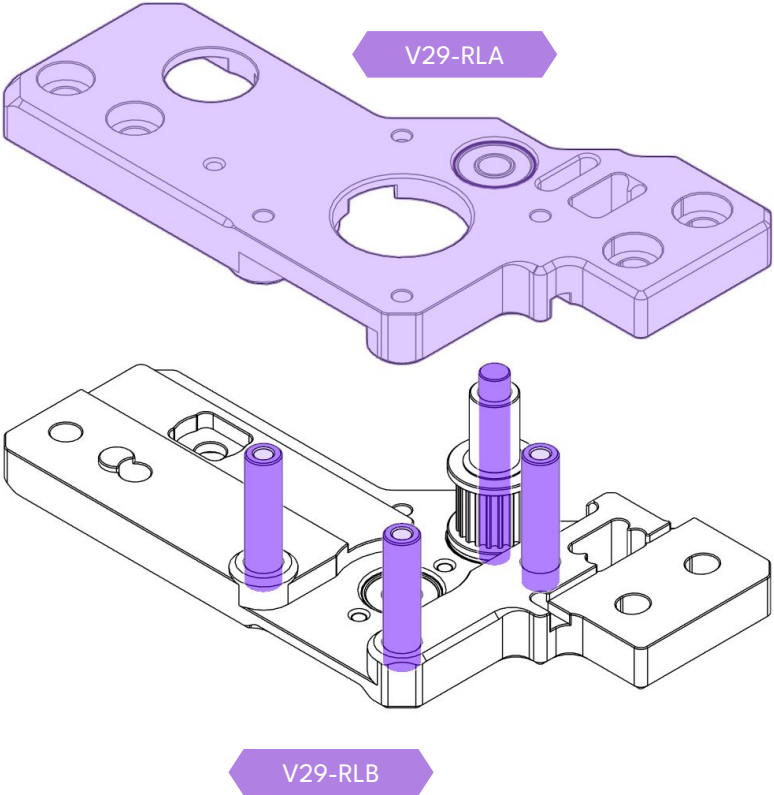


REAR LEFT MOUNT

MONOLITH

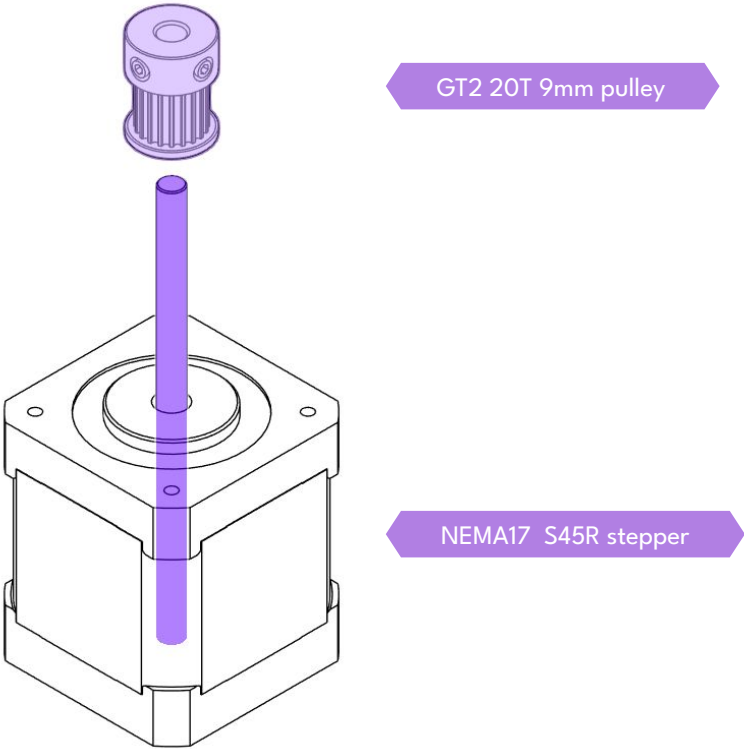
TOP IT OFF

Get the top part of the motor mount and gently put in on the spacers and the live-shaft. The fit will be the tightest at the bearing, so make sure you're not forcing it at an angle.



THE DRIVE PULLEY

Slide the pulley onto the stepper shaft with its hub facing up, but DO NOT tighten the set screws yet.



REAR LEFT MOUNT

MONOLITH

CABLE MANAGEMENT TIP

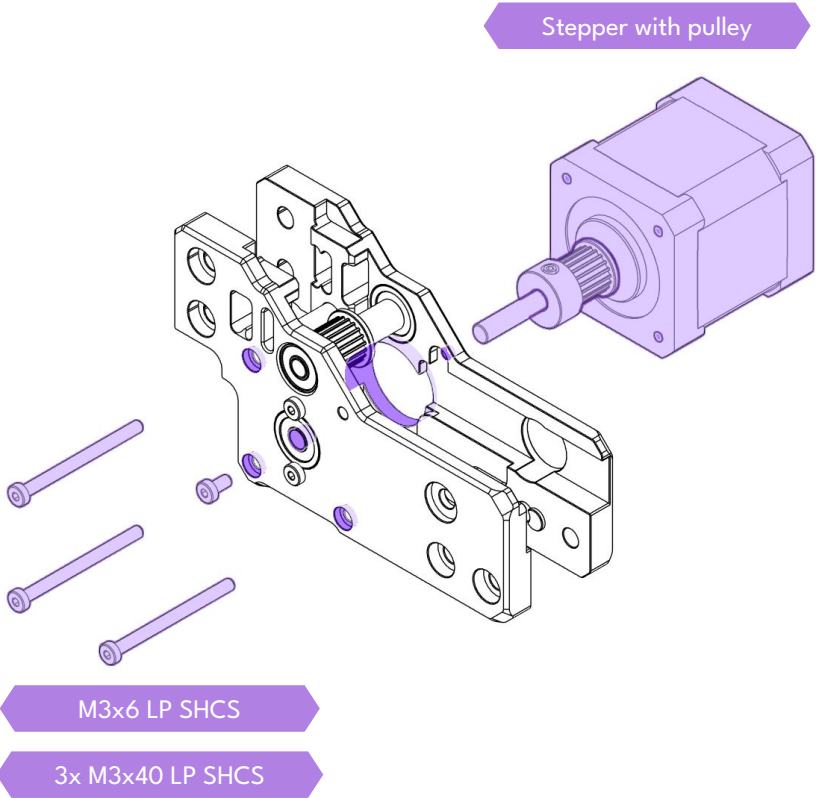
For optimal cable management later, make sure the stepper cable/connector points toward the rear extrusion mounting holes.

MOTORIZE

Put the previous assembly on its side on a flat surface, but be careful, because it can still fall apart. Then secure the stepper with the 4 screws.

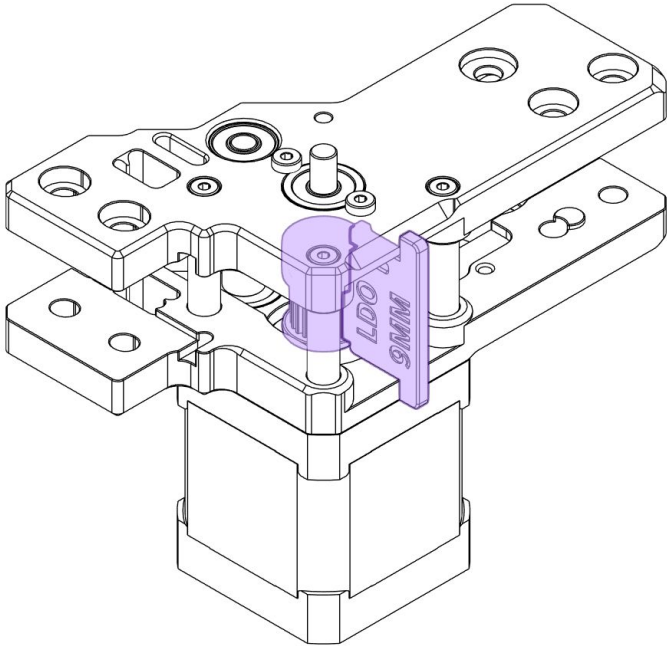
USE WASHERS

Batch 1 kits have excessive thread engagement length. This can cause problems with steppers that have more limited hole depths than the most common 48mm body LDO steppers. Please use spring washers (or any other small OD washers) in the counterbores to avoid this issue.



ALIGN AND SECURE THE PULLEY

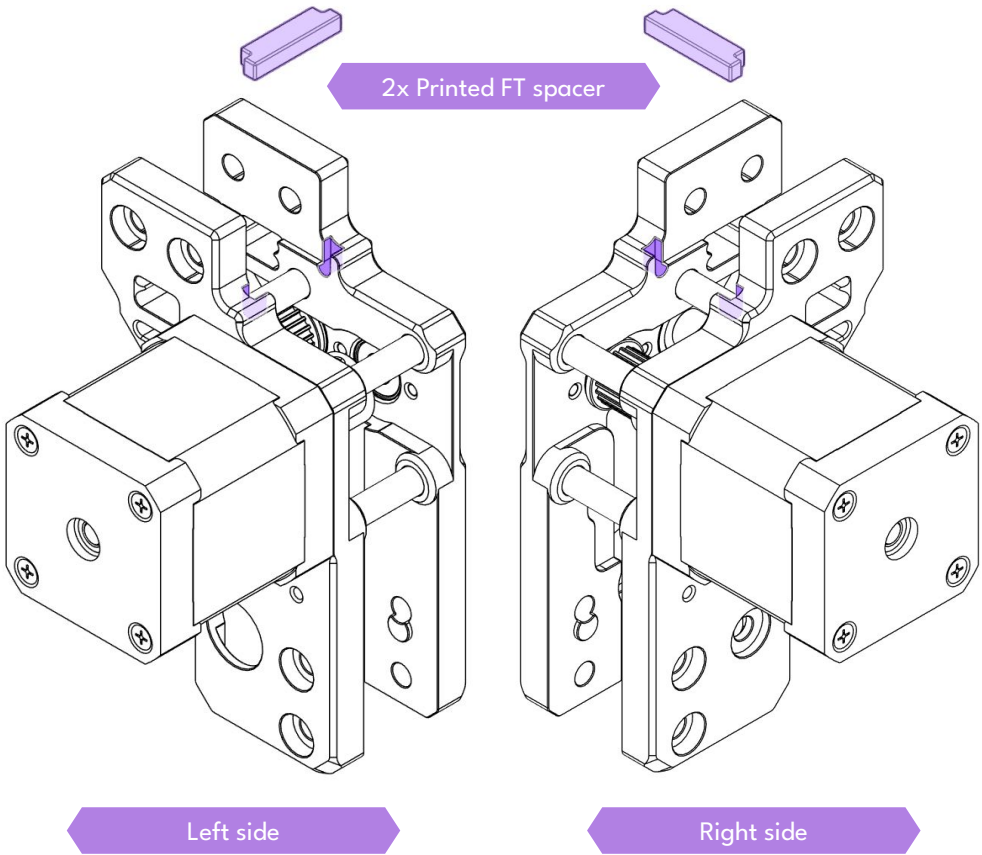
Use the printed tool to help with the pulley alignment. It will keep everything in place while you tighten the pulley set screws. It's a good idea to double-check the pulley height once the tensioners are also in place.



PRINTED SPACERS (optional)

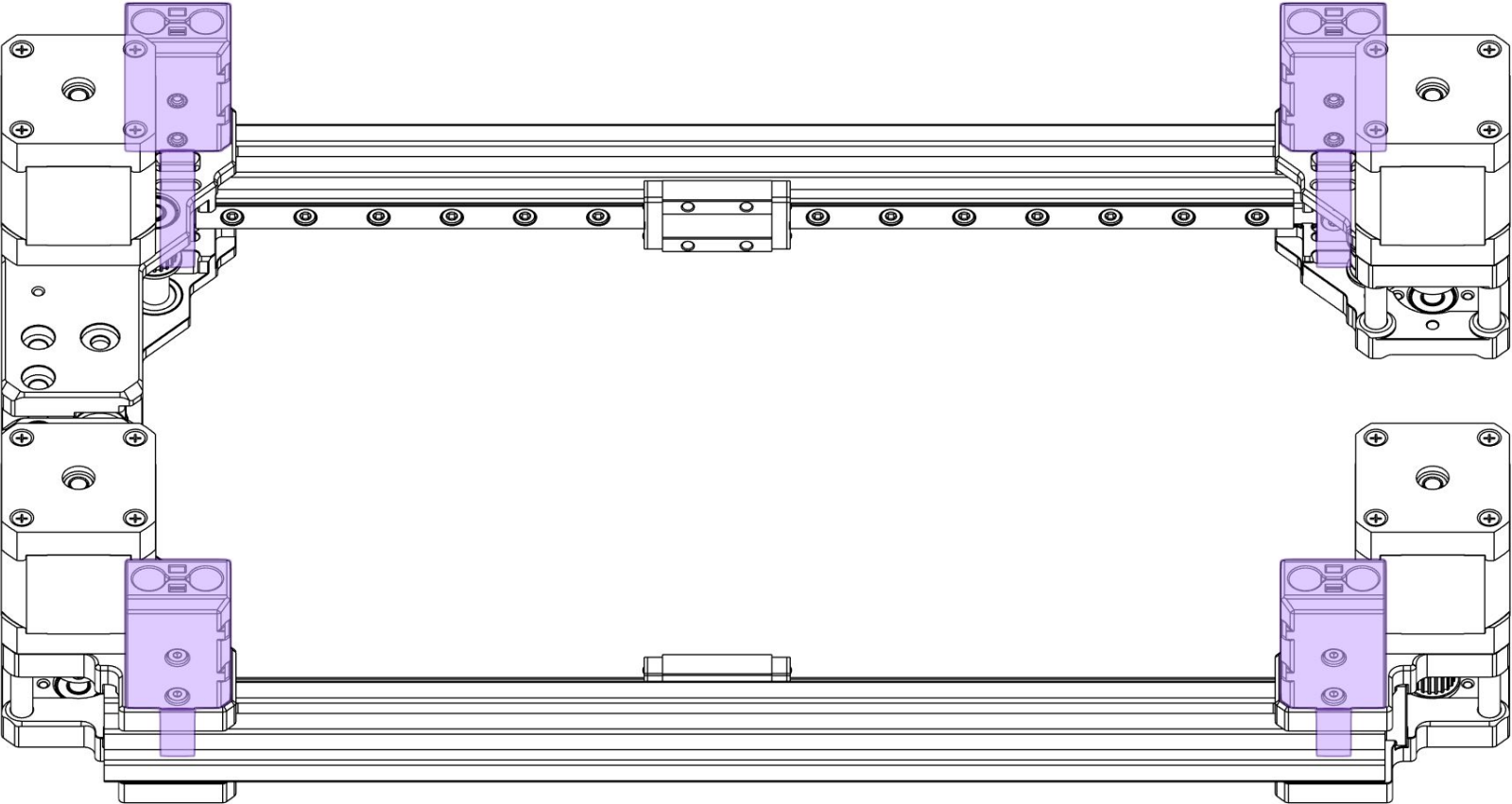
If you chose the FT (Full Travel) configuration for the mounts, now it's time to install the printed spacers which will correctly position the assembly relative to the Y-extrusion. Note the orientation of the printed part.

Skip these if you chose NP (No Protrusion) and shifted the Z-belt path.



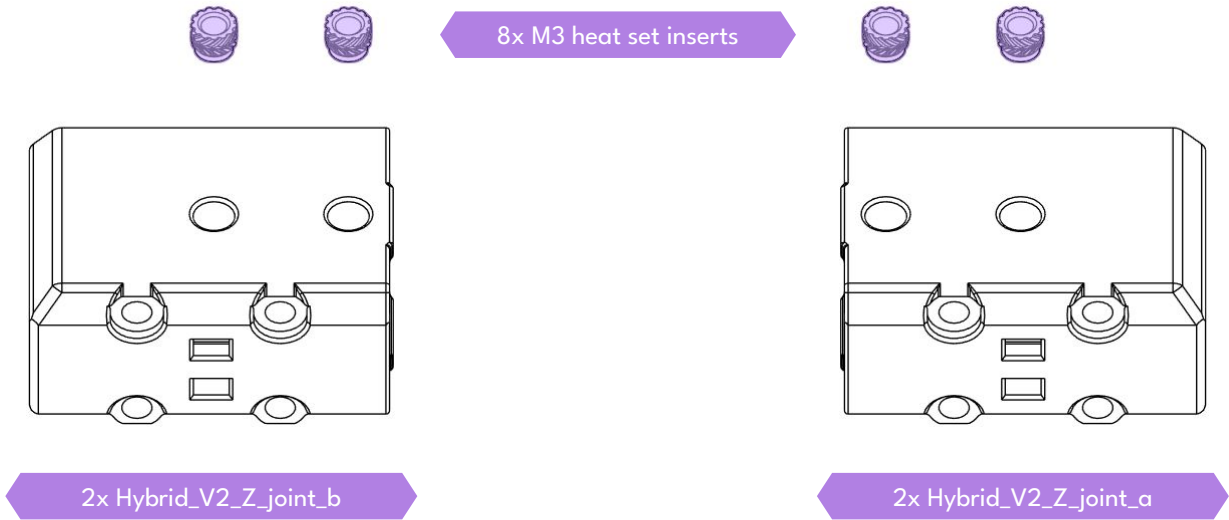
Z-JOINTS

MONOLITH



PREPARE THE Z-JOINTS

The printed Z-joint parts need 2 heat set inserts each. The only difference between the “a” and the “b” parts is that they’re mirrored.



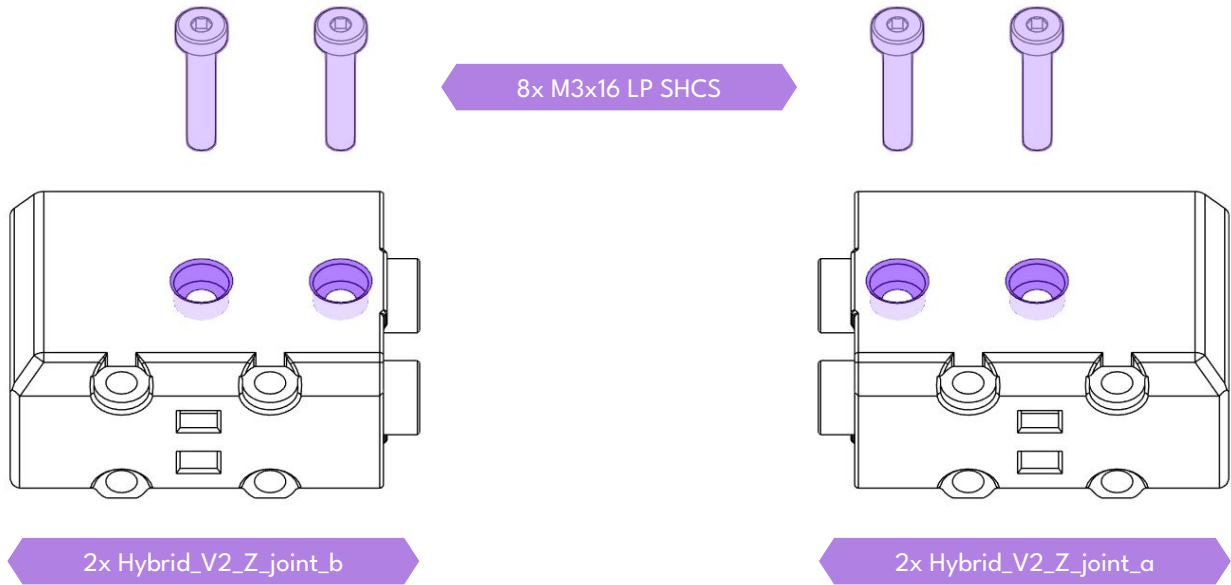
PREPARE THE Z-JOINTS

The hybrid Z-joints have 2 clamped M5 spacers inside them. They should stick out by 2-3mm at this point. The mounting screws will pull them into the counterbores anyway, which will set the correct position.



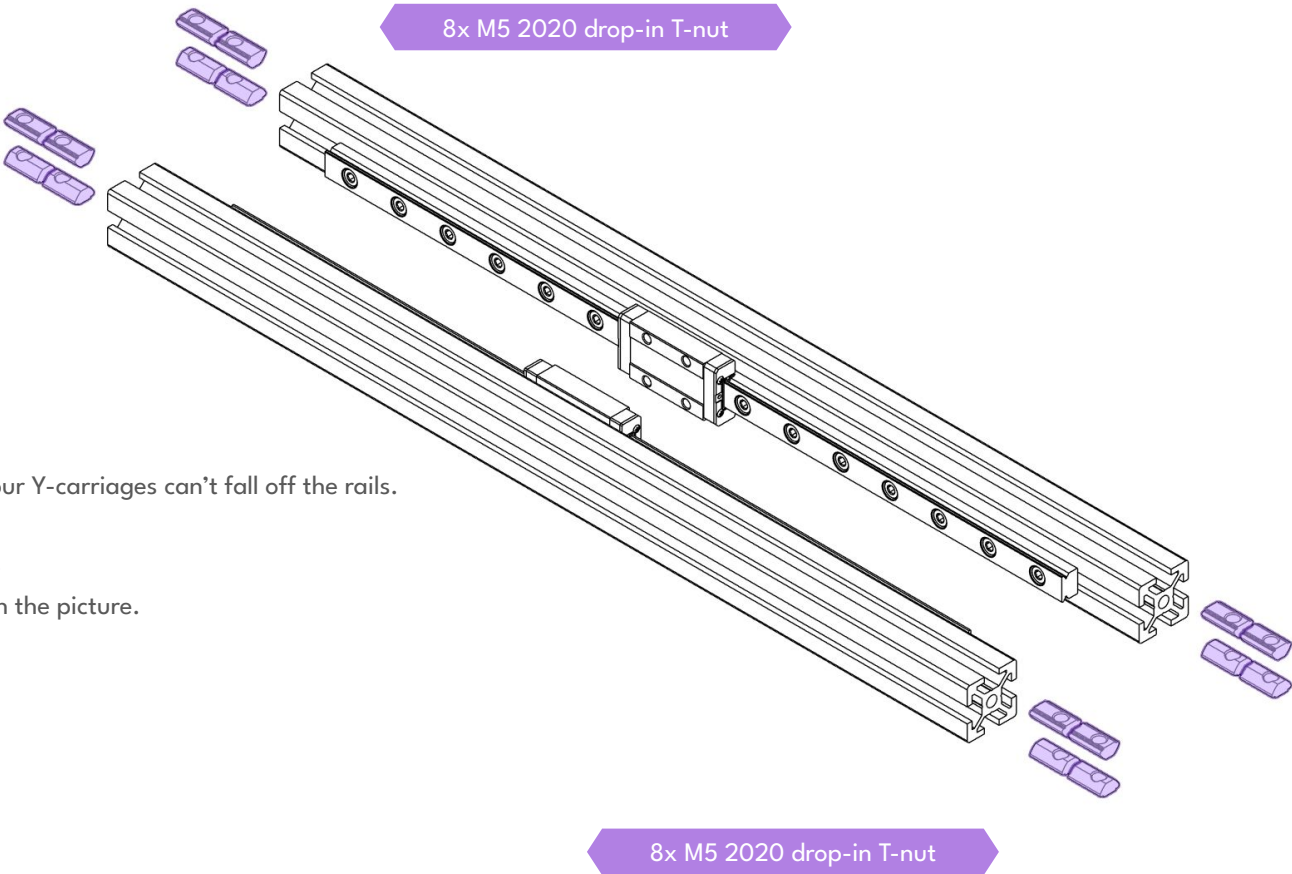
Z-JOINT CLAMPING SCREWS

The hybrid Z-joint design uses clamping screws to firmly hold the M5 spacers inside them. Install these screws.



Y-EXTRUSIONS

MONOLITH

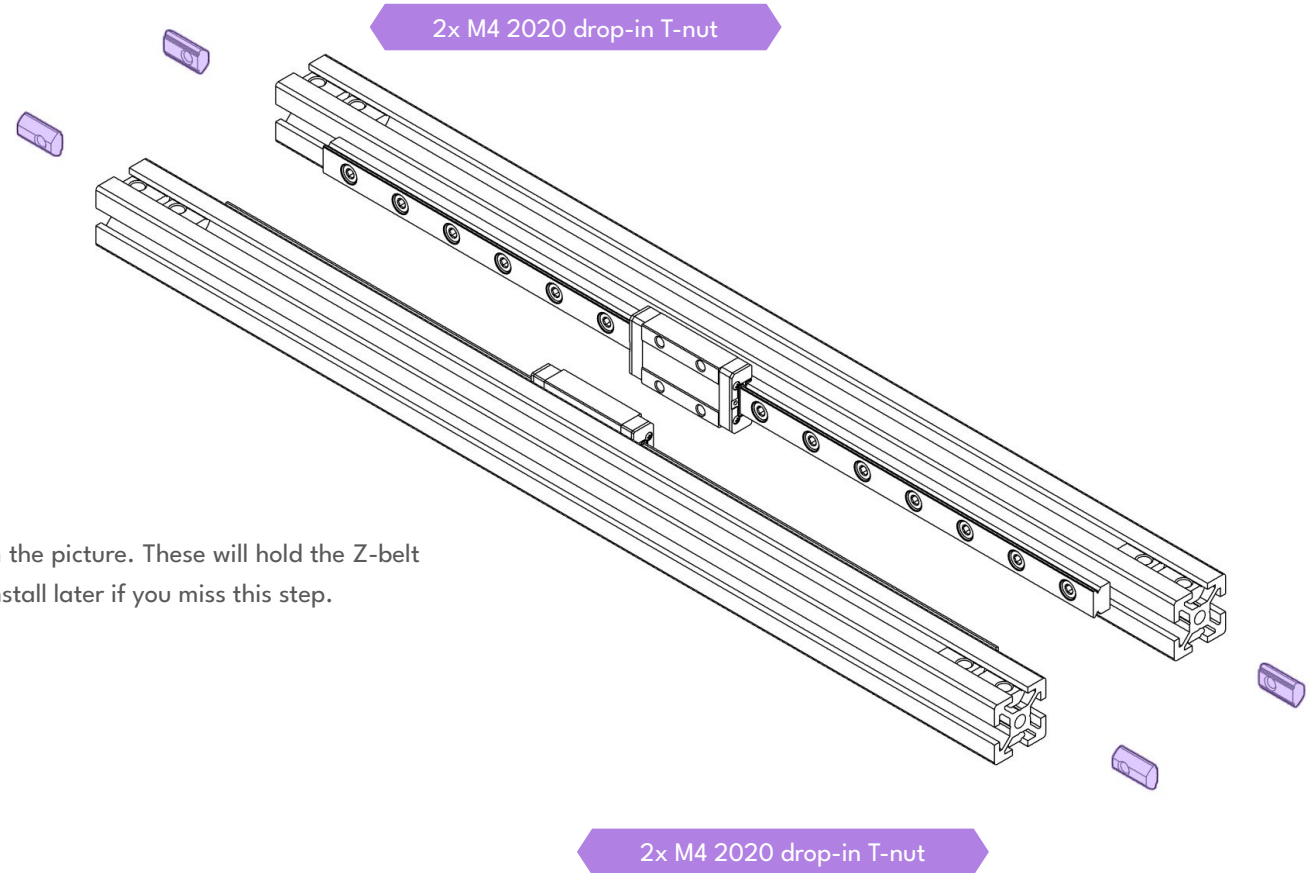


SECURE THE CARRIAGES

It's a good idea to make sure your Y-carriages can't fall off the rails.

PREPARE THE Y-EXTRUSIONS

Insert the M5 T-nuts as shown in the picture.



PREPARE THE Y-EXTRUSIONS

Insert the M4 T-nuts as shown in the picture. These will hold the Z-belt clamps. They're pretty hard to install later if you miss this step.

INSTALLING THE MOUNTS

SLIDE THEM ONTO THE Y-EXTRUSIONS

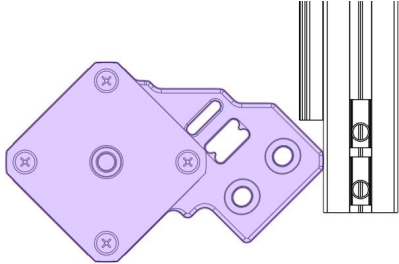
The motor mounts are designed as units, so you don't have to take them apart to install/remove them on/from the extrusions.

This doesn't affect the ease of maintenance, and they don't have to be completely removed to replace the bearings.

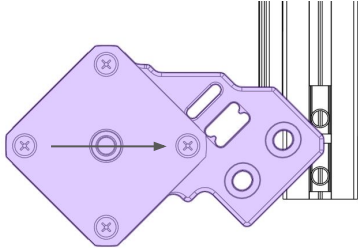
STEP-BY-STEP

- Step 1:** Line up the chamfered corner of the mount with the side of the extrusion.
- Step 2:** Slide the mount onto the extrusion.
- Step 3:** Rotate the mount to the correct orientation.
- Step 4:** Slide the mount along the side of the extrusion until you hit the indexing feature.

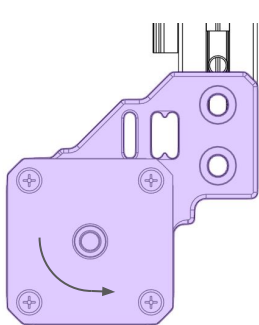
Step 1



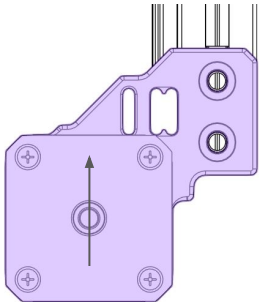
Step 2



Step 3



Step 4

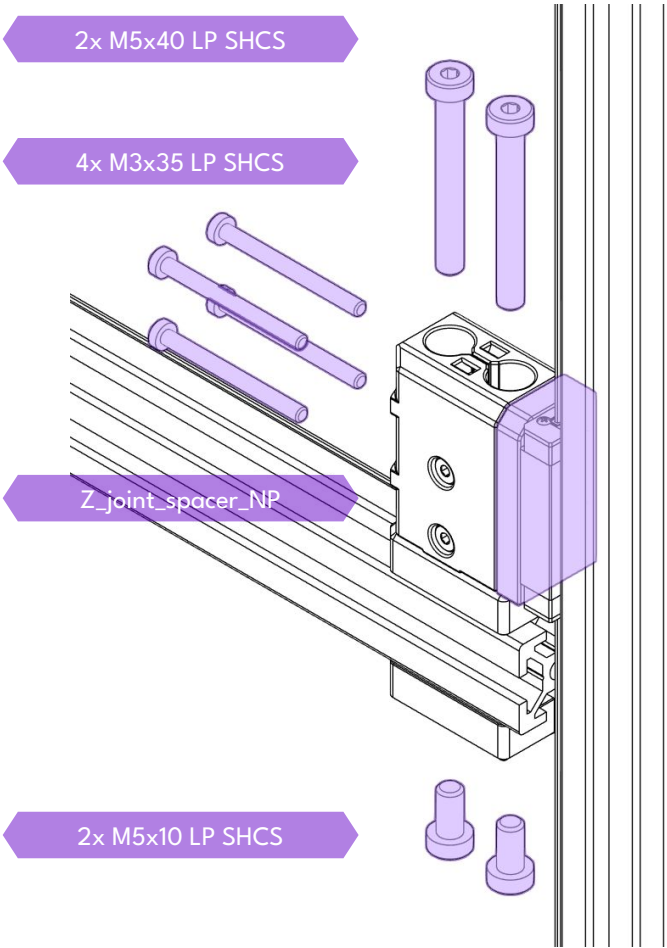


Z-JOINTS (NP mode)

MONOLITH

Z-JOINT CLAMPING SCREWS

The hybrid Z-joint design uses clamping screws to firmly hold the M5 spacers inside them. Install these screws.

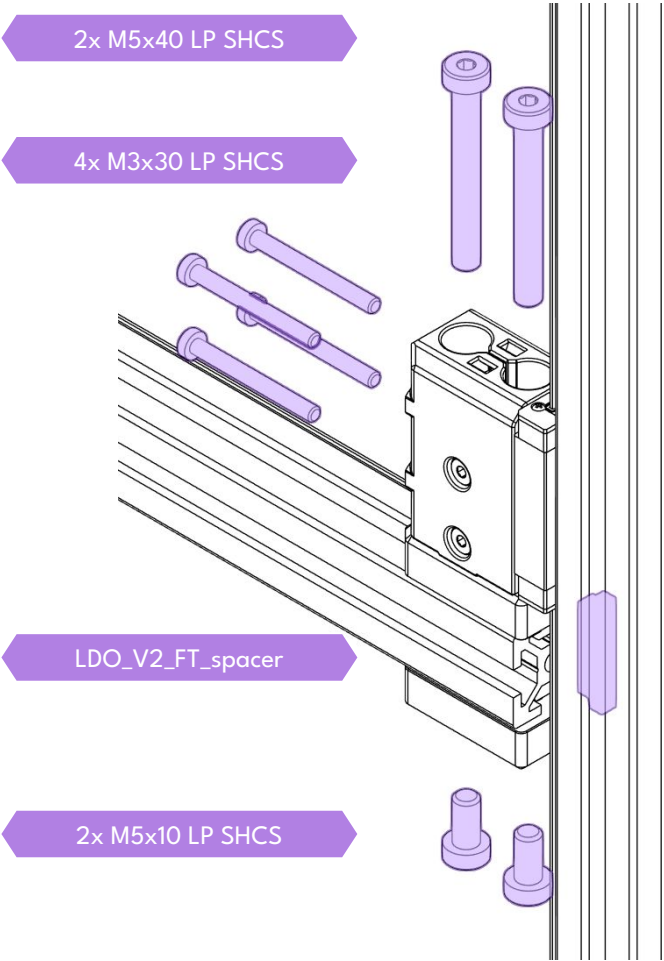


Z-JOINTS (FT mode)

MONOLITH

Z-JOINT CLAMPING SCREWS

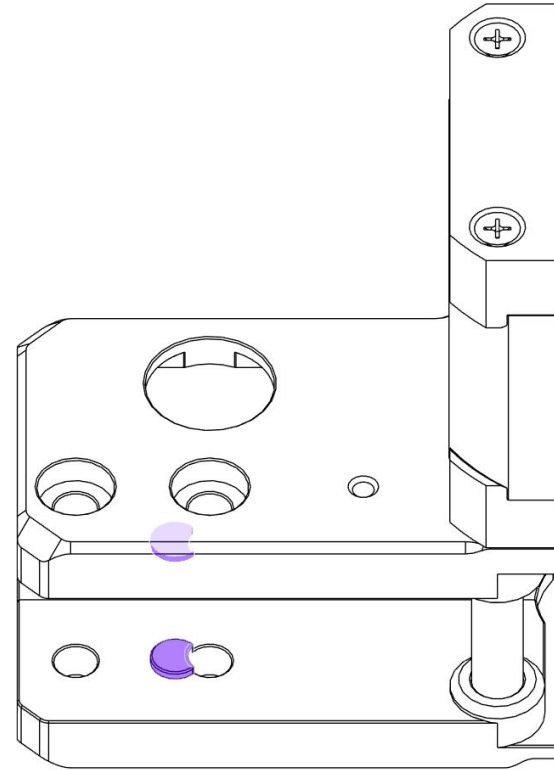
The hybrid Z-joint design uses clamping screws to firmly hold the M5 spacers inside them. Install these screws.



CYLINDRICAL FEATURES

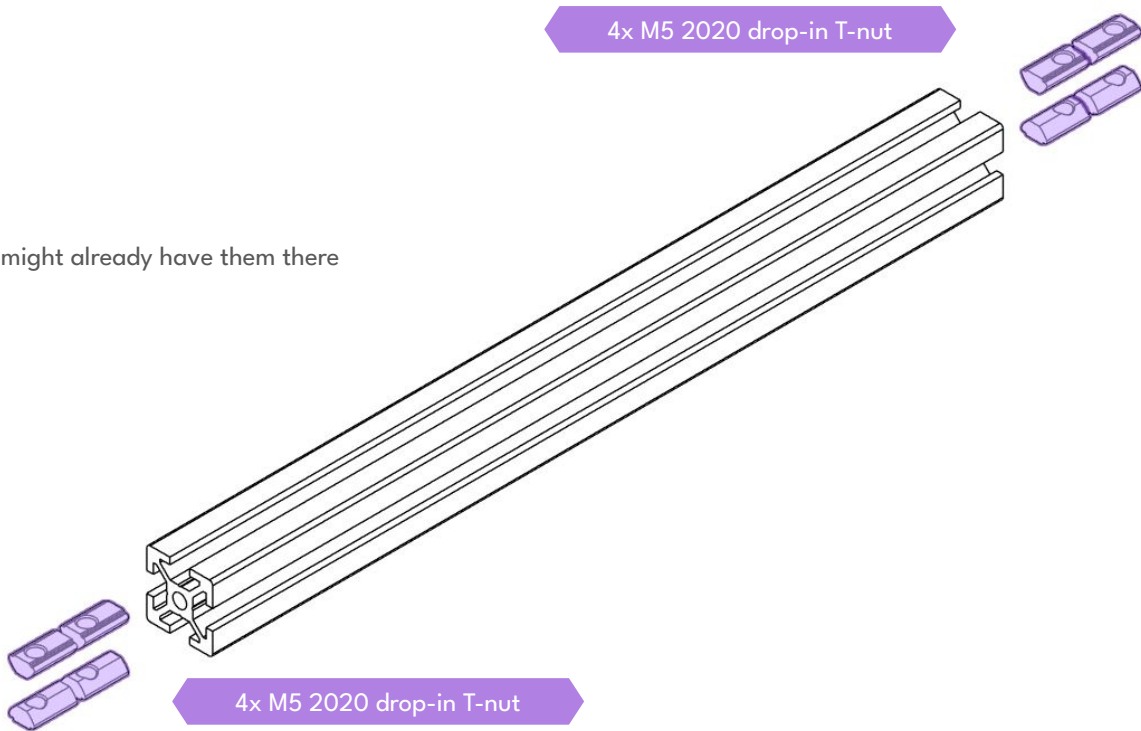
The rear mounts have cylindrical features for the rear extrusion positioning. This means that as long as the tensioners aren't installed, you can remove/install the mounts or the extrusion independently without taking everything apart.

On the V2 this means you can install the right and the left side of the gantry (with the Z-joints) independently, and only then connect them later with the rear extrusion.



T-NUTS

Insert the M5 T-nuts into the rear extrusion. You might already have them there from the stock gantry.



REAR EXTRUSION

MONOLITH

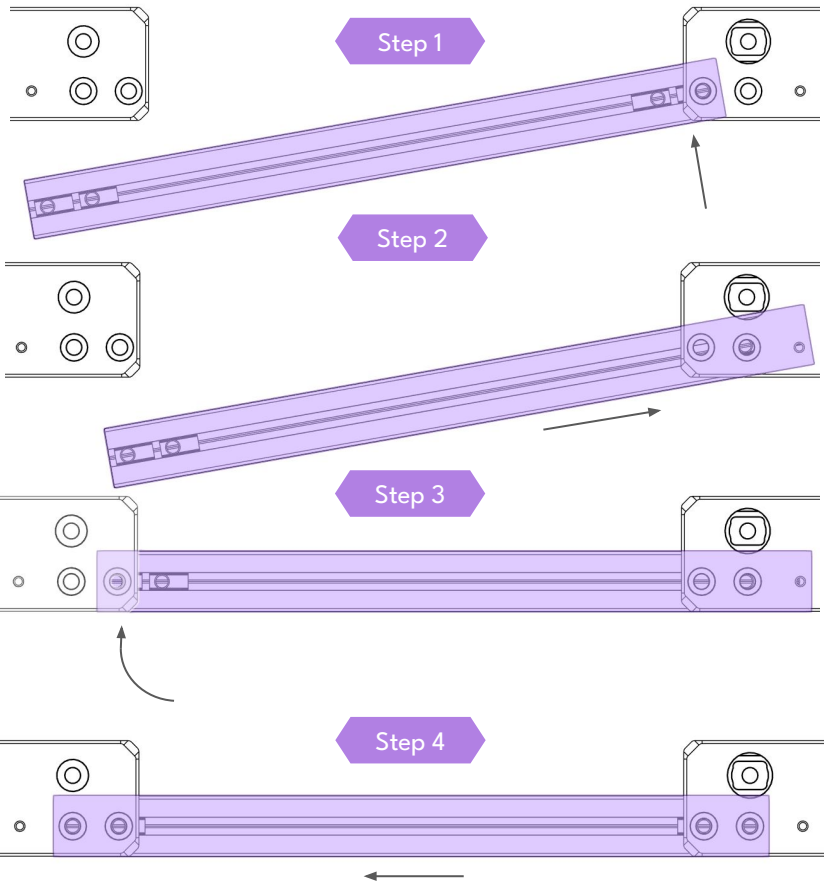
STEP-BY-STEP

Step 1: Push the very end of the extrusion in between the rear mount parts. Use the filleted edge to your advantage.

Step 2: Slide the extrusion over the cylindrical features so that it will clear the ones on the other side.

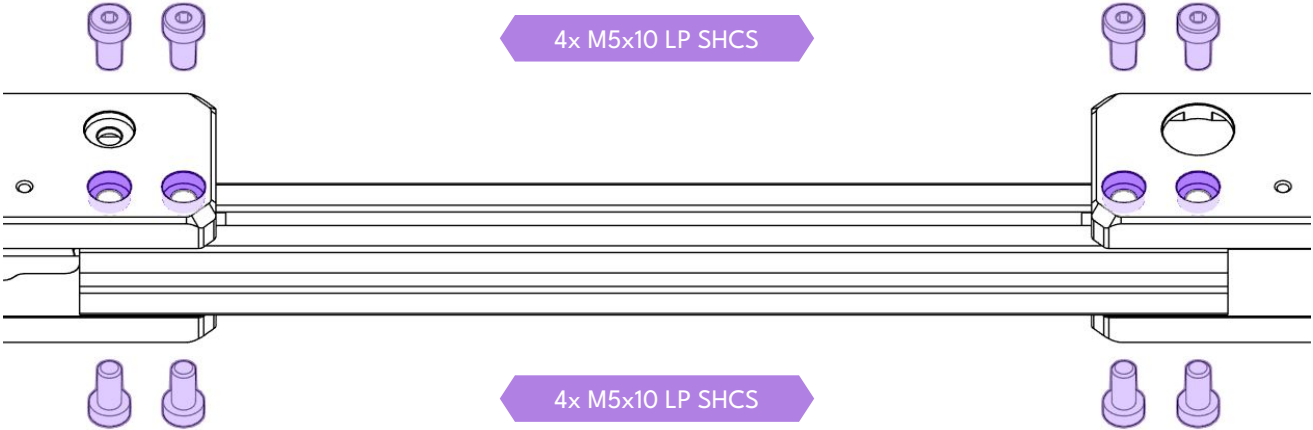
Step 3: Push the other end of the extrusion in between the rear mount parts.

Step 4: Slide the mount along the side of the extrusion until you hit the indexing feature.



SECURE THE EXTRUSION

The rear extrusion is rigidly mounted with eight M5 screws.



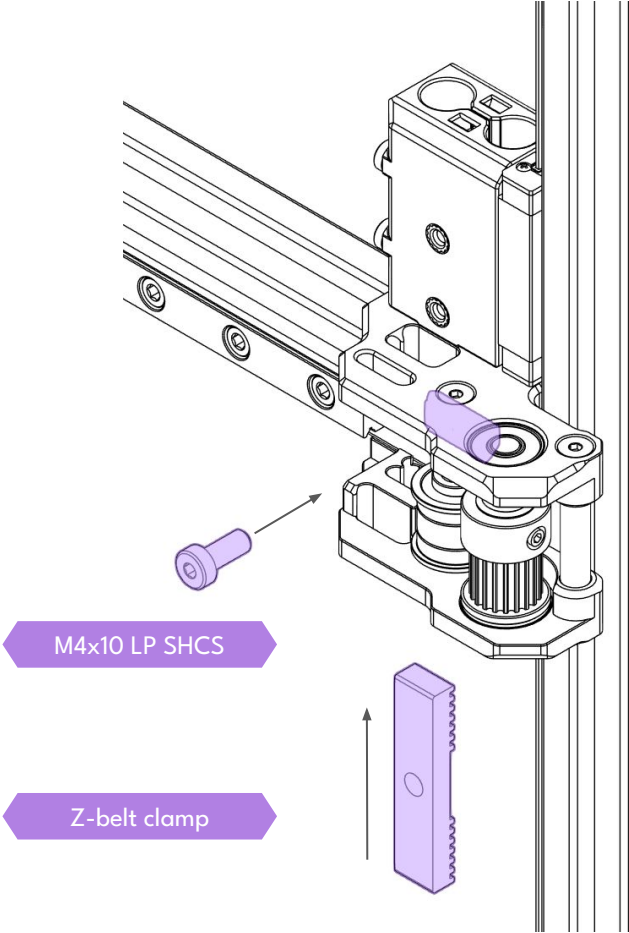
Z-BELTS

Z-BELT CLAMPS

The kits use standard, aluminum GT2 belt clamps. These have sufficient belt-tooth engagement, and they work without tight bends in the Z-belt path. This makes them suitable for higher belt tensions without the belts snapping. This is especially important because of the rigid Z-joints, which can put more stress on the belts and require higher tension to work well.

KEEP THEM LOOSE

Install the Z-belt clamps with the screws, then back off about 1 full turn. This will set a gap for easy belt insertion.



Z-BELTS

MONOLITH

MEASURE TWICE AND CUT ONCE

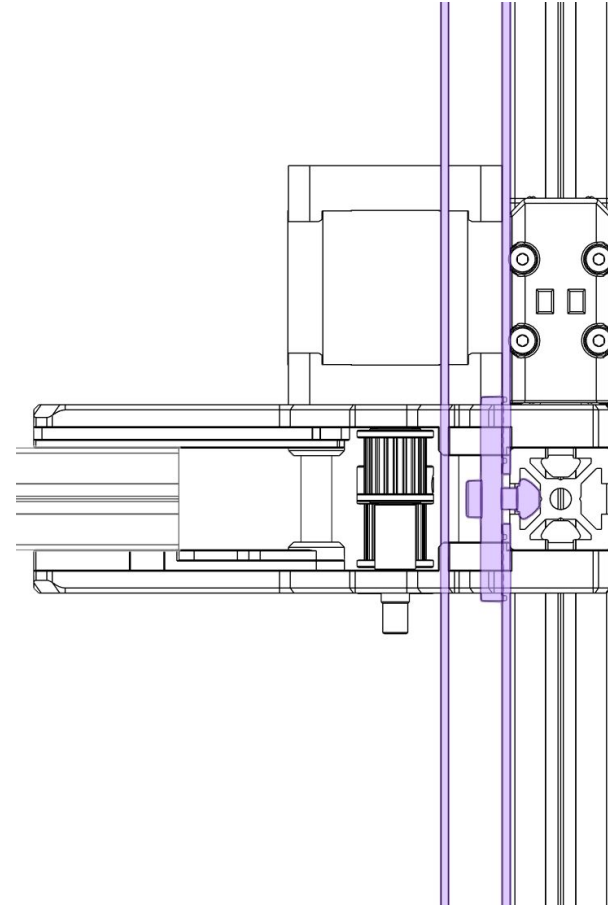
The only real drawback of this setup is that there's no space for a significant amount of excess belt. But the idea is that belts that don't snap are better than long belts that snap too short.

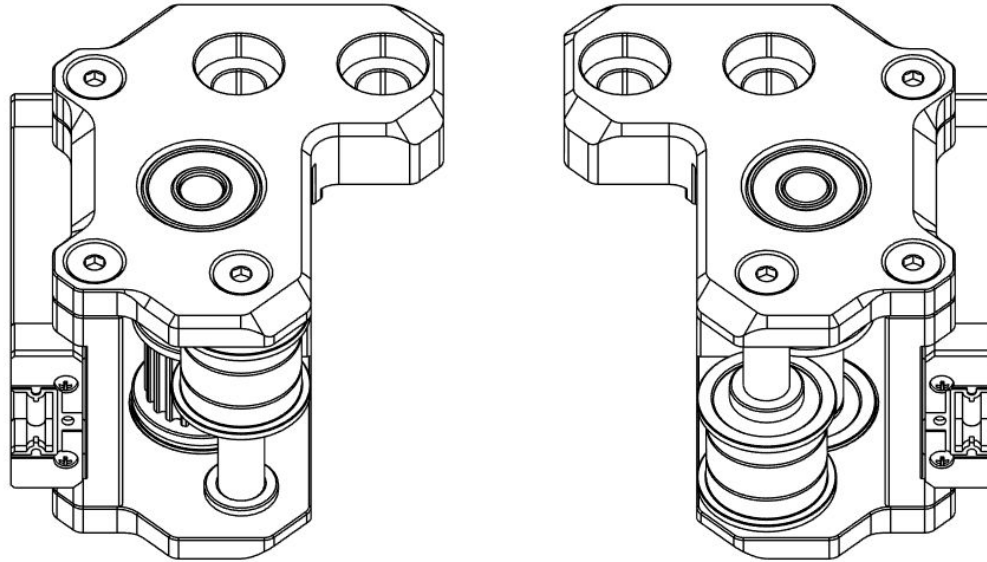
REMOVE THE Z-IDLERS

The easiest way to thread both ends of the belt and tighten the clamp's screw behind it is to remove the Z-idler while you're working on that corner.

DON'T OVERTIGHTEN

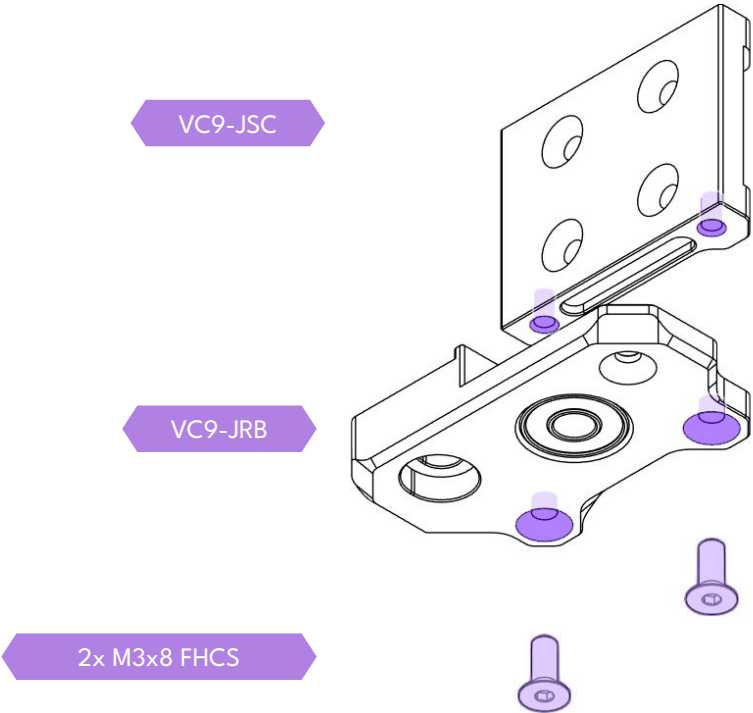
By tightening the screws, you're directly compressing the belt. Too much of this pressure can damage the belts. **A good rule of thumb is to only tighten with the short side of the hex key.**





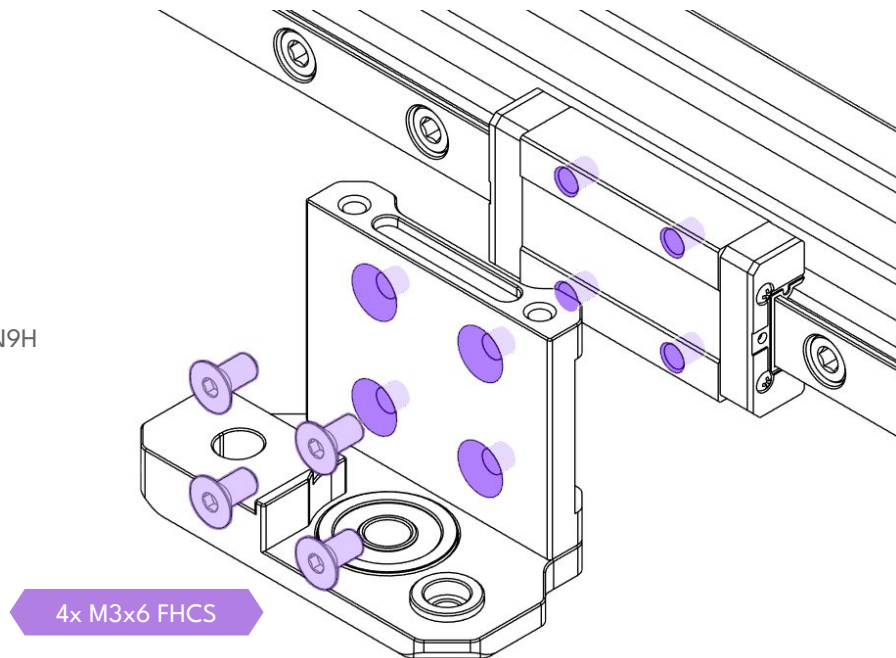
PREPARE THE PARTS

Attach the bottom part of the XY-joint to the middle part with 2 screws.



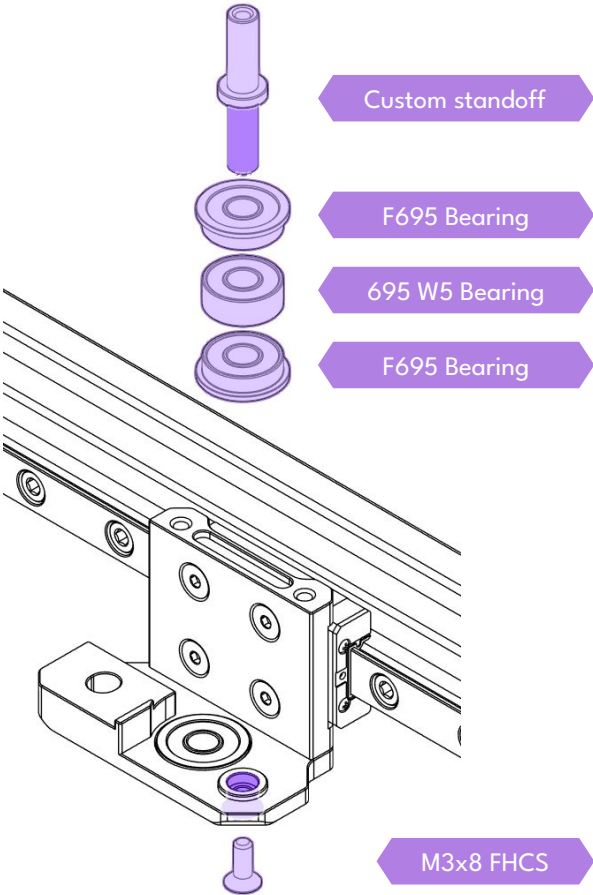
PUT THEM ON THE CARRIAGE

Mount the previously assembled milled part duo onto the MGN9H Y-carriage with 4 screws.



INSTALL THE IDLER STACK

Slide the 3 bearings onto the standoff, then secure them with a screw.

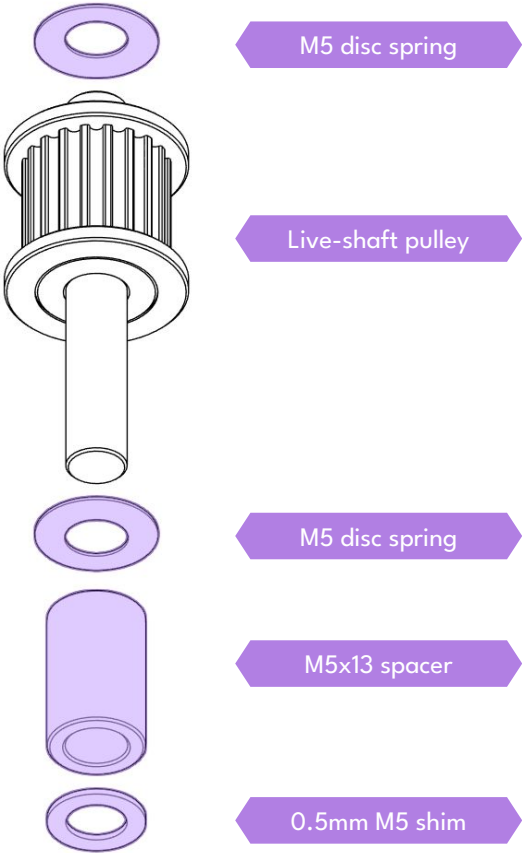
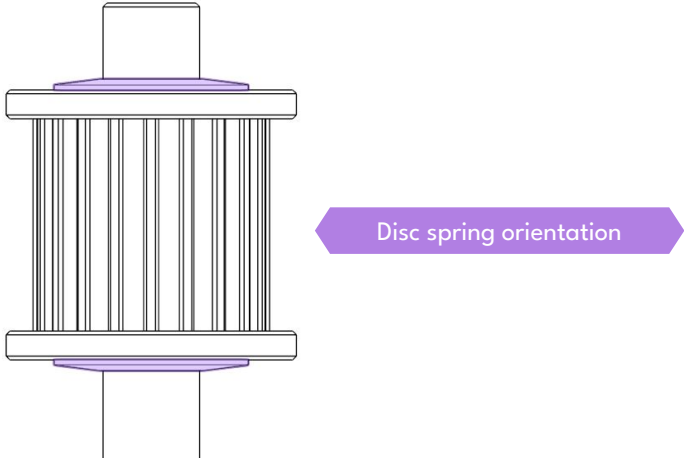


RIGHT 2WD XY-JOINT

MONOLITH

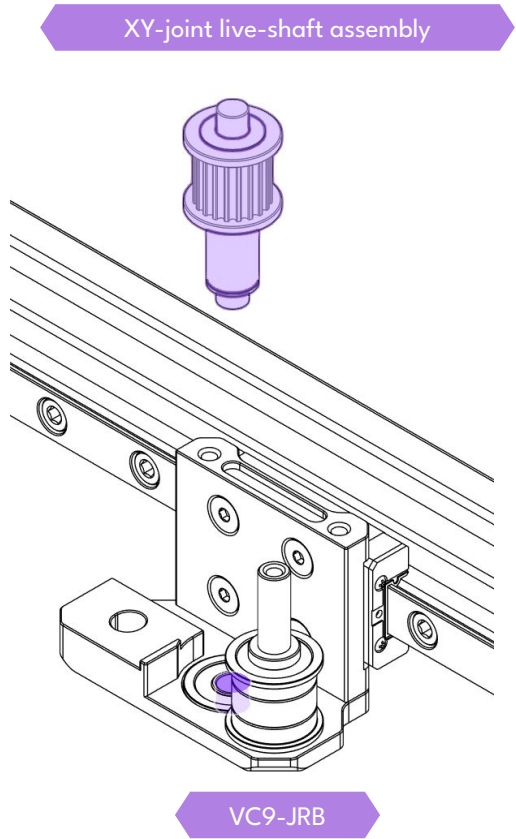
XY-JOINT LIVE-SHAFT ASSEMBLY

Put together the live-shaft assembly for the XY-joint. Make sure the disc springs are in the correct orientation.



LIVE-SHAFT TIME

Insert the previously prepared live-shaft assembly into the pre-installed F695 bearing with the idler on the top.

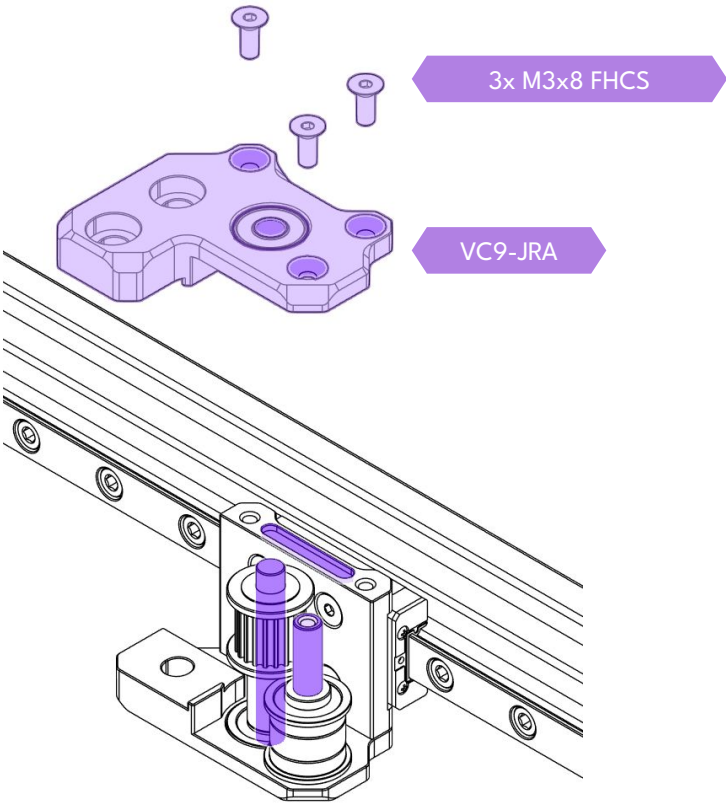


RIGHT 2WD XY-JOINT

MONOLITH

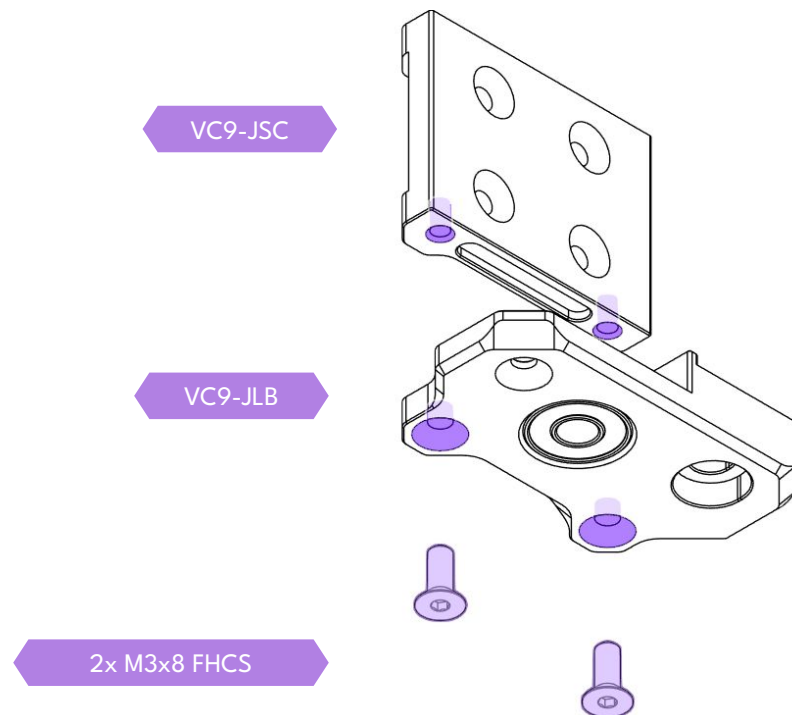
TOP IT OFF

Get the top of the XY-joint and gently put in on the middle part and the live-shaft. The fit will be the tightest at the bearing, so make sure you're not forcing it at an angle.



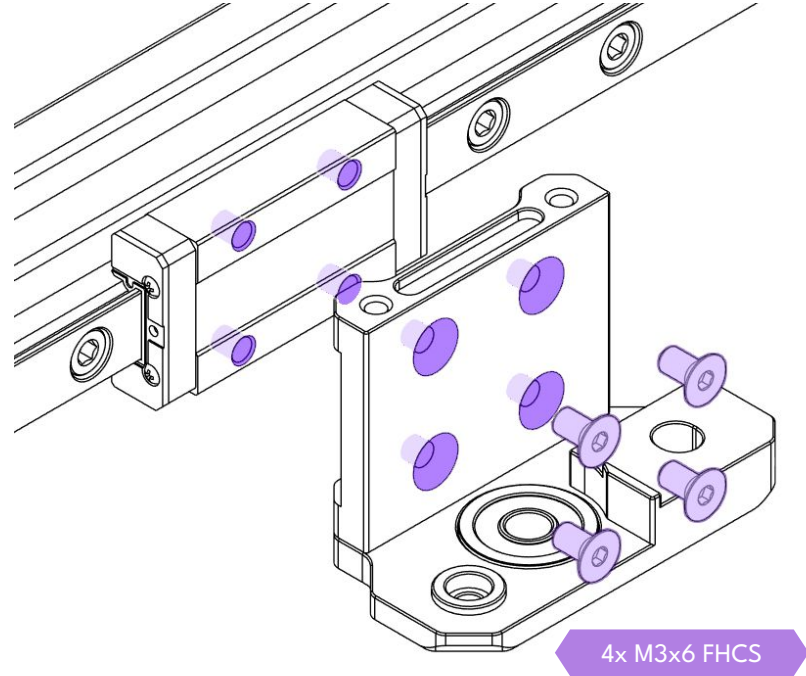
PREPARE THE PARTS

Attach the bottom part of the XY-joint to the middle part with 2 screws.



PUT THEM ON THE CARRIAGE

Mount the previously assembled milled part duo onto the MGN9H Y-carriage with 4 screws.

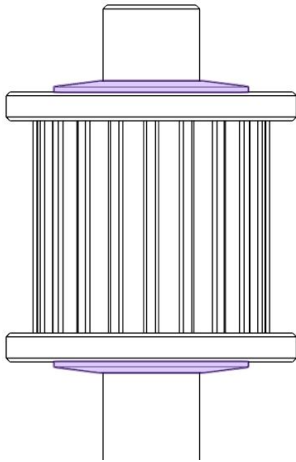


LEFT 2WD XY-JOINT

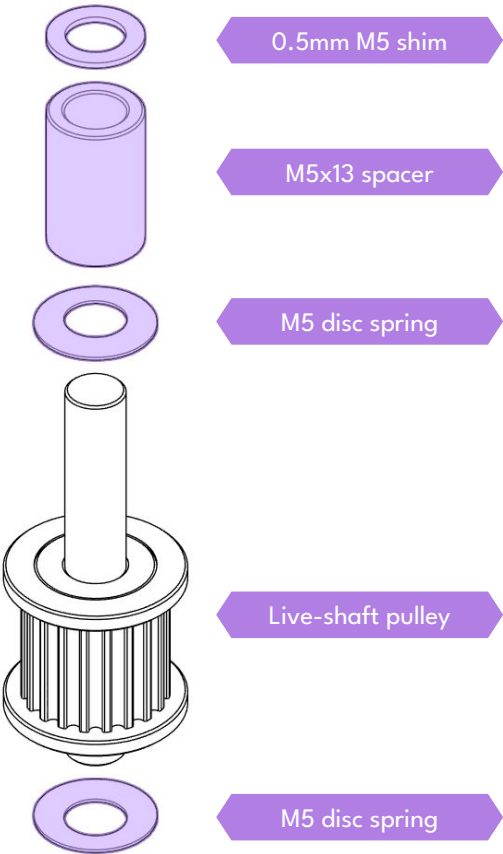
MONOLITH

XY-JOINT LIVE-SHAFT ASSEMBLY

Put together the live-shaft assembly for the XY-joint. Make sure the disc springs are in the correct orientation.

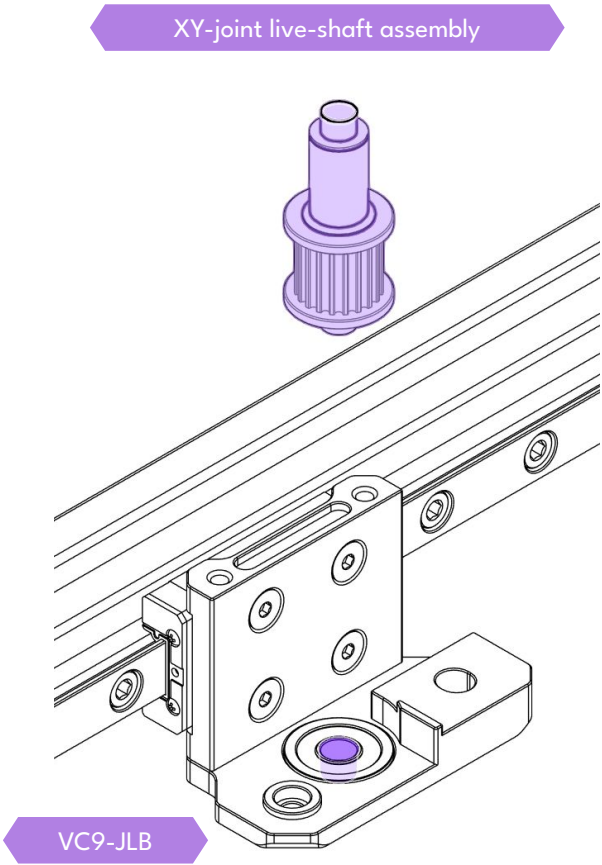


Disc spring orientation



LIVE-SHAFT TIME

Insert the previously prepared live-shaft assembly into the pre-installed F695 bearing with the idler on the top.

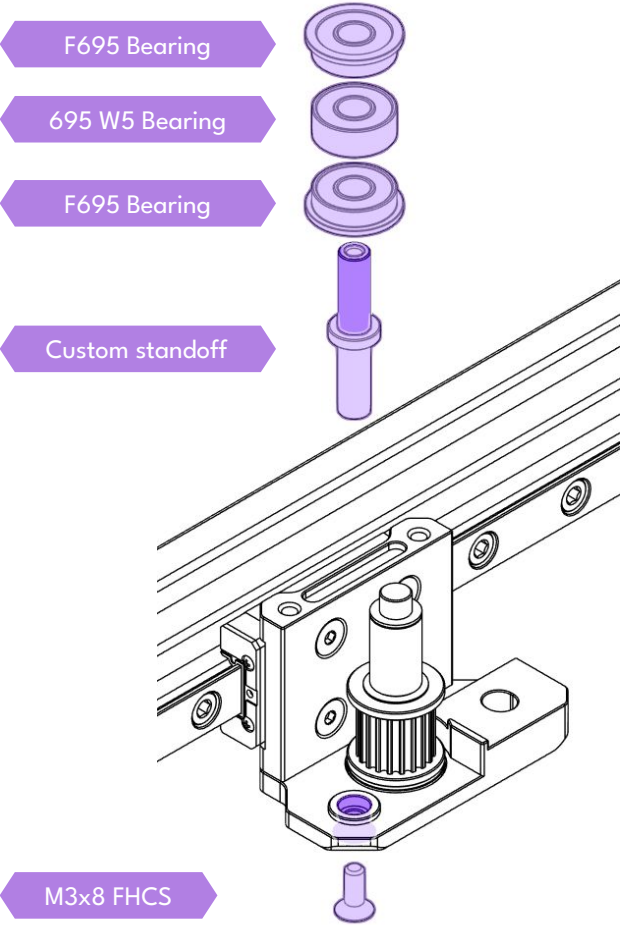


LEFT 2WD XY-JOINT

MONOLITH

INSTALL THE IDLER STACK

Secure the standoff with a screw, then slide the bearings onto it.

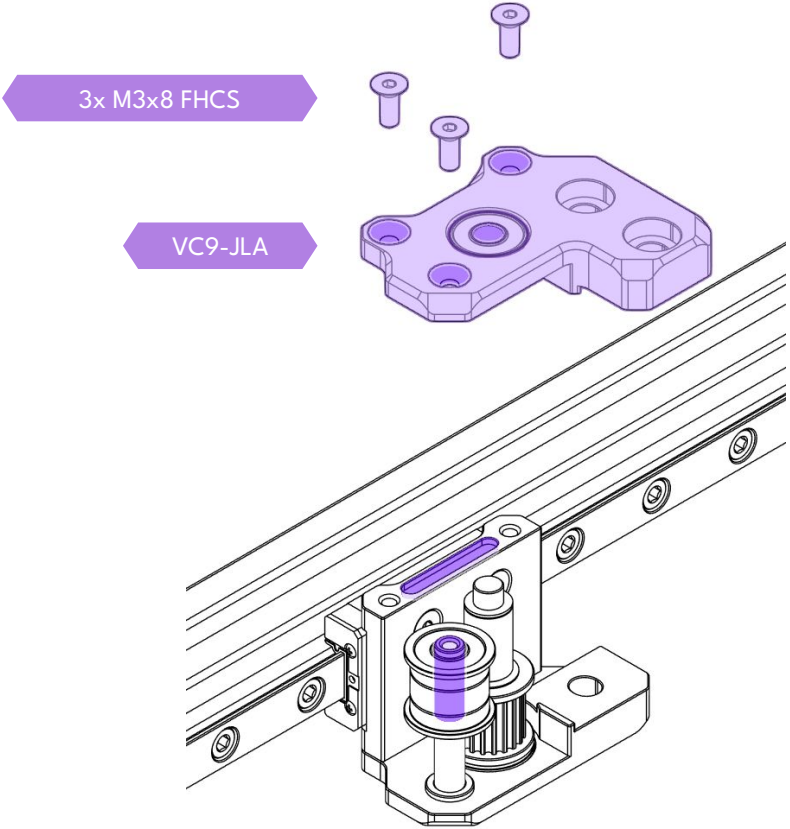


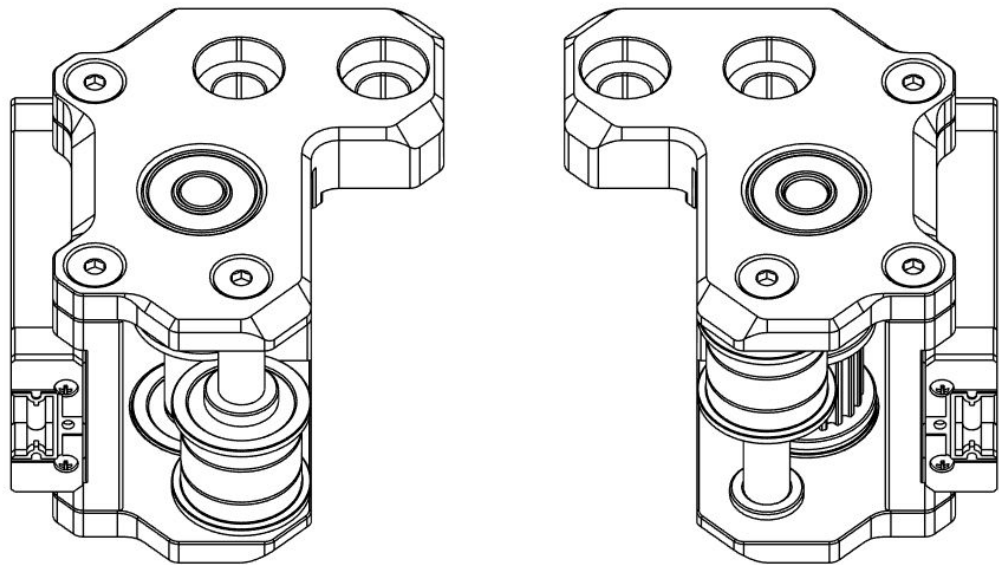
LEFT 2WD XY-JOINT

MONOLITH

TOP IT OFF

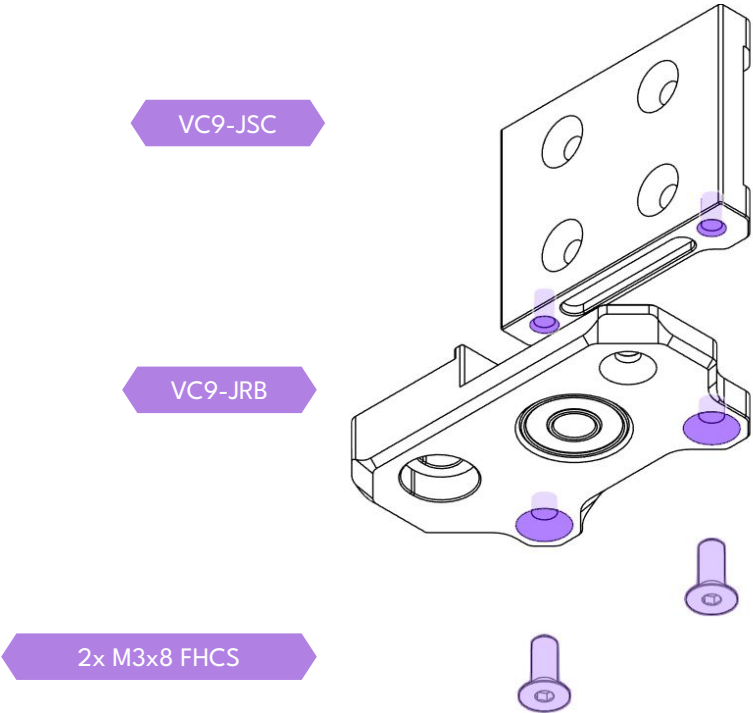
Get the top of the XY-joint and gently put in on the middle part and the live-shaft. The fit will be the tightest at the bearing, so make sure you're not forcing it at an angle.





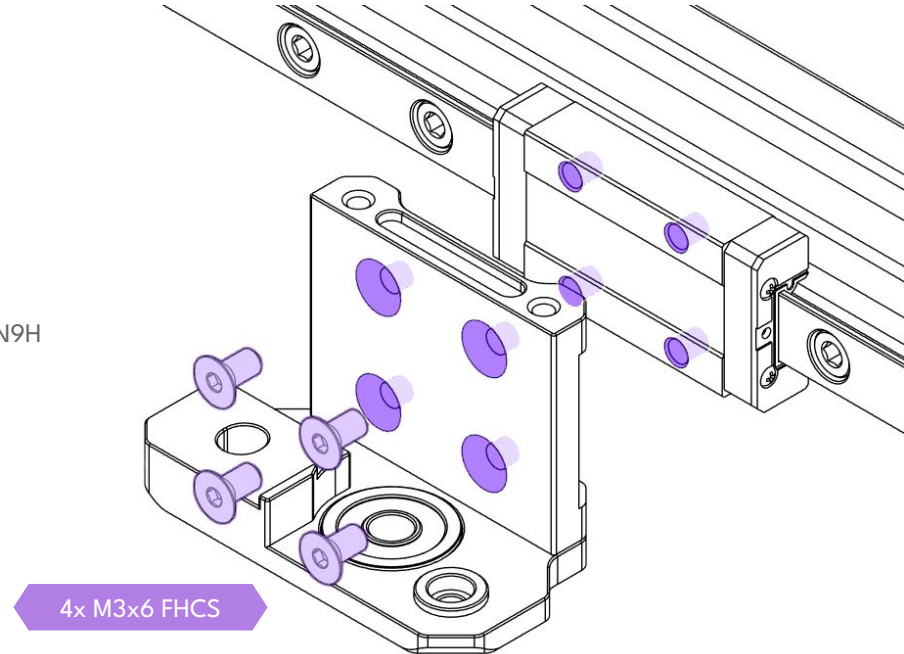
PREPARE THE PARTS

Attach the bottom part of the XY-joint to the middle part with 2 screws.



PUT THEM ON THE CARRIAGE

Mount the previously assembled milled part duo onto the MGN9H Y-carriage with 4 screws.

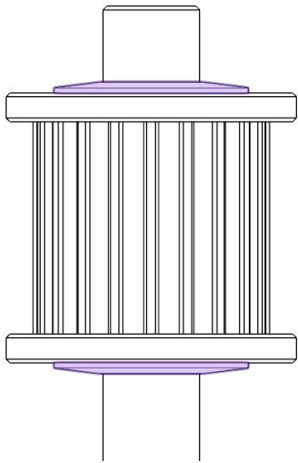


RIGHT AWD XY-JOINT

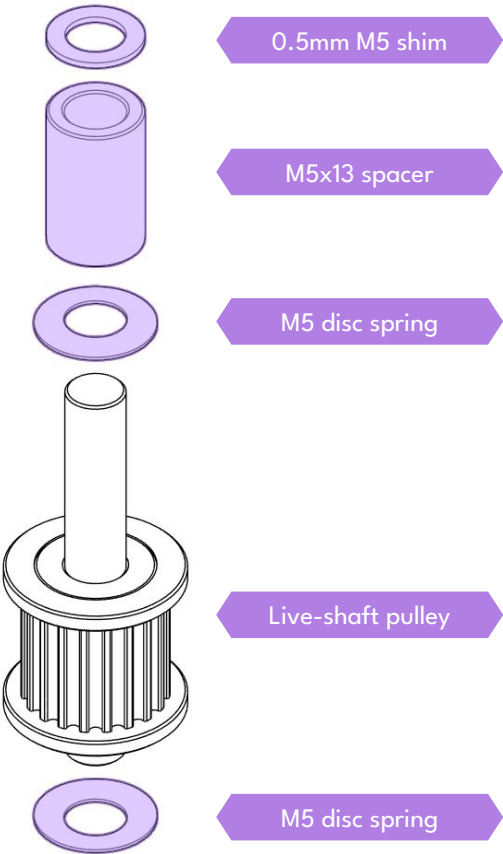
MONOLITH

XY-JOINT LIVE-SHAFT ASSEMBLY

Put together the live-shaft assembly for the XY-joint. Make sure the disc springs are in the correct orientation.

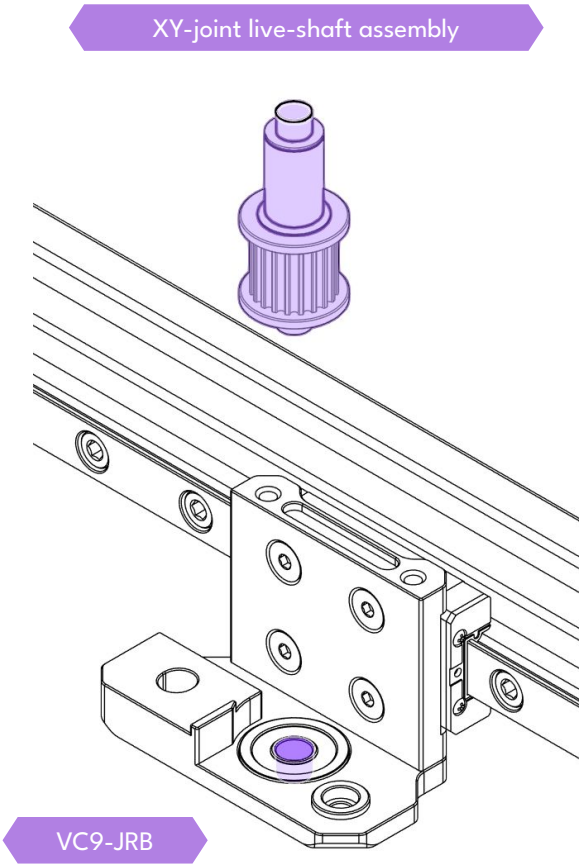


Disc spring orientation



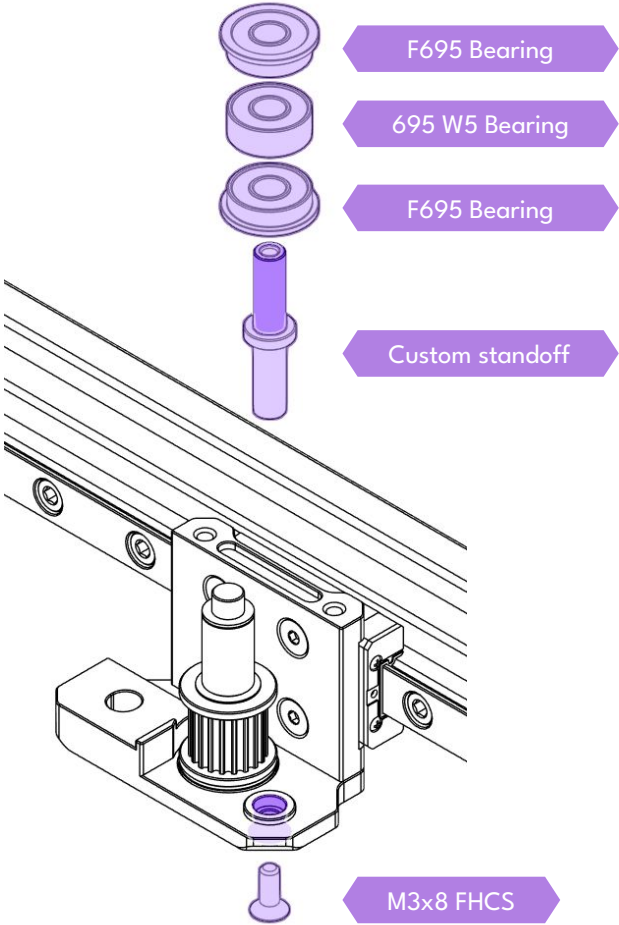
LIVE-SHAFT TIME

Insert the previously prepared live-shaft assembly into the pre-installed F695 bearing with the idler on the top.



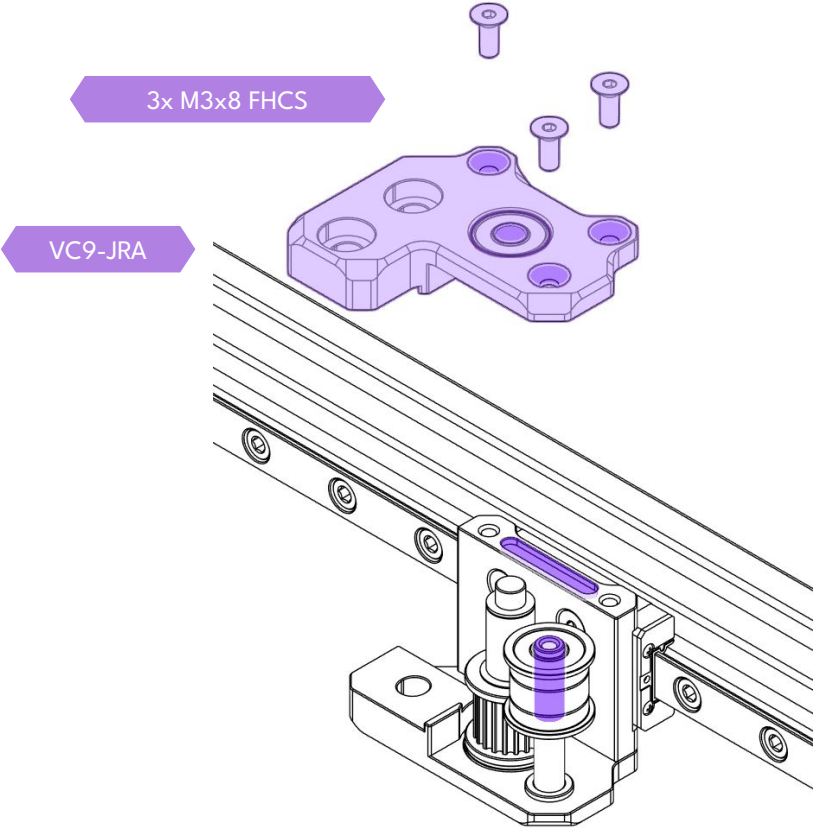
INSTALL THE IDLER STACK

Secure the standoff with a screw, then slide the bearings onto it.



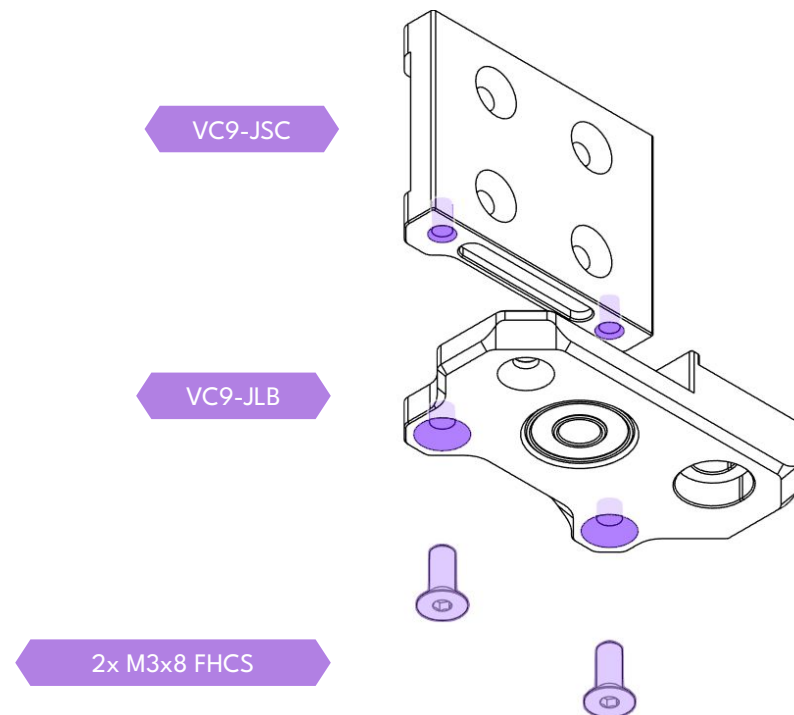
TOP IT OFF

Get the top of the XY-joint and gently put in on the middle part and the live-shaft. The fit will be the tightest at the bearing, so make sure you're not forcing it at an angle.



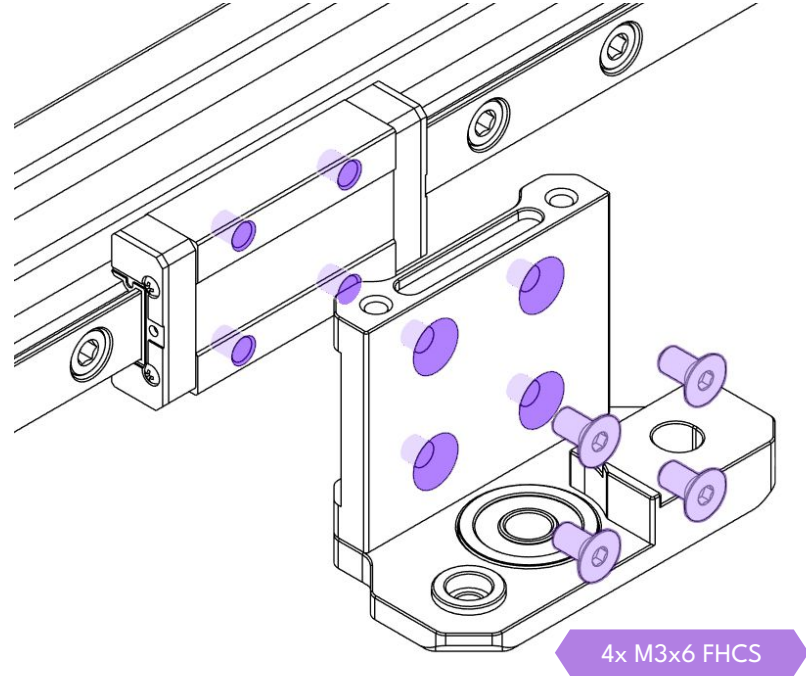
PREPARE THE PARTS

Attach the bottom part of the XY-joint to the middle part with 2 screws.



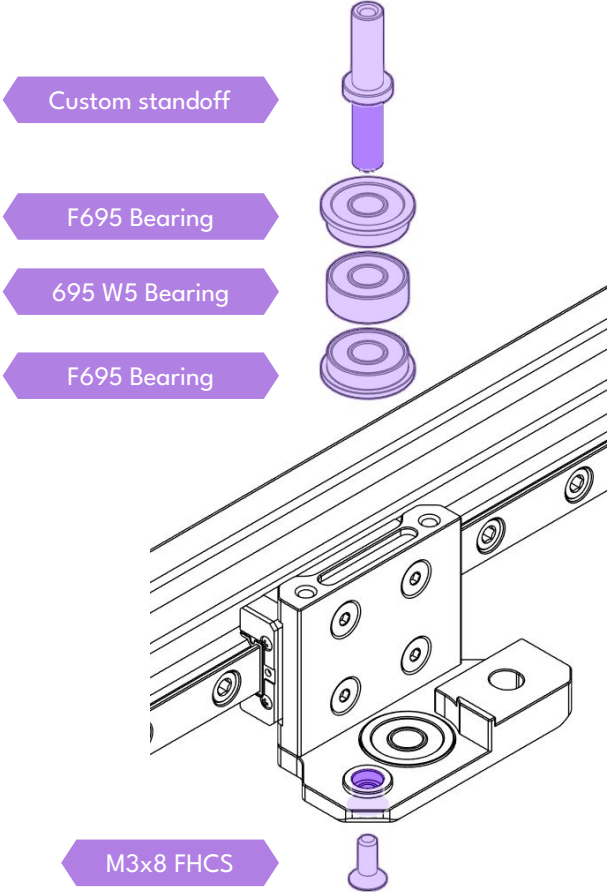
PUT THEM ON THE CARRIAGE

Mount the previously assembled milled part duo onto the MGN9H Y-carriage with 4 screws.



INSTALL THE IDLER STACK

Slide the 3 bearings onto the standoff, then secure them with a screw.

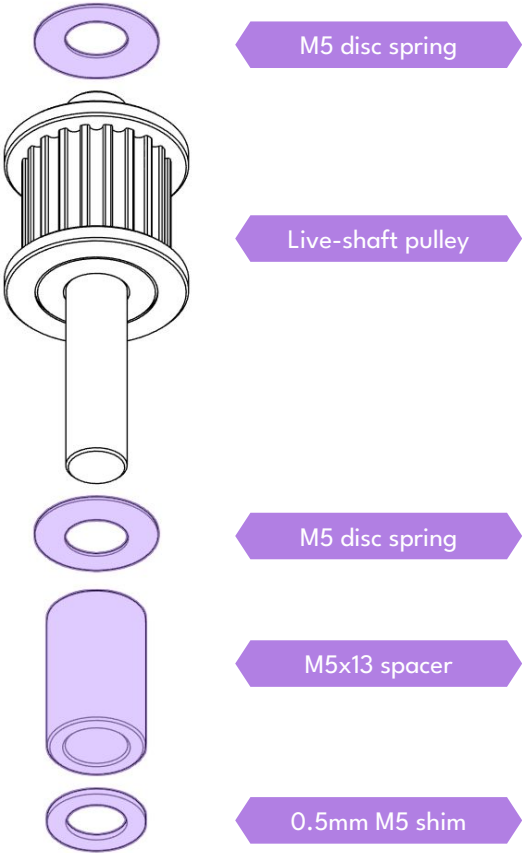
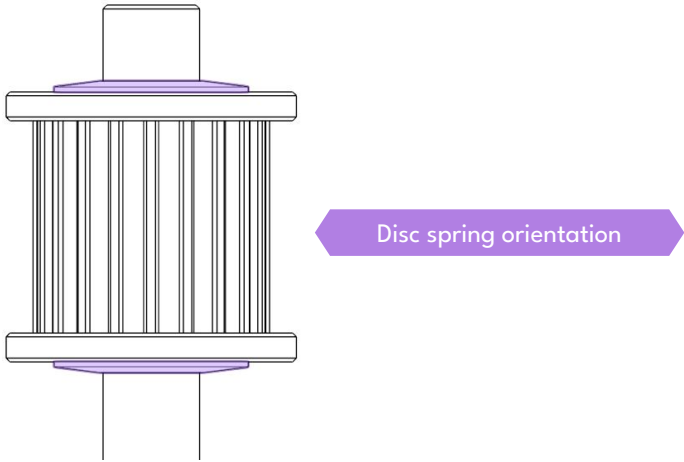


LEFT AWD XY-JOINT

MONOLITH

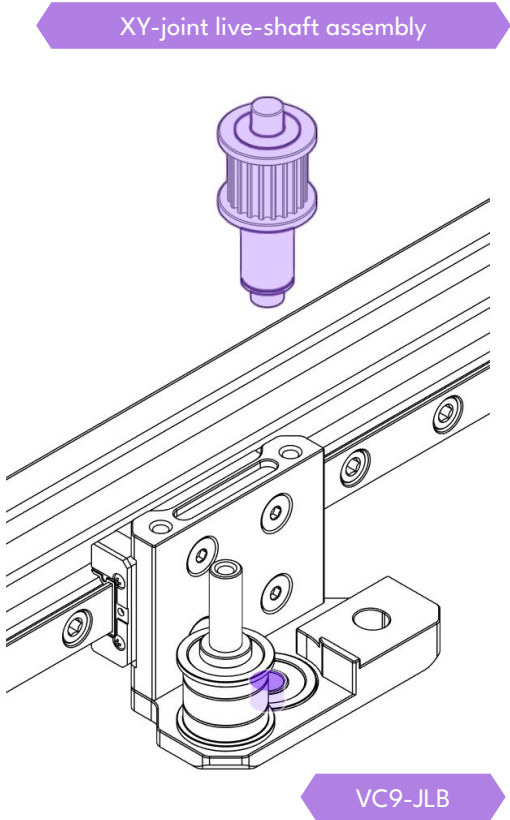
XY-JOINT LIVE-SHAFT ASSEMBLY

Put together the live-shaft assembly for the XY-joint. Make sure the disc springs are in the correct orientation.



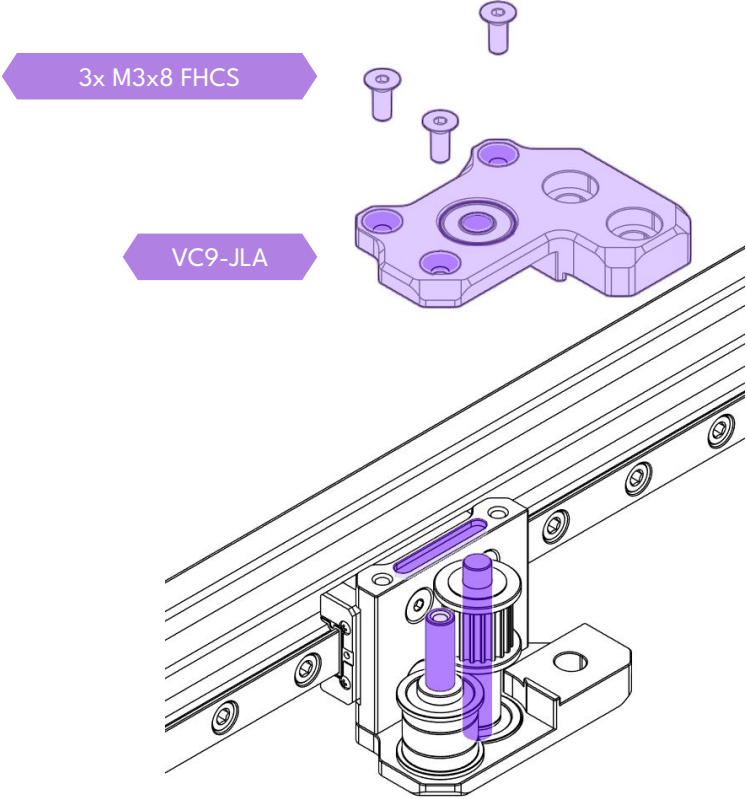
LIVE-SHAFT TIME

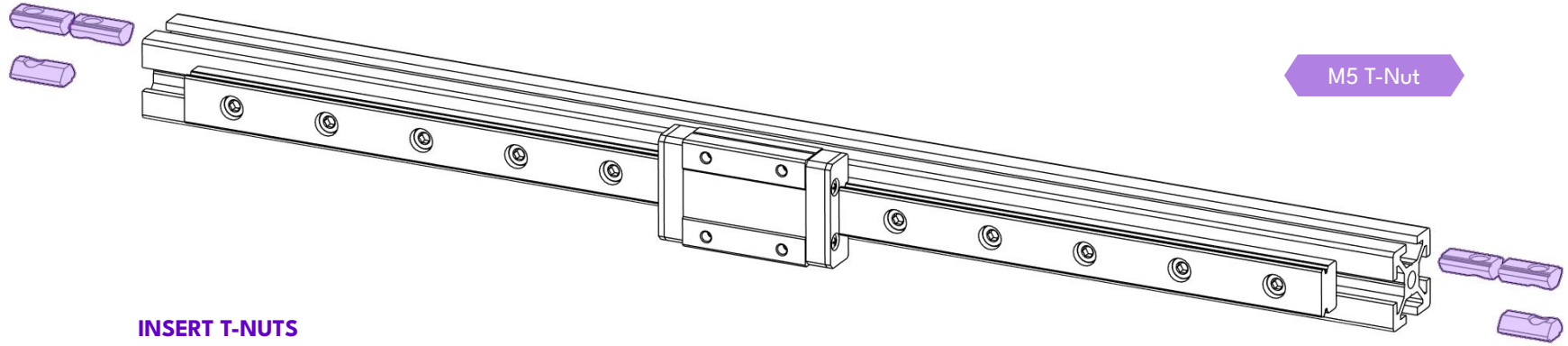
Insert the previously prepared live-shaft assembly into the pre-installed F695 bearing with the idler on the top.



TOP IT OFF

Get the top of the XY-joint and gently put in on the middle part and the live-shaft. The fit will be the tightest at the bearing, so make sure you're not forcing it at an angle.

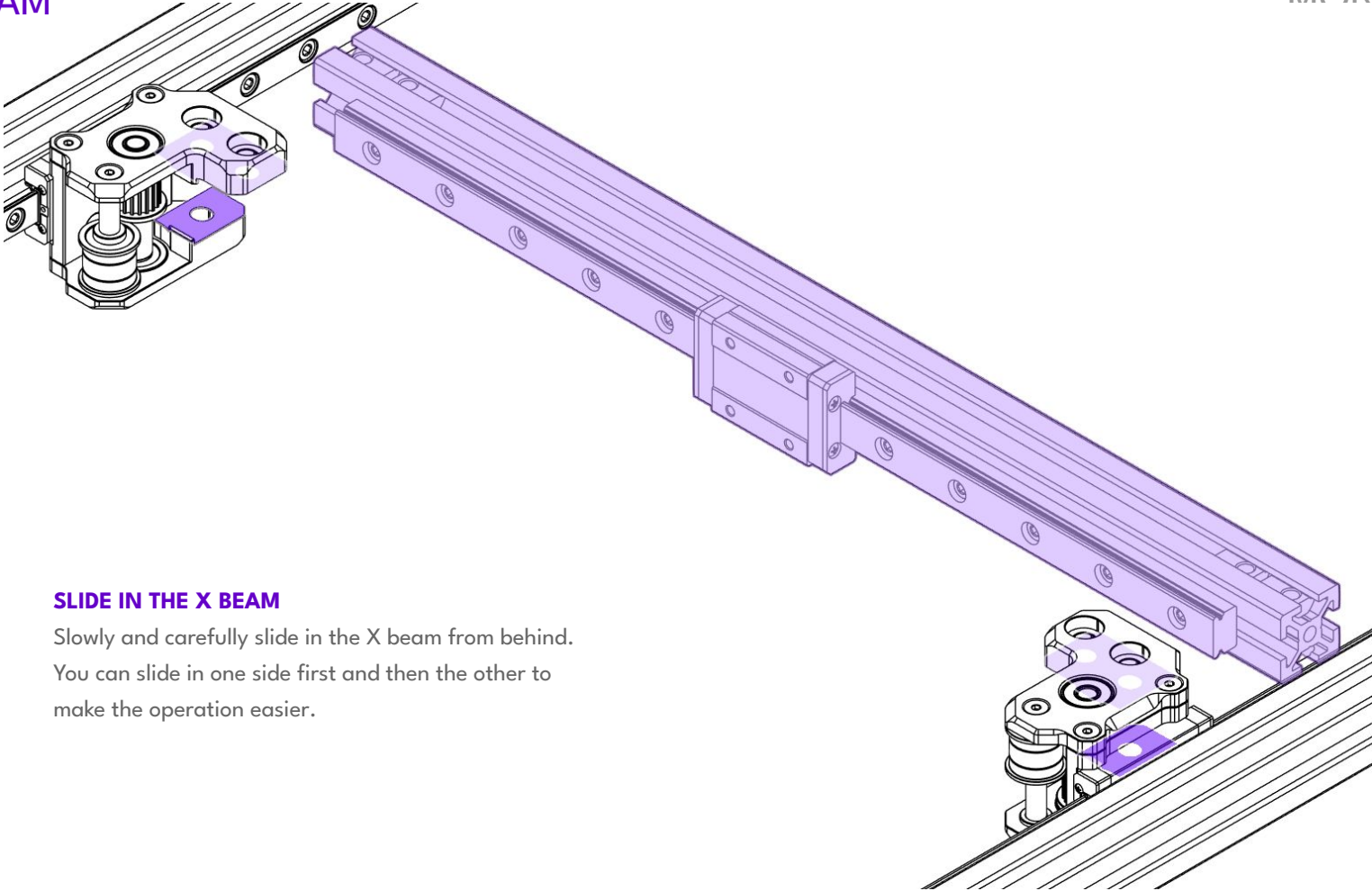


**INSERT T-NUTS**

Take your X extrusion and rail, and slide in two T-nuts per side on top, and one per side at the bottom

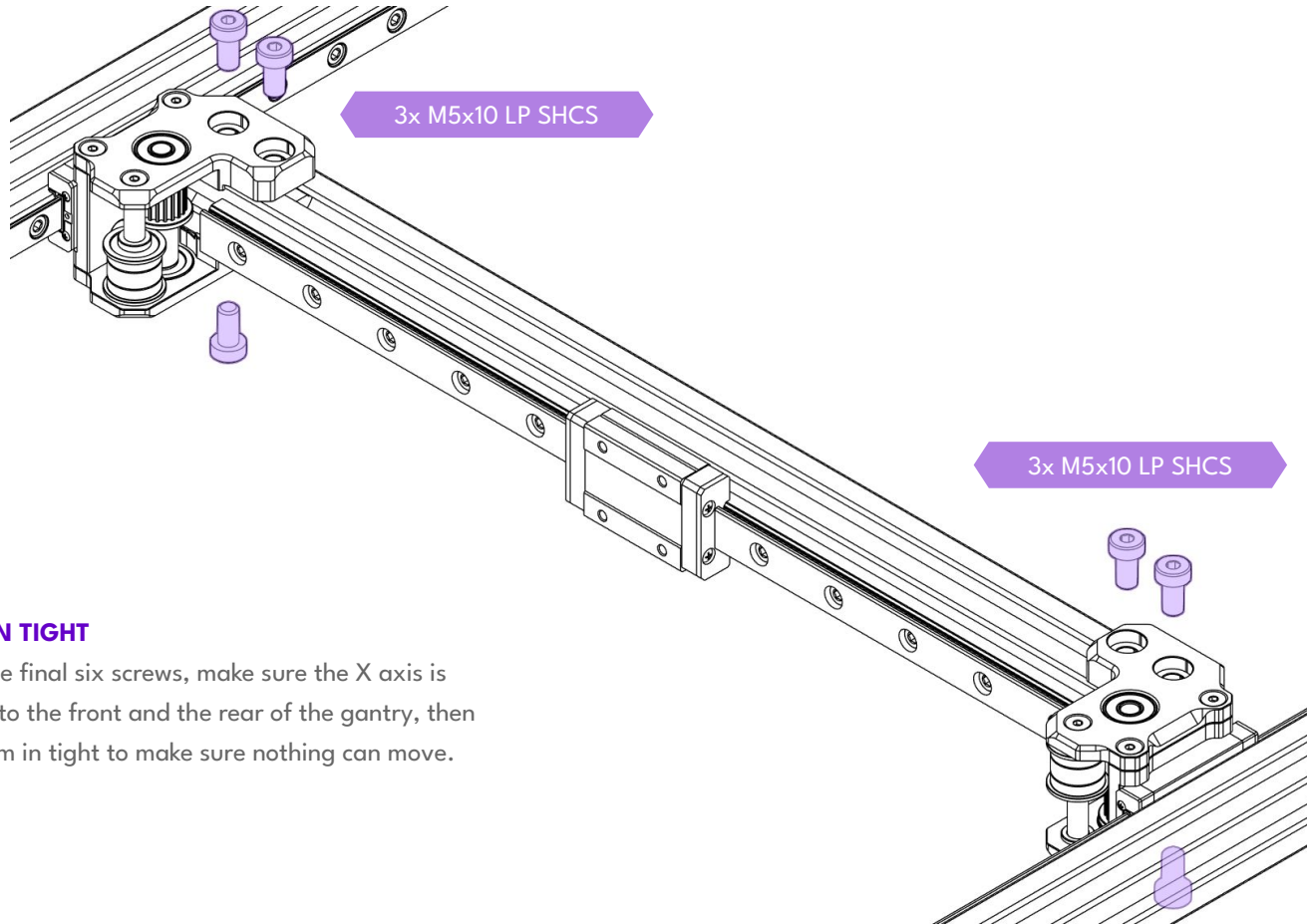
X BEAM

MONOLITH



SLIDE IN THE X BEAM

Slowly and carefully slide in the X beam from behind.
You can slide in one side first and then the other to
make the operation easier.

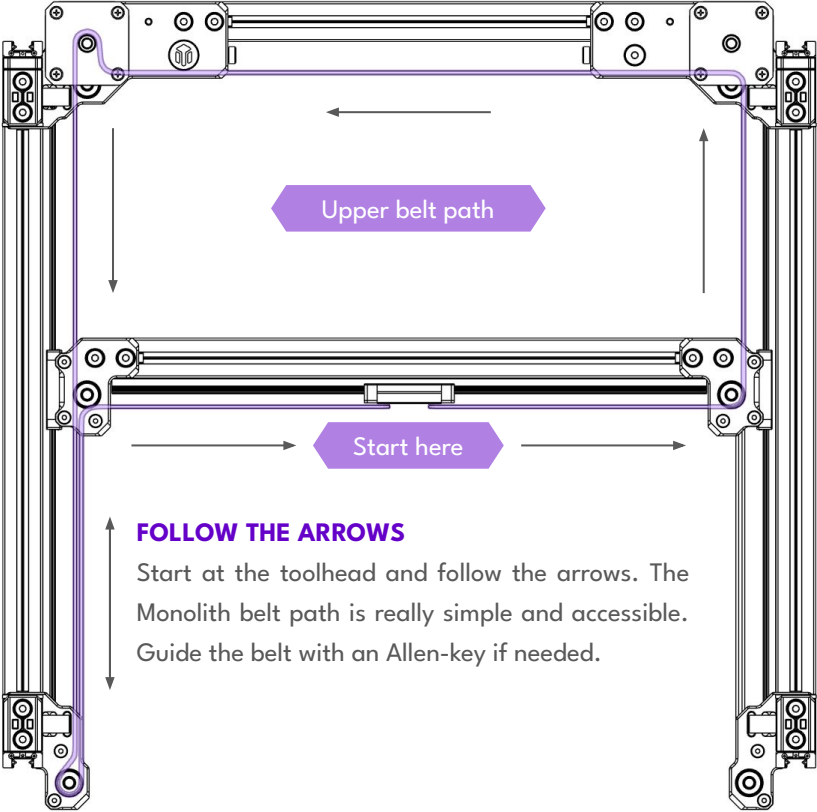


SCREW IN TIGHT

Slide in the final six screws, make sure the X axis is trammed to the front and the rear of the gantry, then screw them in tight to make sure nothing can move.

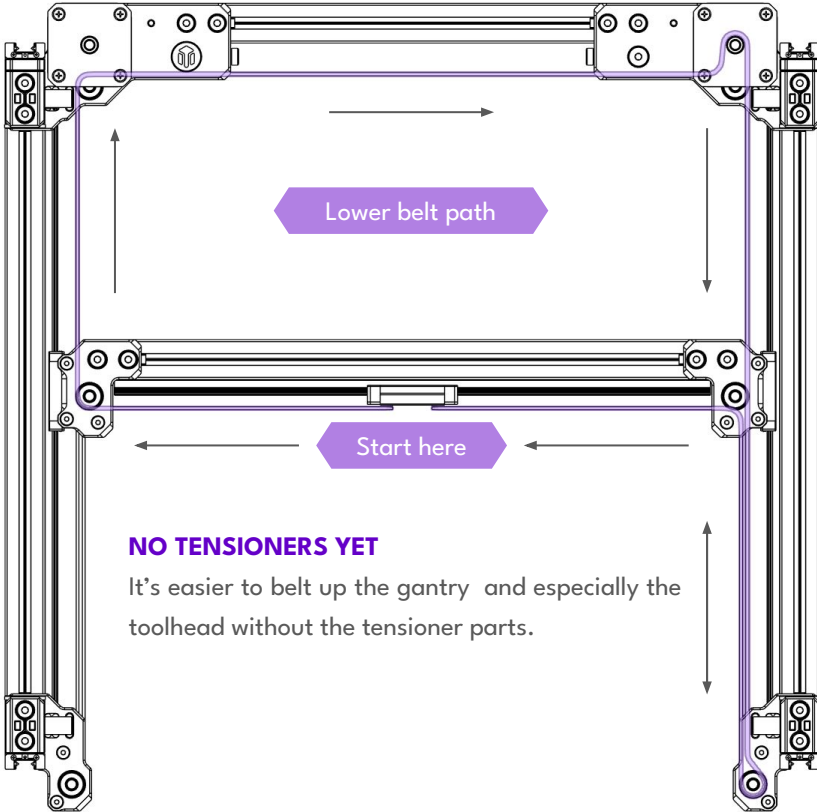
2WD BELT PATH

MONOLITH



FOLLOW THE ARROWS

Start at the toolhead and follow the arrows. The Monolith belt path is really simple and accessible. Guide the belt with an Allen-key if needed.

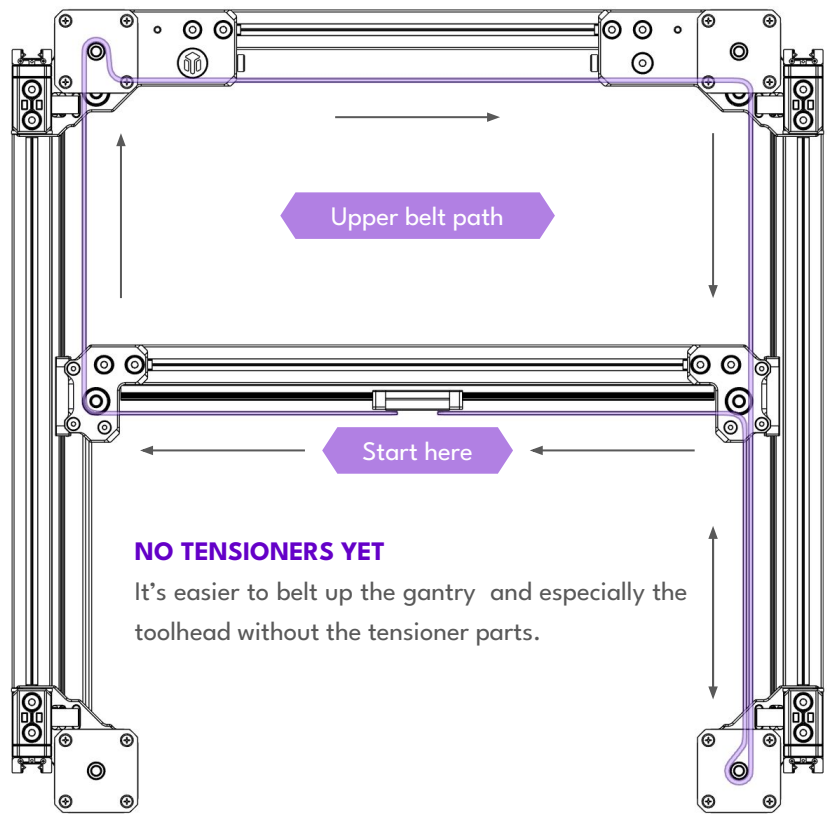
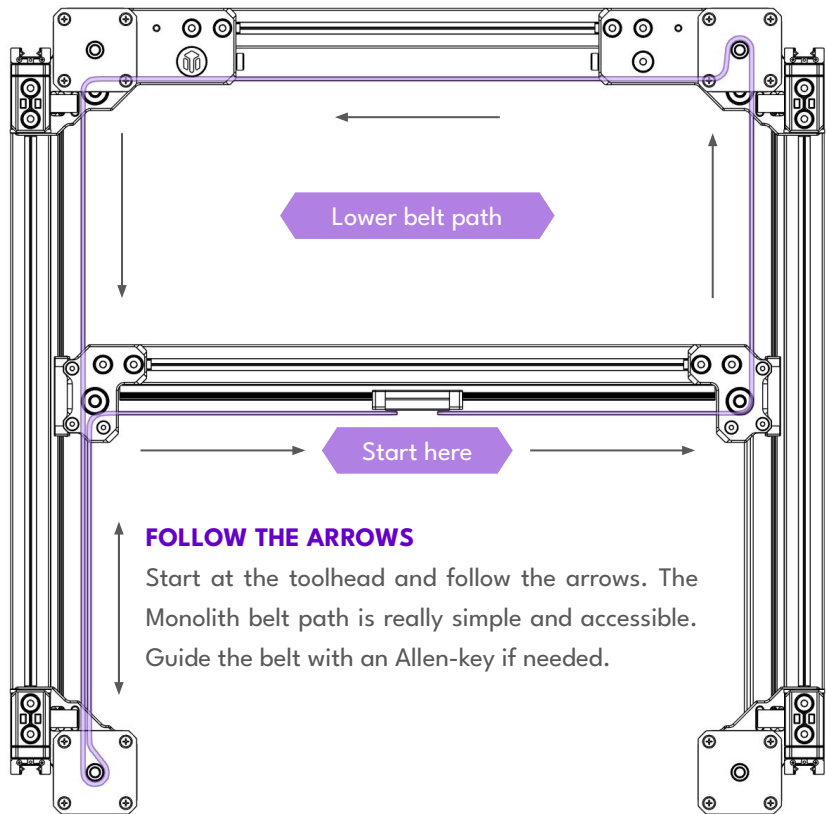


NO TENSIONERS YET

It's easier to belt up the gantry and especially the toolhead without the tensioner parts.

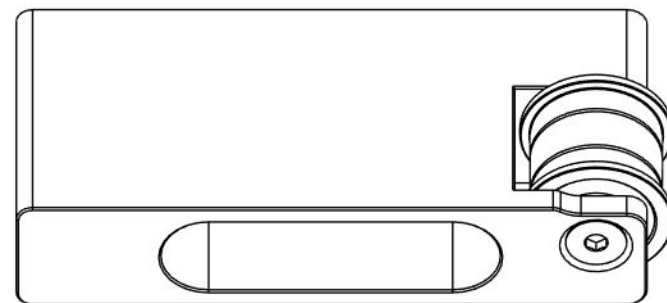
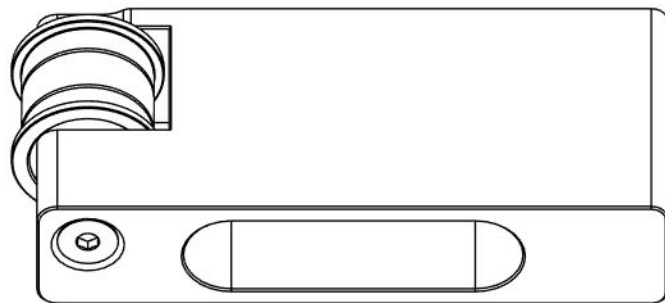
AWD BELT PATH

MONOLITH



TENSIONERS

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TENSIONERS

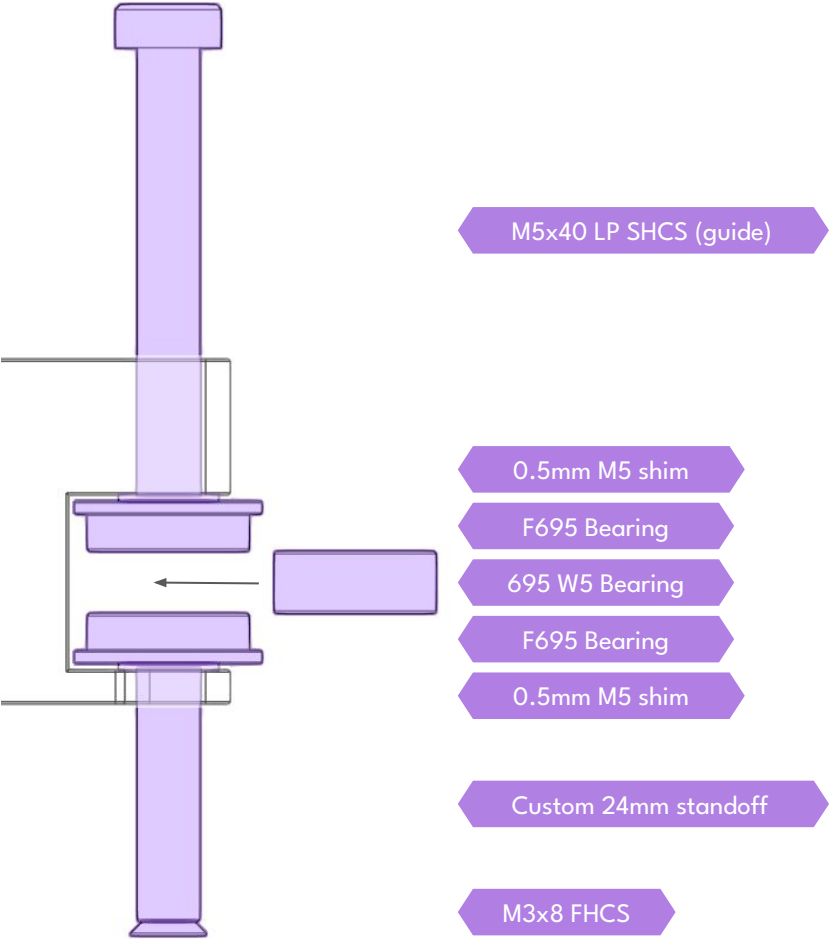
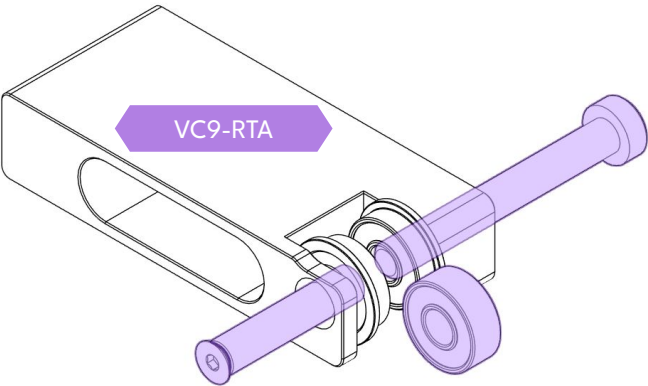
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PRE-ASSEMBLED

The tensioners should come pre-assembled, but if you ever need to replace the bearings, here's how to do it the easy way.

MIDDLE BEARING LAST

By using the M5 tensioner screw as a guide and by putting the whole assembly on a flat surface, you can more easily align the F695s and shims. This leaves the middle 695 W5 bearing last, which then can be easily inserted between the two half-stacks.

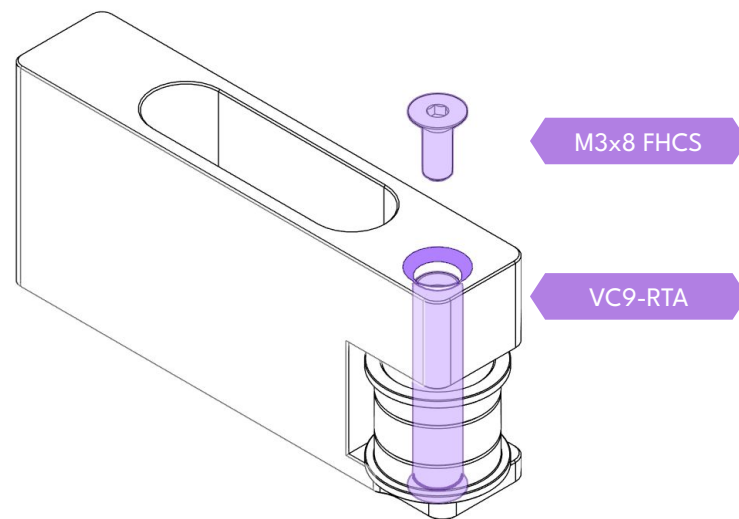


FINISH THE ASSEMBLY

Once the middle bearing is in place, push out the M5 screw that was used as a guide with the standoff. The bottom screw on the should be fully threaded in before you add the second one up top.

IDENTICAL SIDES

The tensioner assemblies are the same for both sides, so repeat the same steps.



TENSIONERS

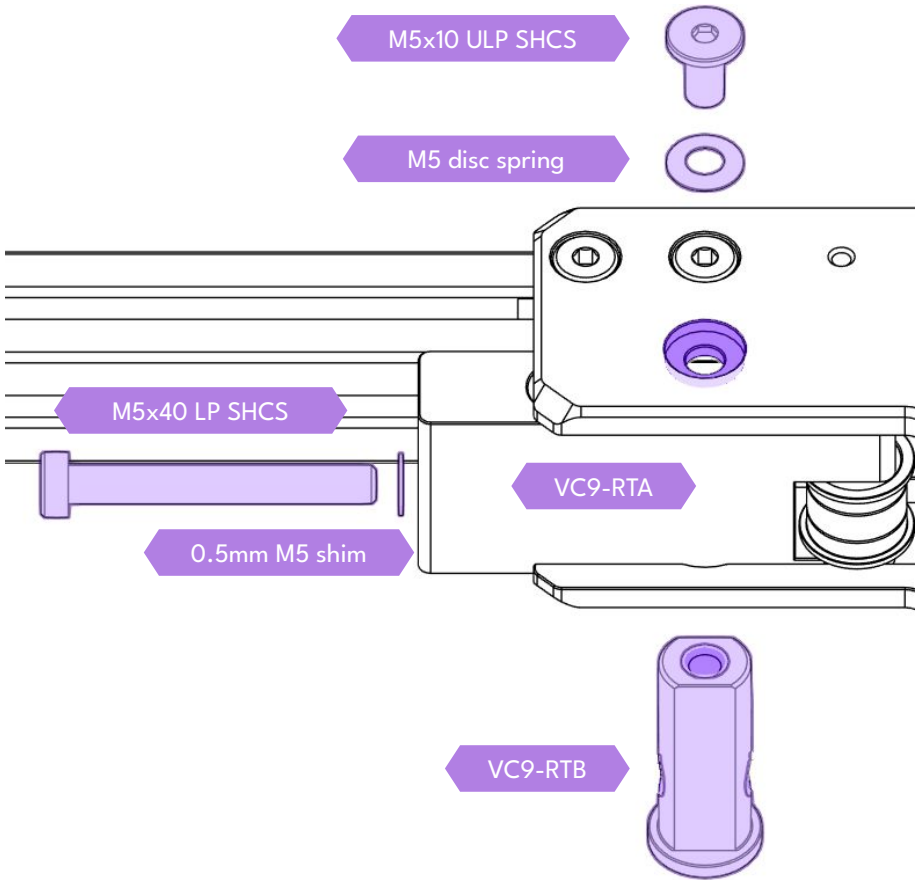
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INSTALL THE TENSIONER

The cylindrical tensioner “key” goes through the middle of the larger tensioner part. With the disc spring and the screw you can set how sticky you want the tensioner to be. You don’t have to loosen/tighten this secondary screw every time you want to adjust the tension.

IDENTICAL SIDES

The tensioner assemblies are the same for both sides, so repeat the same steps but upside down.



THANK YOU

Join us on Discord!

discord.gg/monolith3d

